# MAMMA ADA CENTRE

In Collaboration with FACM ITALY

PG 1st Year

*From Intelligence to Impact Beyond Automation*

*Research Build Innovate for a Smarter Tomorrow*

# GAIO

# Global AI Olympiad

**Think | Explore | Create | A Better Tomorrow**

## 6 KEY LEARNING AREAS (PG 1st Year)

### 1 Modern Intelligence Explained

- AI Beyond Automation
- Core Concepts, Trends & Real-World Applications

### 2 Develop with Intelligence

- Design Smart Workflows
- Build Intelligent Applications

### 3 AI for Advanced Research

- Research Methods
- Explore New Frontiers in AI
- Publications & Innovation

### 4 AI Specialist Roadmap

- Research to Profession
- Skills for AI Careers
- Domain Specialization

### 5 AI Research Workbench

- Intelligent Prototype Lab
- Experiment
- Build & Test Real Projects

### 6 Transparent AI

- Trustworthy AI Systems
- Explainability
- Data Privacy Matters

Knowledge | Research | Innovation | Collaboration | Global Perspective | Real-World Impact

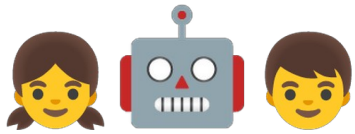
# This book belongs to

**Name:** _____

**Class & Section:** _____

**School:** _____

# 📖 PREFACE



AI is transforming every profession, from engineering and medicine to commerce, law and media. As a university student, you must understand how AI works, judge its outputs critically and apply it responsibly in your discipline. This book builds exactly those capabilities.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of PG 1st Year students, with university-level depth.

Every topic opens with an in-depth "Concept Explained" chapter, followed by a Concept Deep Dive, an AI Story, a case study, a lab activity and a Python Code Lab, with worksheets, quizzes, challenge problems and university-style exam preparation.

Responsibility runs through every chapter: you will examine AI's limits, bias, errors, privacy risks and impact on work, and the ethical duties of everyone who builds or uses it.

### 🤝 A note for teachers and parents

Faculty can use the Concept Explained chapters for lectures, the Code Labs and lab activities for practical sessions, and the exam-prep pages for assessment. Encourage students to verify AI outputs, cite sources and reflect on ethics. An answer key is provided at the end of the book.

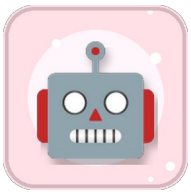
*We hope this book helps you think critically, build confidently and lead responsibly in an AI-powered world. Your ideas today can change tomorrow!*

With best wishes,

**The GAIO Academic Team**

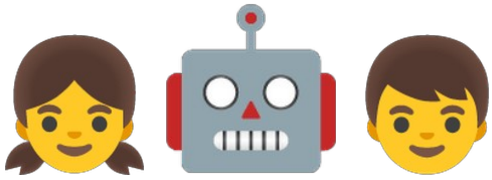
Mamma Ada Centre • In collaboration with FACM, Italy

# Welcome, Innovators!



**MEET BOLT!**

Welcome to university-level AI! I am Bolt, your AI guide. This year you will write real Python, use AI as a coding partner, study AI in your profession and practise exam-style answers. Let's build your AI edge!



## Look for these pictures in your book:

**CONCEPT EXPLAINED** Concept explained in depth

**CONCEPT DEEP DIVE** Detailed explanation & code

**SCENARIO** Real-life scenario

**CASE STUDY** Analyse a real case

**LAB ACTIVITY** Hands-on lab

**WORKSHEET** Practice exercises

**PUZZLE & DEFINE** Puzzles & definitions

**QUIZ TIME** MCQ practice

**AI STORY** An AI story to think about

**CHALLENGE PROBLEMS** Higher-order thinking

**CODE LAB** Python code lab

**EXAM PREP** University-style exam prep

# My 6-Month AI Journey

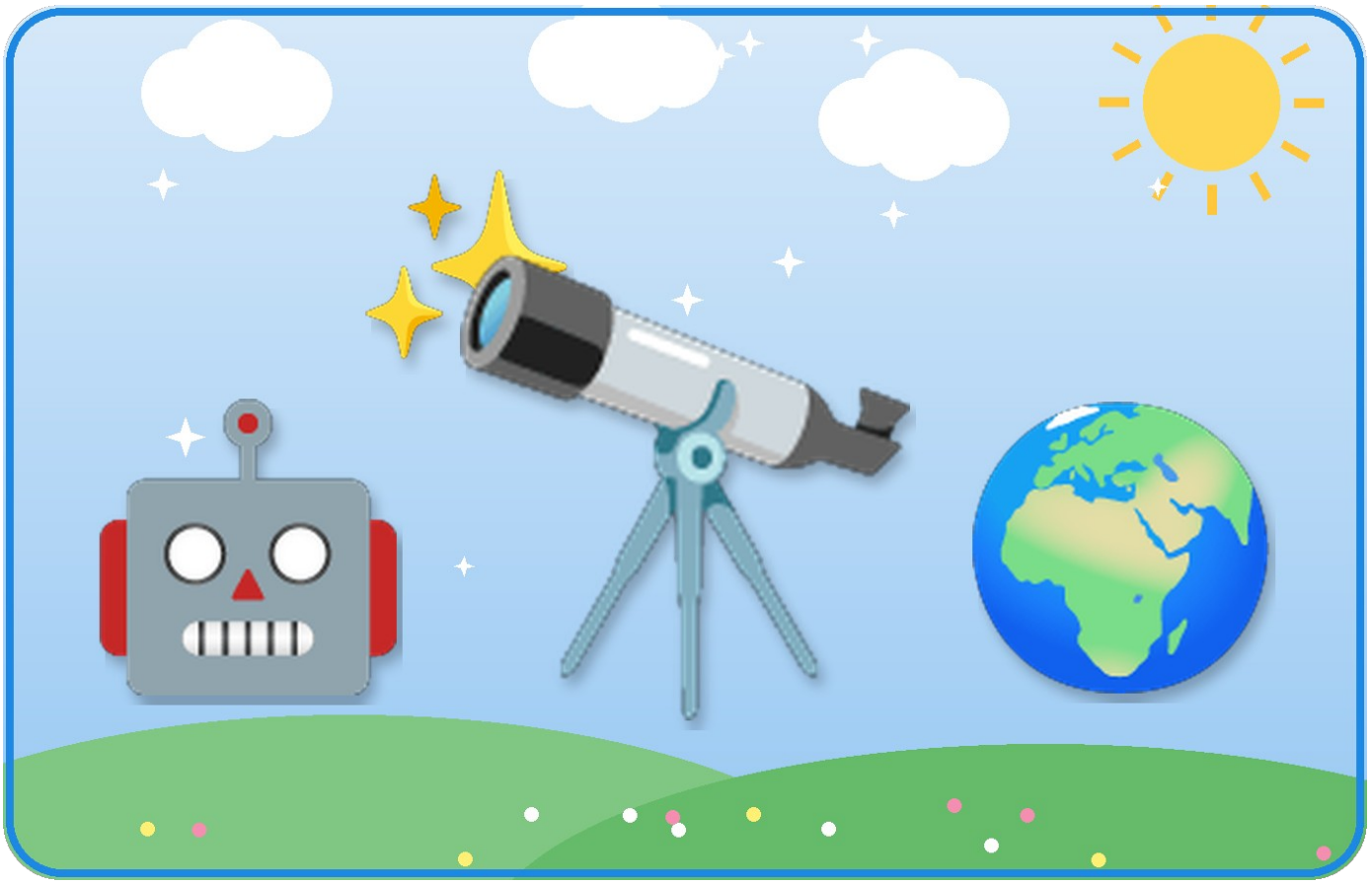
Follow the path. One month at a time!

| | | |
|---|---|---|
| | **MONTH 1**<br>**AI DISCOVER**<br>AI Basics | **Modern Intelligence Explained • AI Beyond Automation** |
| | **MONTH 2**<br>**AI CONNECT**<br>Communicate with AI | **Develop with Intelligence • Design Smart Workflows** |
| | **MONTH 3**<br>**AI SOLVE**<br>AI Applications | **AI for Advanced Research • AI for Professional Innovation** |
| | **MONTH 4**<br>**AI RISE**<br>AI & Career | **AI Specialist Roadmap • Research to Profession** |
| | **MONTH 5**<br>**AI CREATE**<br>AI Tools | **AI Research Workbench • Intelligent Prototype Lab** |
| | **MONTH 6**<br>**AI CARE**<br>Responsible AI | **Transparent AI • Trustworthy AI Systems** |

MONTH 1

# AI DISCOVER

See • Understand • Explore



**Area:** AI Basics

## In this unit, you will explore:

Topic 1

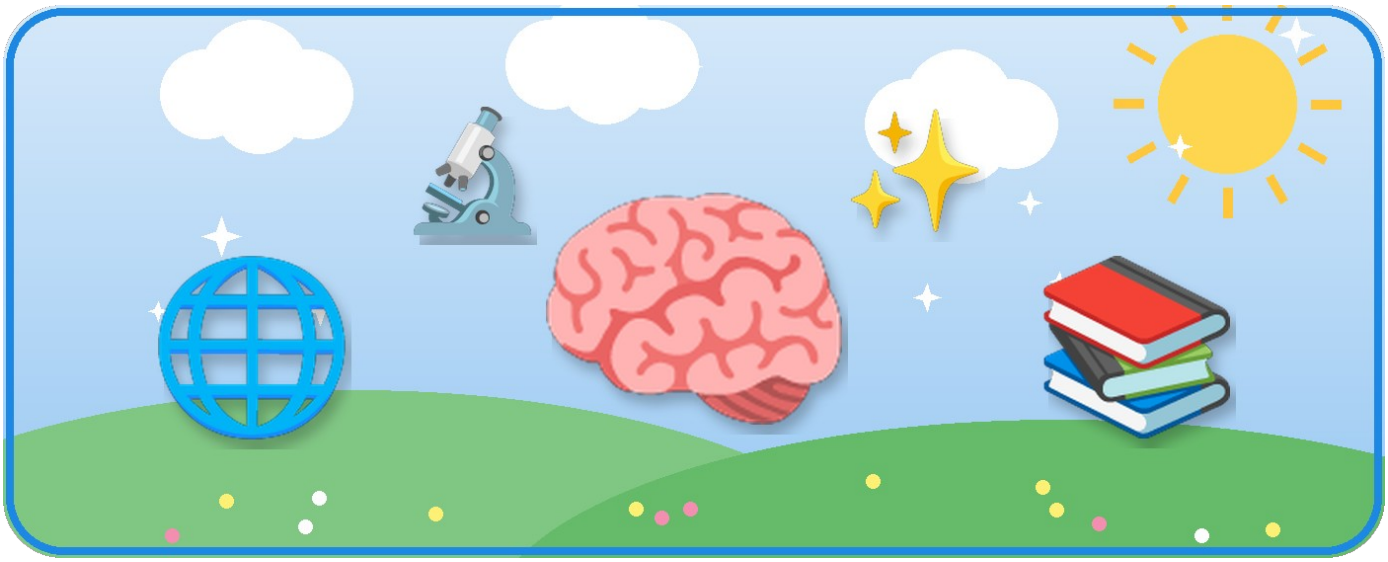
### Modern Intelligence Explained

Topic 2

### AI Beyond Automation

TOPIC 1 • MONTH 1 • AI BASICS

# Modern Intelligence Explained



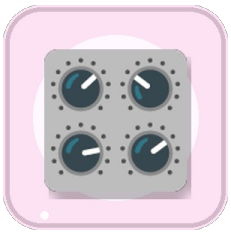
## LET'S LEARN

- Modern AI learns representations from data at scale.
- Self-supervised, multimodal and RL methods dominate.
- Capability is jagged: superhuman here, brittle there.

### Pillars of modern AI

**Representations**

**Self-supervision**



**Multimodality**

**Reinforcement**

## KEY TERMS

**SSL** Self-supervised learning

**Embedding** Learned vector

**Multimodal** Many data types

# CONCEPT EXPLAINED: Modern Intelligence Explained

### In this chapter

Modern AI is not a single algorithm but a family of methods that learn representations from very large datasets. Foundation models, trained once on broad data and adapted to many tasks, now underpin language, vision, speech and scientific applications. At postgraduate level you need more than a user's view: you must understand how these systems learn, what their internal representations capture, and why their abilities are powerful yet uneven.

## 1. Learning signals: from labels to self-supervision

Classical supervised learning depends on human labels, which are expensive and limited. Self-supervised learning creates the training signal from the data itself: predict the next token, fill in masked words, reconstruct hidden image patches, or match an image to its caption. Because the internet supplies vast unlabelled text, images and code, self-supervision lets models learn from orders of magnitude more data than any labelled dataset could provide.

Other signals complement it. Reinforcement learning optimises behaviour from rewards, as in game-playing systems and in fine-tuning language models with human or AI feedback. Contrastive learning pulls related pairs together and pushes unrelated ones apart. A modern pipeline often chains these: large-scale self-supervised pre-training, supervised instruction tuning, then preference optimisation to shape helpful, safe behaviour.

**WORKED EXAMPLE**

Pipeline sketch: pre-train on trillions of tokens (predict next token) → instruction-tune on curated prompt–response pairs → preference-tune using rankings of alternative answers → evaluate on held-out tasks and safety tests.

## 2. Representations and embeddings

What a network learns is best understood as a representation: an internal vector space where similar inputs lie close together. Embeddings map words, sentences, images or molecules to vectors of hundreds or thousands of dimensions. Distances and directions in this space carry meaning, which is why cosine similarity of embeddings powers search, recommendation, clustering and retrieval-augmented generation.

Good representations transfer. Features learned while predicting text help with classification, translation or question answering after light adaptation. This is why fine-tuning, adapters such as LoRA, and even simple prompting can repurpose one foundation model for many tasks. Research on probing and interpretability tries to discover which concepts are encoded and where, and how reliably the model uses them.

🤔 **PAUSE & THINK**

Two sentences have an embedding cosine similarity of 0.92 but mean opposite things ('the drug helped' vs 'the drug barely helped'). What does this reveal about relying on similarity alone?

## 🎛️ 3. Architectures and modalities

The transformer, built on self-attention, is the dominant architecture because it models long-range relationships and scales efficiently on parallel hardware. Vision transformers treat image patches as tokens; audio models treat short slices of sound as tokens. Diffusion models, which learn to reverse a gradual noising process, lead image, audio and video generation. State-space and mixture-of-experts designs aim to improve efficiency.

Multimodal models connect these streams in a shared space, so a single system can read a chart, describe a photograph, transcribe speech or answer questions about a video. Combined with search and reinforcement learning, as in AlphaGo, modern systems can also plan. The broad lesson is convergence: similar architectures and training recipes now work across very different data types.

✏️ **WORKED EXAMPLE**

AlphaGo (2016) combined a policy network to suggest moves, a value network to judge positions and Monte Carlo tree search to look ahead. Its famous Move 37 against Lee Sedol emerged from this mix of learning and search.

## ⚡ 4. The jagged frontier of capability

Capability is uneven. The same model can draft a competent legal memo yet miscount letters in a word, solve olympiad problems yet fail a simple spatial puzzle phrased unusually. Researchers describe this as a jagged frontier: tasks just inside it are done well, tasks just outside it fail, and the boundary is not obvious from human intuitions about difficulty.

For postgraduate work this has methodological consequences. Never infer general ability from a few impressive demos. Evaluate on the exact distribution you care about, include adversarial and out-of-distribution cases, report failure modes alongside averages, and remember that benchmark scores can be inflated by contamination when test items appear in training data.

🖐 **TRY IT**

Pick a model you can access. Design ten tasks in your discipline, five you expect it to ace and five you expect it to fail. Run them, record results and map where your intuition about its frontier was wrong.

## ✅ Key takeaways

- Self-supervision lets models learn from vast unlabelled data; RL and preference tuning shape behaviour.
- Embeddings are learned vector representations; similarity in that space carries meaning.
- Transformers and diffusion models dominate; multimodal models share one representation space.
- Capability is jagged: strong and weak tasks sit side by side unpredictably.
- Evaluate on your target distribution and watch for benchmark contamination.

## ❓ Quick check: answer in one or two lines.

**1.** Explain how next-token prediction is a form of self-supervision.

**2.** Why do embeddings transfer across tasks?

**3.** Give two research practices that guard against the jagged frontier.

# 🔬 CONCEPT DEEP DIVE: Modern Intelligence Explained

🔷 Representation learning: models learn embeddings where similar inputs lie close, enabling transfer to new tasks.

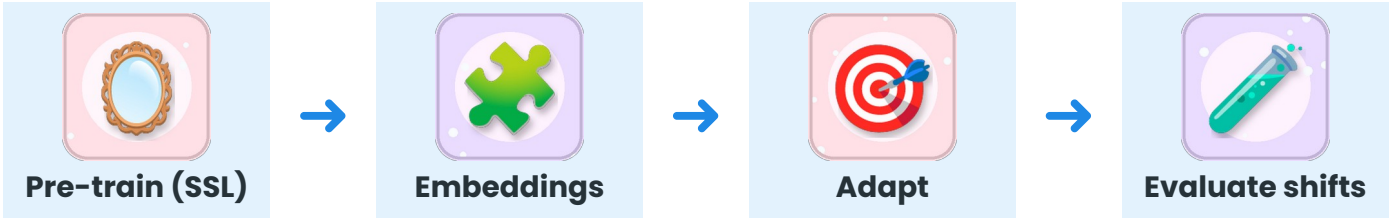
🔷 Self-supervised learning creates labels from data (masked tokens, next tokens, contrastive pairs), unlocking unlabelled corpora.

🔷 Foundation models are pre-trained broadly, then adapted via fine-tuning, adapters (e.g., LoRA) or prompting.

🔷 Multimodal models align text, images, audio and video in shared spaces; reinforcement learning optimises sequential decisions.

🔷 Capabilities are jagged: strong on some benchmarks, brittle on distribution shift, rare cases and reasoning traps.

## 🔄 How it works




## 💻 Cosine similarity of embeddings

```
import numpy as np
a, b = np.array([1, 0, 1]), np.array([1, 1, 1])
cos = a @ b / (np.linalg.norm(a) * np.linalg.norm(b))
print(round(cos, 2))   # 0.82
```

## ✍️ In your own words: explain this topic to a Class 10 student.

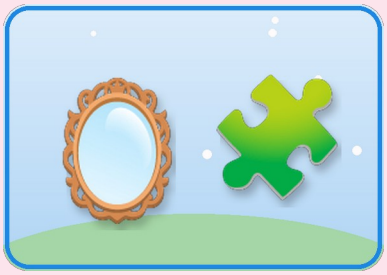
# AI STORY: The Model That Knew Too Little

**Characters:** *Meera (MSc Data Science), her supervisor Dr. Iyer, labmate Faizal.*

SCENE 1

Meera's thesis aimed to classify Tamil farmer queries for an agriculture helpline. She had only 500 labelled messages, and her first classifier reached 61% accuracy. Dr. Iyer asked what the model was actually learning from so little data.

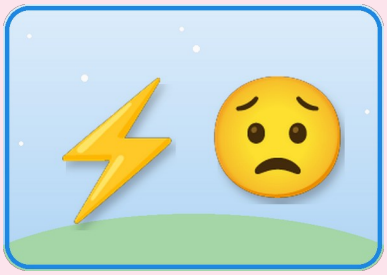
**Pause & predict:** Why do small labelled datasets limit supervised models?



SCENE 2

Faizal suggested starting from a multilingual model pre-trained with self-supervision on billions of words. Meera fine-tuned it on her 500 examples. Accuracy jumped to 84%, and she proudly drafted her results chapter that night.

**Pause & predict:** Why did pre-training help so much?



SCENE 3

Dr. Iyer tested it on messages written in Tamil script mixed with English words. Accuracy fell to 58%. Queries about rare pests were confidently misclassified. The model's strengths and weaknesses were jagged, not uniform.

**Pause & predict:** What does this reveal about evaluating on one test set?

SCENE 4

Meera built a test set with code-mixed text and rare categories, reported results per slice and added a 'route to expert' option for low-confidence cases. Her thesis became more modest and far more useful.

⏸ **Pause & predict:** How does slice-based reporting improve research honesty?

## 🌟 The big idea

*Pre-training gives powerful representations, but capability is jagged; evaluate where it matters.*

## 💬 Discuss with your classmates

**1.** What slices of data matter most in your own research area?

**2.** When should a model abstain rather than answer?

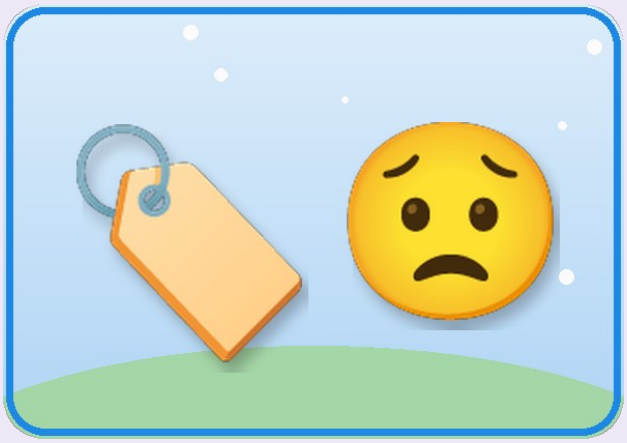
## SCENARIO: Modern Intelligence Explained

Read the scenario. Think about the choices made at each step.

**1**

**Meera's PG thesis needs Tamil text classification.**

**2**

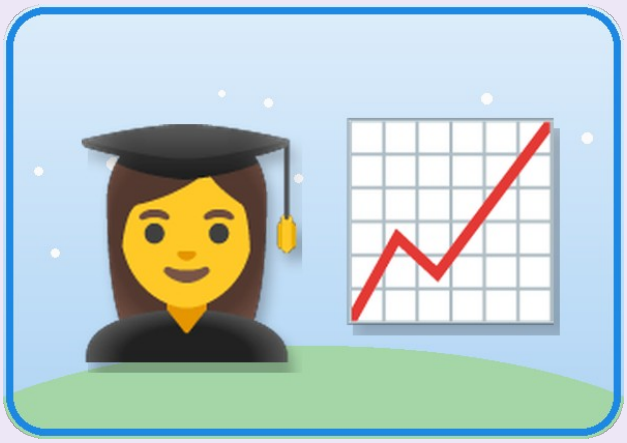


**She has only 500 labelled examples.**

**3**

**A self-supervised model gives rich embeddings.**

**4**



**Fine-tuning on 500 labels beats her baseline.**

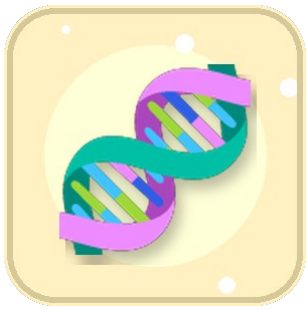
💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Where modern AI shines

**Game-playing**



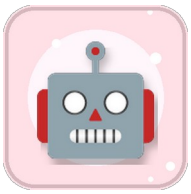
**Protein science**

**Speech & language**

**Vision**

**Robotics**

**Drug discovery**



**BOLT SAYS:**

I don't follow hand-written rules. I learn features from oceans of data!

**FUN FACT!**

In 2016 AlphaGo's 'Move 37' stunned Go experts; it was a move few humans would play.

**Reflect: where have YOU seen this in real life? Note it down.**

# CASE STUDY: AlphaGo's Move 37

**CASE FILE**

In March 2016, DeepMind's AlphaGo played world champion Lee Sedol in Seoul. In game two, its 37th move placed a stone in a position that commentators initially thought was a mistake; AlphaGo's own estimate suggested a human would play it about 1 in 10,000 times. The move proved strong, and AlphaGo won the match 4–1, a landmark for learning-based AI combining deep networks, search and reinforcement learning.

## Analyse the case

**1. Why couldn't experts predict Move 37?**

**2. Which learning methods did AlphaGo combine?**

**3. Is a superhuman Go player 'generally' intelligent? Why?**

**My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Mask & Predict

## You need:

Text passages

Markers

Pairs

## Steps:

1.
   Blank out 1 in 6 words of a passage.
2.
   Partner predicts the hidden words.
3.
   Score accuracy; note easy vs hard words.
4.
   Discuss what context gave the answer.

**THINK & TALK**

Why does masking teach meaning?

## Record your observations & results:

# 💻 CODE LAB: Embedding Similarity

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  import numpy as np
 2  a = np.array([1, 0, 1]); b = np.array([1, 1, 1]); c =
np.array([0, 1, 0])
 3  cos = lambda x, y: x @ y / (np.linalg.norm(x) *
np.linalg.____(y))
 4  print(round(cos(a, b), 2), round(cos(a, c), 2))
```

**COMPLETE** **T1. Fill the blank.**

**PREDICT** **T2. Both cosine values?**

**EXPLAIN** **T3. What does cosine 0 mean?**

**EXTEND** **T4. Find the most similar pair among 4 vectors.**

# WORKSHEET: Modern Intelligence Explained (Part A)

Name: _____ Date: _____

## 1. Match the paradigm to its learning signal.

**Supervised** ●

● **Rewards from actions**

**Self-supervised** ●

● **Human labels**

**Reinforcement** ●

● **Structure without labels**

**Unsupervised** ●

● **Labels made from the data itself**

## 2. Narrow or general capability? Tick ✓.

| | Narrow | General |
|---|---|---|
| Plays only Go | | |
| Writes, codes and translates | | |
| Reads one scan type | | |
| Captions, answers, draws | | |

# ✏️ WORKSHEET: Modern Intelligence Explained (Part B)

Name: _____ Date: _____

## 1. Masking arithmetic. Use the word box.

**WORD BOX: 15 ✿ 85 ✿ 40**

Mask 15% of 100 tokens → ____ masked

Visible tokens = ____

Predict 6 of 15 → ____%

## 2. Match the adaptation method to its idea.

**Fine-tuning** ● ●

**Add small trainable matrices**

**LoRA adapter** ● ●

**Retrieve documents at answer time**

**Prompting** ● ●

**Update model weights**

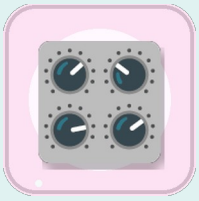
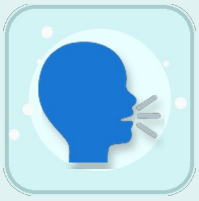
**RAG** ● ●

**Instructions only, no training**

# PUZZLE & DEFINE: Modern Intelligence Explained

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | |  | |
| B | | | | |
| C | | |  | |

## 2. Define it! Write each term in your own words with an example.

**SSL**

Meaning: _____

Example: _____

**Embedding**

Meaning: _____

Example: _____

**Multimodal**

Meaning: _____

Example: _____

# ? QUIZ TIME: Modern Intelligence Explained

**Circle the right answer!**

**Q1. Labels created from the data itself:**

a) Self-supervised

b) Hand-labelled

c) Random

**Q2. An embedding is a learned...**

a) Picture

b) Vector

c) Rule

**Q3. Learning from rewards is...**

a) Supervised

b) Clustering

c) Reinforcement

**Q4. Uneven AI ability is called...**

a) Jagged

b) Linear

c) Circular

**Q5. Move 37 was played by...**

a) A human

b) AlphaGo

c) A phone

**MY SCORE: _____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: Modern Intelligence Explained

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | Self-supervised learning creates labels from data. | | |
| | Foundation models need no data. | | |
| | Embeddings place similar items close. | | |
| | Multimodal models handle several data types. | | |
| | A high benchmark score proves general intelligence. | | |
| | AlphaGo won its 2016 match 4–1. | | |

**HOME CONNECT**

Ask a chatbot to explain a concept in 3 languages you know. Where did quality drop, and why might that be?

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Modern Intelligence Explained

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does self-supervision reduce labelling needs?**

COMPARE **Q2. Fine-tuning vs prompting a foundation model.**

APPLY **Q3. Cosine of [1,0,1] and [1,1,1]?**

EVALUATE **Q4. Do high benchmark scores prove understanding?**

CREATE **Q5. Design an SSL task for Tamil speech audio.**

# EXAM PREP: Modern Intelligence Explained

Name: _____ Date: _____

**Time: 30 min Max marks: 13 Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** Define self-supervised learning.

**2.** What is a foundation model?

**3.** Explain 'jagged' capability.

**4.** Name two adaptation methods.

## PART B • Long answer (5 marks)

**5.** Explain the paradigms behind modern AI and critically assess claims of general intelligence.

TOPIC 2 • MONTH 1 • AI BASICS

# AI Beyond Automation

## LET'S LEARN

- Beyond automating tasks, AI augments experts.
- It aids discovery, creativity and decisions.
- Best results pair AI strengths with human judgement.

### Four roles beyond automation

**Augment**

**Discover**

**Co-create**

**Decide**

## KEY TERMS

**Augment** Boost human work

**2nd reader** AI double-check

**DSS** Decision support

# 📖 CONCEPT EXPLAINED: AI Beyond Automation

### 💡 In this chapter

Automation replaces a human step with a machine step. Much of AI's value, however, lies elsewhere: augmenting expert judgement, accelerating discovery, generating options and supporting decisions. Postgraduates, who will often be the experts using these systems, need to design human–AI collaboration deliberately, so that the combination outperforms either alone rather than inheriting the weaknesses of both.

## 🤝 1. From automation to augmentation

A useful framing is a spectrum of roles. AI can automate a task end to end; augment a person by drafting, summarising or flagging; act as a second reader that checks human work; or serve as a discovery engine that proposes hypotheses humans test. Each role implies different requirements for accuracy, explanation and oversight, and the same model may play several roles in one workflow.

Augmentation tends to win when tasks are varied, stakes are high and context matters. In such settings the human supplies goals, judgement and accountability, while AI supplies speed, recall and consistency. Designing for augmentation means deciding who sees what, in which order, and who has the final say when the AI and the human disagree.

### ✏️ WORKED EXAMPLE

Mammography in Sweden (MASAI trial): AI triaged screening exams by risk, so low-risk scans had one radiologist and high-risk scans had two. Cancer detection was maintained or improved while radiologist screen-reading workload fell substantially.

## 🔭 2. AI for discovery and creativity

In science, AI now proposes rather than merely processes. Protein structure prediction, materials screening, drug candidate generation and automated literature synthesis expand the space of ideas researchers can explore. The human role shifts toward framing questions, selecting promising candidates and designing experiments that confirm or refute AI suggestions.

In creative and design work, generative models offer many variants quickly. The risk is homogenisation: if everyone uses the same model and prompts, outputs converge. Strong practitioners treat AI output as raw material, combine it with domain insight and

personal style, and keep a record of which ideas came from where for honesty and credit.

🤔 **PAUSE & THINK**

If an AI system suggests a hypothesis that you then confirm experimentally, how should authorship and credit be described in your paper?

## ⚖️ 3. Decision support and its pitfalls

Decision support systems present predictions, risks or recommendations to human decision-makers. Well-designed systems show calibrated probabilities, relevant evidence and uncertainty, and fit into existing workflows. Poorly designed ones overwhelm users with alerts, hide their reasoning or present guesses with false precision.

Two opposite failures recur. Automation bias leads people to accept AI advice even when it is wrong, especially under time pressure. Algorithm aversion leads them to ignore AI after seeing one mistake, even when it is usually right. Mitigations include showing confidence, asking users to commit to their own judgement before seeing AI advice, and training them on cases where the AI is known to fail.

✏️ **WORKED EXAMPLE**

Clinical alerts: a sepsis model firing on 20% of patients will be ignored; one that fires on 3% with a clear explanation and recommended next step is far more likely to change outcomes.

## 🧪 4. Measuring human–AI teams

The right question is not 'is the AI accurate?' but 'does the team perform better?'. Evaluate the human alone, the AI alone and the human with AI on the same cases. Measure accuracy, time, confidence, error types and user trust. Sometimes the combination underperforms the AI alone because humans override correct advice; sometimes it beats both.

Field studies matter because lab results rarely transfer cleanly. Randomised trials, staggered rollouts and careful monitoring of workload, burnout and equity reveal effects that offline metrics miss. Report results by subgroup so improvements for some users do not hide harms for others.

**TRY IT**

Design a three-arm study for an AI writing-feedback tool in your department: students alone, AI alone, students with AI. Define the cases, metrics, sample size and how you will avoid learning effects.

## Key takeaways

- AI roles: automate, augment, second reader, discovery engine.
- Augmentation suits varied, high-stakes, context-rich work.
- In discovery, humans frame questions and validate AI-generated candidates.
- Watch for automation bias and algorithm aversion in decision support.
- Evaluate human, AI and human+AI on the same cases, by subgroup.

## Quick check: answer in one or two lines.

**1.** Contrast automation and augmentation with one example each.

**2.** What is automation bias and how can interface design reduce it?

**3.** Why compare three arms when evaluating a decision support tool?

# 🔬 CONCEPT DEEP DIVE: AI Beyond Automation

🔷 Automation replaces task execution; augmentation improves human performance; AI can also enable new discoveries and creative work.

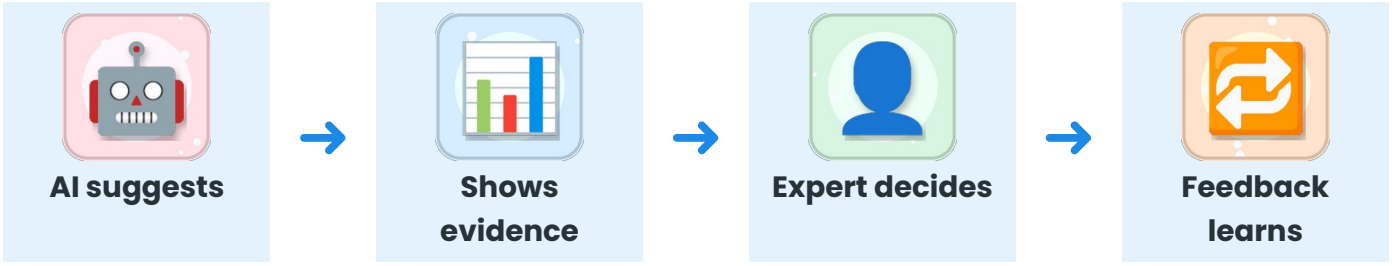
🔷 Human–AI teams outperform either alone when roles are clear and AI uncertainty is visible.

🔷 Automation bias (over-trust) and algorithm aversion (under-trust) both harm outcomes; calibration of trust is key.

🔷 Decision support: present evidence, alternatives and confidence, leaving accountable humans in charge.

🔷 Evaluate augmentation with outcome metrics (accuracy, time, safety), not just model accuracy.

## 🔄 How it works




## 💻 Uncertainty-aware triage

```
def triage(p):
    if p >= 0.9: return 'priority read'
    if p >= 0.3: return 'standard read + AI note'
    return 'single read'   # low risk
```

## ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: The Second Reader

**Characters:** *Dr. Kavya (radiology resident), Prof. Menon (head of department), the hospital's AI tool.*

SCENE 1

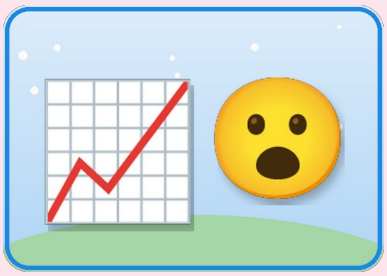
Kavya's hospital piloted an AI tool for chest X-rays. At first it simply replaced the junior doctor's read on 'normal' scans. Prof. Menon worried: who would catch the AI's mistakes?

**Pause & predict:** What risks come with pure automation here?

SCENE 2

They redesigned the workflow. Kavya read each scan first and recorded her judgement; then the AI's view appeared. Disagreements went to Prof. Menon. The AI became a second reader rather than a replacement.

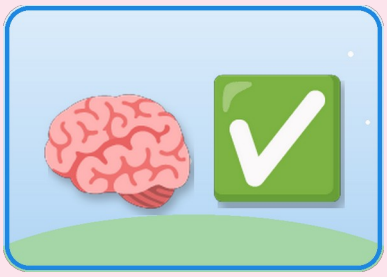
**Pause & predict:** Why read first, then see the AI?



SCENE 3

After three months, missed nodules fell noticeably and reporting time barely changed. But Kavya noticed she trusted the AI more on busy nights, sometimes changing a correct read to match it.

**Pause & predict:** What bias is Kavya describing?



SCENE 4

The team added training on known AI failure cases and showed confidence scores. They now measured three arms: doctor alone, AI alone and doctor with AI. The combined team beat both.

⏸ **Pause & predict:** Why measure all three arms?

## 🌟 The big idea

*AI adds most value when it augments experts and the team, not the tool, is evaluated.*

## 💬 Discuss with your classmates

**1.** Where in your field could AI act as a second reader?

**2.** How would you detect automation bias in a team?

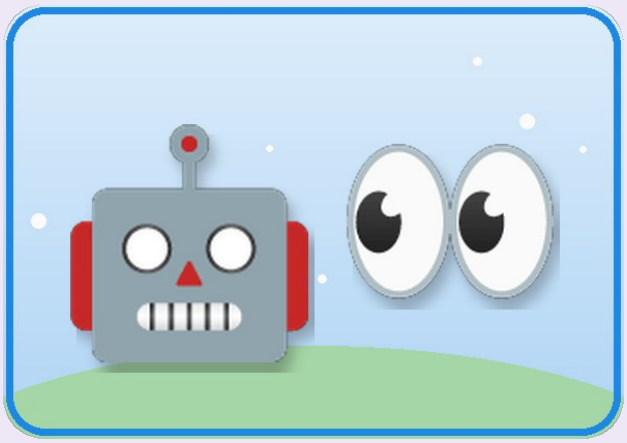
## SCENARIO: AI Beyond Automation

Read the scenario. Think about the choices made at each step.

1

**Rohan studies AI in a hospital radiology unit.**

2



**AI triages scans; doctors read the flagged ones.**

3

**He finds doctors over-trust confident AI outputs.**

4

**New interface shows uncertainty; errors drop.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Augmentation in action

**Second-reader screening**

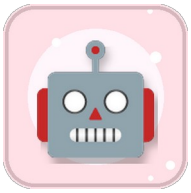
**Legal research**

**Lab hypothesis**

**Design exploration**

**Policy analysis**

**Tutoring**



**BOLT SAYS:**
I'm not here to replace experts. I'm a second pair of eyes that never tires!

**FUN FACT!**
A 2023 Swedish trial found AI-supported mammography found 20% more cancers with 44% less reading.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: AI as a Second Reader in Sweden

**CASE FILE**

In the randomised MASAI trial in Sweden, published in The Lancet Oncology in 2023, about 80,000 women were screened either by two radiologists or with AI-supported screening, where AI triaged exams and highlighted findings. The AI-supported group detected about 20% more cancers while reducing radiologists' screen-reading workload by about 44%, without increasing false positives. Researchers called for longer follow-up before wide adoption.

## 🔍 Analyse the case

**1. Why is a randomised trial strong evidence?**

**2. How did AI augment rather than replace radiologists?**

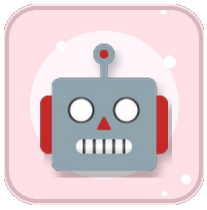
**3. What further evidence is needed before national rollout?**

**⚖️ My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Human + AI Challenge

## You need:

**Puzzle set**



**Chatbot**

**3 groups**

## Steps:

| | | |
|---|---|---|
| 1 | | Group A solves puzzles alone. |
| 2 | | Group B lets AI solve them. |
| 3 | | Group C uses AI as a helper. |
| 4 | | Compare accuracy, time and errors. |

**THINK & TALK**

When did human + AI beat both?

## Record your observations & results:

# 💻 CODE LAB: Screening Workload

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  scans = 10000
2  double = int(scans * ____)          # high-risk share
3  reads = (scans - double) * 1 + double * 2
4  print(reads, round(1 - reads / (scans * 2), 2))
```

**COMPLETE** **T1. 10% are high-risk. Fill the blank.**

**PREDICT** **T2. Reads and saving?**

**MODIFY** **T3. High-risk share 20%.**

**EXTEND** **T4. Add 1 minute per read; hours saved?**

# WORKSHEET: AI Beyond Automation (Part A)

Name: _____ Date: _____

## 1. Automation or augmentation? Tick ✓.

| | Automate | Augment |
|---|---|---|
| Auto-file invoices | | |
| Flag scans for a doctor | | |
| Auto-sort spam | | |
| Suggest cases to a lawyer | | |

## 2. Screening workload. Use the word box.

**WORD BOX:** **20000** ✿ **11000** ✿ **45**

10,000 scans × 2 readers = ____ reads

AI triage: 9,000 single + 1,000 double = ____

Saving = ____%

# WORKSHEET: AI Beyond Automation (Part B)

Name: _____ Date: _____

**1. Automation bias or algorithm aversion? Tick ✓.**

| | Bias | Aversion |
|---|---|---|
| Accepting every AI flag unchecked | | |
| Ignoring accurate AI advice | | |
| Skipping review when AI agrees | | |
| Distrusting AI after one error | | |

**2. Team accuracy. Use the word box.**

**WORD BOX: 8 ✿ 4 ✿ 96**

Human alone 92% of 100 → errors = ____

Human + AI halves errors → ____

Team accuracy = ____%

# PUZZLE & DEFINE: AI Beyond Automation

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**Augment**

Meaning: _____

Example: _____

**2nd reader**

Meaning: _____

Example: _____

**DSS**

Meaning: _____

Example: _____

# ? QUIZ TIME: AI Beyond Automation

**Circle the right answer!**

**Q1. Boosting experts rather than replacing them:**

a) Augmentation

b) Removal

c) Chance

**Q2. An AI double-check is a...**

a) Boss

b) Second reader

c) Speaker

**Q3. DSS stands for decision...**

a) Sound

b) Style

c) Support system

**Q4. The Swedish trial studied...**

a) Mammography

b) Cars

c) Crops

**Q5. Best results often combine AI with...**

a) Luck

b) Human judgement

c) Rest

**MY SCORE: ____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: AI Beyond Automation

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | Augmentation improves human performance. | | |
| | Automation bias means distrusting AI. | | |
| | Showing uncertainty can improve decisions. | | |
| | The Swedish trial was randomised. | | |
| | AI alone is always better than human + AI. | | |
| | Accountable humans should make final calls. | | |

**HOME CONNECT**

Find a task where AI could be your 'second reader'. Describe how you would check its suggestions.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: AI Beyond Automation

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. What is automation bias?**

**COMPARE** **Q2. Automation vs augmentation.**

**APPLY** **Q3. 20,000 reads cut by 44%. Reads saved?**

**EVALUATE** **Q4. Should AI hide its confidence to avoid confusion?**

**CREATE** **Q5. Design a human–AI workflow for legal research.**

# EXAM PREP: AI Beyond Automation

Name: _____ Date: _____

**Time: 30 min Max marks: 13 Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** Distinguish automation and augmentation.

**2.** What is automation bias?

**3.** Why show AI uncertainty?

**4.** How should augmentation be evaluated?

## PART B • Long answer (5 marks)

**5.** Design and evaluate a human–AI decision-support system in a high-stakes domain.

# UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

**Circle the right answer!**

**Q1.** Learning labels from data itself:

a) Self-supervised

b) Hand rules

c) Random

**Q2.** Text + image + audio models are...

a) Closed

b) Multimodal

c) Linear

**Q3.** AlphaGo beat Lee Sedol in...

a) 2012

b) 2020

c) 2016

**Q4.** AI as a second reader means it...

a) Assists experts

b) Replaces all

c) Hides

**Q5.** AI capability is often...

a) Uniform

b) Jagged

c) Perfect

**Q6.** Beyond automation, AI can...

a) Only copy

b) Rest

c) Aid discovery

# WORD SEARCH: AI DISCOVER

🔍 **Find these words. They go across → or down ↓.**

SSL

EMBEDDING

MULTIMODAL

AUGMENT

2ND READER

DSS

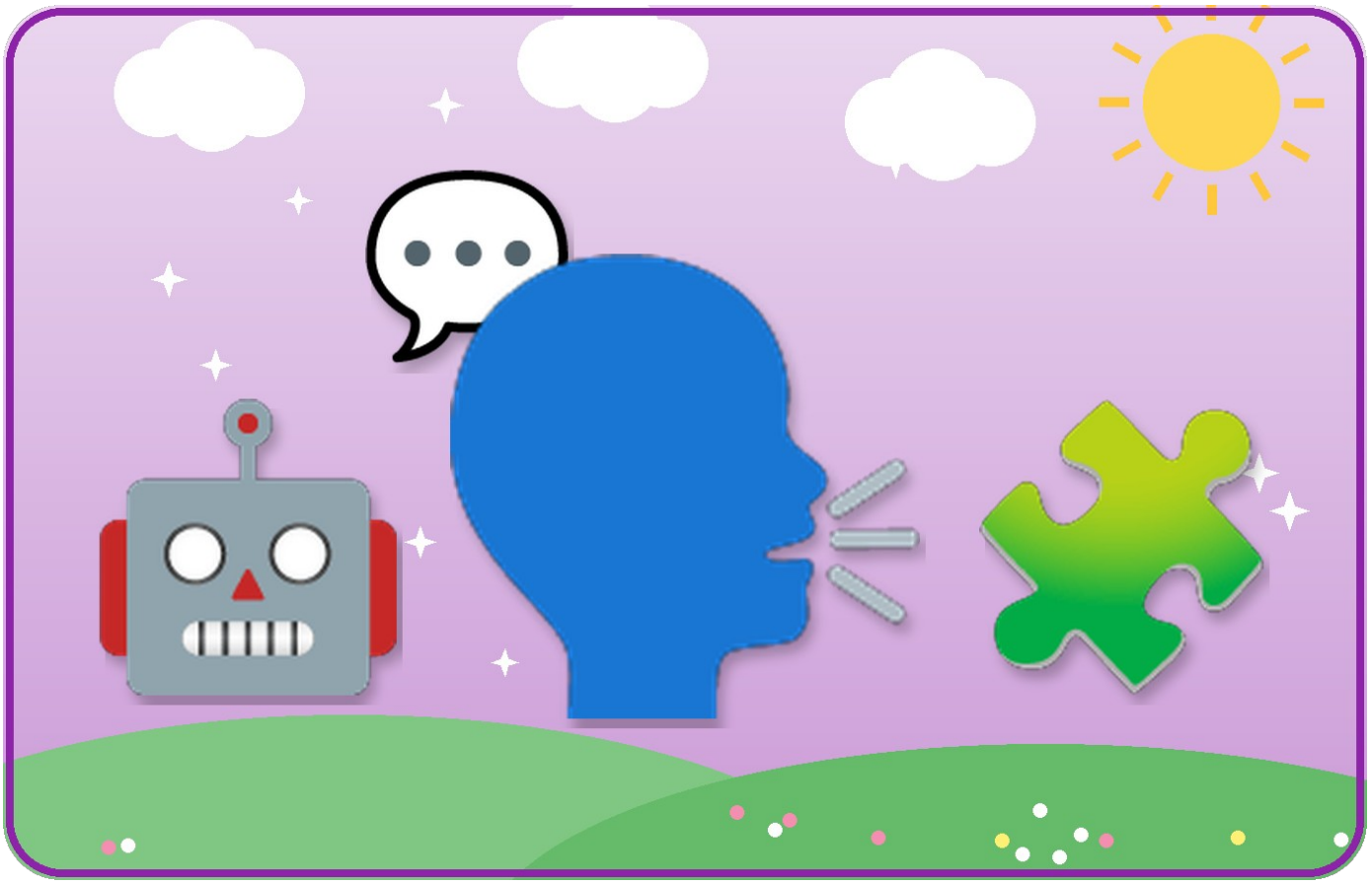
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| C | C | A | V | E | C | T | O | R | P |
| M | U | L | T | I | M | O | D | A | L |
| A | I | H | I | D | W | F | M | S | K |
| A | U | G | M | E | N | T | C | S | F |
| D | F | I | S | F | S | C | A | L | E |
| S | I | K | P | E | X | P | E | R | T |
| S | 2 | N | D | R | E | A | D | E | R |
| L | E | M | B | E | D | D | I | N | G |
| I | N | S | I | G | H | T | R | R | D |
| A | K | N | L | N | E | U | R | A | L |

**I finished AI DISCOVER! Teacher's sign: _____**

MONTH 2

# AI CONNECT

Think • Instruct • Code

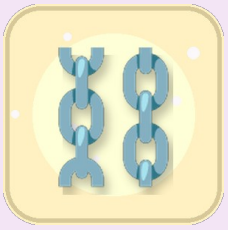


**Area:** Communicate with AI

## In this unit, you will explore:

Topic 3

### Develop with Intelligence



Topic 4

### Design Smart Workflows

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

# Develop with Intelligence

## LET'S LEARN

- ML systems are mostly data, pipelines and infra.
- Version code, data, models; track experiments.
- Test data and models, not just code.

### ML engineering essentials

**Data versioning**

**Experiment tracking**

**ML testing**

**Packaging**

## KEY TERMS

**DVC** Data version control

**Tracking** Log runs & results

**Tech debt** Costly shortcuts

# 📖 CONCEPT EXPLAINED: Develop with Intelligence

### 💡 In this chapter

Building a model in a notebook is the easy part. Real ML systems involve data pipelines, feature stores, training infrastructure, deployment, monitoring and feedback loops. As postgraduates you will be expected to develop AI with the discipline of software engineering and the rigour of science, so that results are reproducible, systems are maintainable and failures are caught early.

## 🧱 1. Hidden technical debt

A 2015 Google paper, 'Hidden Technical Debt in Machine Learning Systems', observed that model code is a small box in a large diagram of data collection, verification, feature extraction, configuration, serving and monitoring. Technical debt accumulates when shortcuts in these surrounding parts make future changes costly or risky.

ML-specific debt includes entanglement (changing one feature changes everything), undeclared consumers that silently depend on model outputs, unstable data dependencies, glue code, pipeline jungles and dead experimental code paths. Feedback loops, where model predictions influence future training data, are especially hard to detect and can amplify bias over time.

### ✏️ WORKED EXAMPLE

A recommender trained on clicks promotes popular items; users click what they see, so future data shows those items as even more popular. Without logging exposure and randomised exploration, the loop narrows recommendations.

## 🗂️ 2. Versioning and experiment tracking

Reproducibility requires versioning everything that affects a result: code with Git, data with tools such as DVC or dataset hashes, models in a registry, and environments with lock files or containers. A result is only meaningful if you can say exactly which data, code, configuration and seed produced it.

Experiment trackers such as MLflow or Weights & Biases log parameters, metrics, artefacts and system details for every run. This turns tuning from folklore into a searchable record, makes comparisons fair, and lets a supervisor or reviewer audit your claims. Naming conventions and short run notes save enormous time later.

🤔 **PAUSE & THINK**

Your best model from three months ago cannot be reproduced; accuracy is now 2 points lower. List every source of difference you would investigate, in order.

## 🧪 3. Testing ML systems

Unit tests for code are necessary but insufficient. Data tests check schemas, ranges, missing values, duplicates and distribution shifts before training. Model tests check performance on critical slices, invariances (changing a name should not change a loan decision), directional expectations (more income should not lower approval) and minimum functionality on simple cases.

Infrastructure tests confirm that training is reproducible, that serving produces the same outputs as offline evaluation, and that rollbacks work. Google's ML Test Score rubric offers a checklist across data, model, infrastructure and monitoring; scoring your project against it is a quick way to find gaps.

✏️ **WORKED EXAMPLE**

Invariance test: for 500 resumes, swap first names across genders and confirm that the screening score changes by less than 0.01 for at least 99% of cases.

## 🔄 4. Deployment and monitoring

Deployment strategies limit risk: shadow mode runs a new model silently alongside the old; canary releases expose it to a small fraction of traffic; A/B tests compare outcomes statistically. Each stage should have explicit go or no-go criteria agreed before launch.

After launch, monitor input distributions, prediction distributions, latency, cost, and when possible ground-truth performance. Alerts should be actionable and owned by named people. Plan retraining triggers and data retention policies, and record every model version served so incidents can be traced.

🖐️ **TRY IT**

Take a model you trained for a course. Write a one-page ML system design: data sources, versioning, tests, deployment stages with go or no-go criteria, monitoring signals and owners.

## ✅ Key takeaways

- Model code is a small part of an ML system; debt hides in data and pipelines.
- Feedback loops and undeclared consumers are ML-specific risks.
- Version code, data, models and environments; track every experiment.
- Test data, model behaviour on slices and invariances, and infrastructure.
- Deploy via shadow, canary and A/B stages; monitor with owned alerts.

## ❓ Quick check: answer in one or two lines.

**1.** Name three forms of ML technical debt.

**2.** What must be versioned to reproduce a result?

**3.** Describe an invariance test for a model in your field.

## 🔬 CONCEPT DEEP DIVE: Develop with Intelligence

🔷 Real ML systems include data collection, validation, feature pipelines, configuration, serving and monitoring around the model.

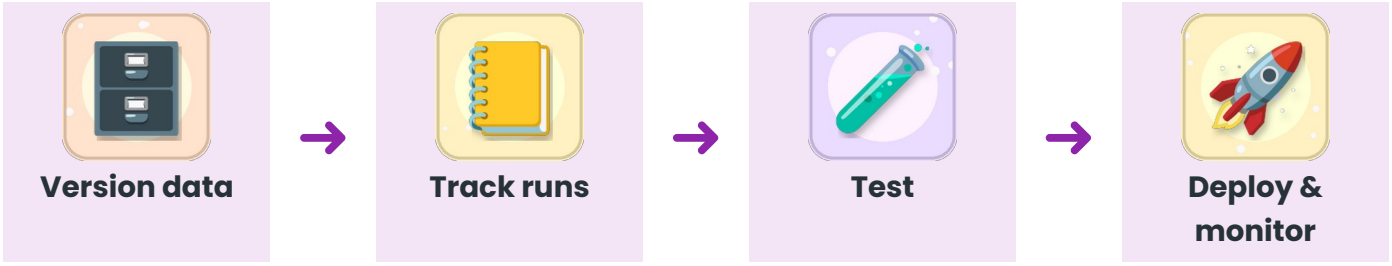
🔷 Reproducibility: version code (Git), data and models (DVC or similar), environments (containers) and seeds.

🔷 Experiment tracking logs parameters, metrics, artefacts and code versions for every run, enabling comparison and audit.

🔷 Testing ML: unit tests for code, data validation tests, model performance gates, and checks for training–serving skew.

🔷 CI/CD for ML automates testing, training and deployment; monitoring catches drift after release.

### 🔄 How it works



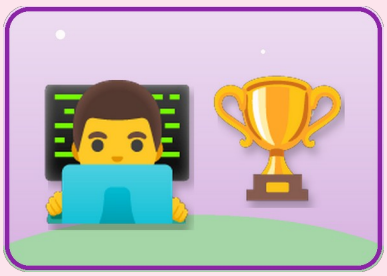

### 💻 Tracking a run

```
import mlflow
with mlflow.start_run():
    mlflow.log_params({'lr': 0.01, 'epochs': 10})
    mlflow.log_metric('val_acc', 0.86)
```

### ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: The Notebook That Could Not Be Repeated

**Characters:** *Arjun (MTech AI), teammate Riya, industry mentor Mr. Shah.*



SCENE 1

Arjun's crop-yield model scored an impressive error of 4% in his Jupyter notebook. Mr. Shah's company wanted to deploy it across three districts and asked for the training pipeline.

**Pause & predict:** What does a company need beyond a notebook?

SCENE 2

Arjun could not reproduce the 4%. He had changed the data cleaning twice, never fixed the random seed and overwritten the dataset file. Retraining gave 9%, then 7%.

**Pause & predict:** Which practices would have prevented this?

SCENE 3

Riya set up Git, DVC for data, MLflow for runs and a config file for every experiment. They also wrote data tests that caught 300 duplicate rows that had leaked across train and test.

**Pause & predict:** How did leakage inflate the original 4%?

SCENE 4

The honest result was 6.5%, still useful. The company launched a canary in one district with monitoring and a rollback plan. Mr. Shah told Arjun that a reproducible 6.5% beats an unrepeatable 4%.

⏸ **Pause & predict:** Why is a reproducible result more valuable?

## 🌟 The big idea

*Reliable AI is engineered: version everything, test the data and deploy in stages.*

## 💬 Discuss with your classmates

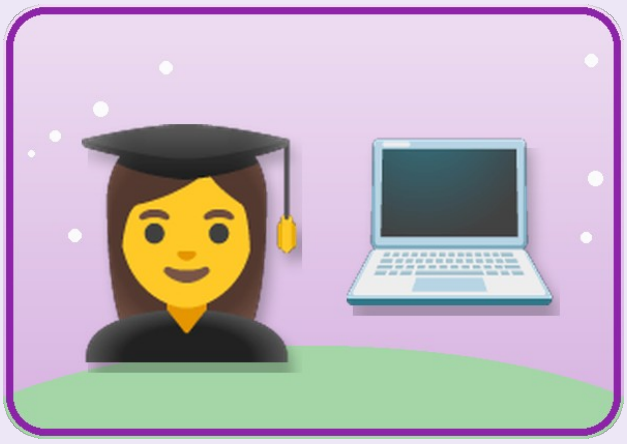
**1.** Which of your past results could you reproduce today?

**2.** What is the first test you would add to your pipeline?

## SCENARIO: Develop with Intelligence

Read the scenario. Think about the choices made at each step.

**1**



**Ananya's research model can't be reproduced.**

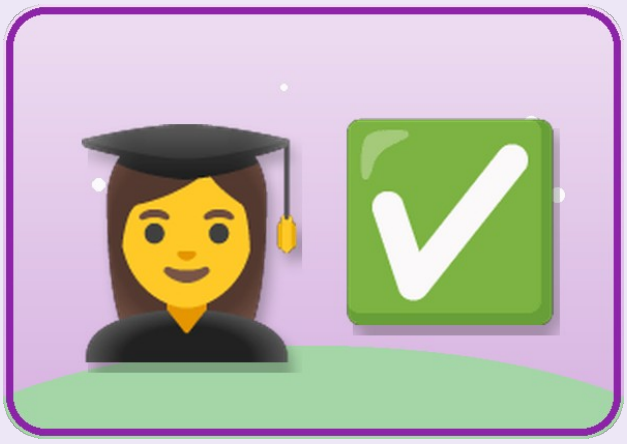
**2**

**Data changed; library versions drifted.**

**3**

**She adds DVC, MLflow tracking and Docker.**

**4**



**A labmate reproduces her result in an hour.**

**Talk about it:**

What decision was made? What would YOU have done differently?

**What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Tools of ML engineering

Git

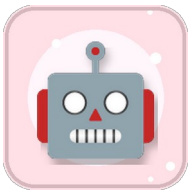
DVC

MLflow

Docker

CI/CD

Monitoring



**BOLT SAYS:**

The model is the tip of the iceberg. Data and pipelines are the rest!

**FUN FACT!**

Google's 2015 'Hidden Technical Debt' paper: ML code is a small part of real ML systems.

**Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: Hidden Technical Debt in ML Systems

**CASE FILE**

In a 2015 NeurIPS paper, Google engineers described 'hidden technical debt' in machine-learning systems. They illustrated that the ML code in a real system is a small box surrounded by much larger components: data collection, verification, feature extraction, configuration, serving infrastructure and monitoring. They warned that shortcuts such as undeclared data dependencies and glue code create costly maintenance problems over time.

## 🔍 Analyse the case

**1. Why is model code only a small part of an ML system?**

**2. Give one example of hidden technical debt.**

**3. How do versioning and testing reduce this debt?**

### ⚖️ My verdict: Was AI used well here? Why?

# LAB ACTIVITY: Iceberg Mapping

## You need:

System diagram

Sticky notes

Teams of 4

## Steps:

1.
   Draw an iceberg; put 'model code' on top.
2.
   Add hidden parts: data, config, serving.
3.
   Mark parts that broke in past projects.
4.
   Pick one fix per hidden part.

**THINK & TALK**

Which hidden part is riskiest?

**Record your observations & results:**

# 💻 CODE LAB: Best Run Finder

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  runs = [(0.1, 0.78), (0.01, 0.86), (0.001, 0.83)]   # (lr,
val_acc)
 2  best = max(runs, key=lambda r: r[____])
 3  print(best)
```

**COMPLETE** **T1. Pick by validation accuracy.**

**PREDICT** **T2. Best run?**

**EXPLAIN** **T3. Why use validation, not test?**

**EXTEND** **T4. Log runs to a CSV file.**

# WORKSHEET: Develop with Intelligence (Part A)

Name: _____ Date: _____

## 1. Match the tool to its job.

Git ●

● Package environment

DVC ●

● Version code

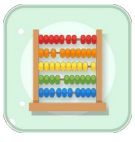
MLflow ●

● Track experiments

Docker ●

● Version data & models

## 2. Code test or data/model test? Tick ✓.

| | Code | Data/ model |
|---|---|---|
| Function returns right type | | |
| No nulls in key columns | | |
|  Unit test for a parser | | |
| Accuracy ≥ baseline on test set | | |

# WORKSHEET: Develop with Intelligence (Part B)

Name: _____ Date: _____

## 1. Order an ML CI pipeline. Write 1–4.

Deploy if gates pass ☐

Run unit & data tests ☐

Train model ☐

Check performance gates ☐

## 2. Circle good reproducibility practices.

Versioned datasets

Untracked config edits

Container image

Unknown random seed

Logged experiments

Results only in memory

# PUZZLE & DEFINE: Develop with Intelligence

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**DVC**

Meaning: _____

Example: _____

**Tracking**

Meaning: _____

Example: _____

**Tech debt**

Meaning: _____

Example: _____

# ? QUIZ TIME: Develop with Intelligence

**Circle the right answer!**

**Q1. Version control for data:**

a) DVC

b) CSS

c) SMTP

**Q2. Logging runs and results is...**

a) Mixing

b) Tracking

c) Cleaning

**Q3. Packages the environment:**

a) Git

b) MLflow

c) Docker

**Q4. Costly shortcuts in systems:**

a) Technical debt

b) Profit

c) Gift

**Q5. Most of an ML system is...**

a) Model code

b) Data & infra

c) UI only

**MY SCORE: ____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: Develop with Intelligence

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | Data should be versioned like code. | | |
| | Model code is most of an ML system. | | |
| | Experiment tracking aids reproducibility. | | |
| | Containers capture software environments. | | |
| | Seeds don't affect results. | | |
| | The 2015 paper described hidden technical debt. | | |

**HOME CONNECT**

Take an old project. Could someone else reproduce it? List what's missing and fix one item.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Develop with Intelligence

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why must data be versioned, not just code?**

**COMPARE** **Q2. Unit tests vs model performance gates.**

**APPLY** **Q3. Runs: lr 0.1 → 0.78, 0.01 → 0.86, 0.001 → 0.83. Best?**

**EVALUATE** **Q4. Is 'it works on my laptop' enough?**

**CREATE** **Q5. List 5 items for an ML reproducibility checklist.**

# 📝 EXAM PREP: Develop with Intelligence

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is hidden technical debt in ML?

**2.** List four reproducibility artefacts.

**3.** What does experiment tracking record?

**4.** What is training–serving skew?

## PART B • Long answer (5 marks)

**5.** Describe an end-to-end ML engineering workflow that ensures reproducibility and reliability.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

# Design Smart Workflows

## LET'S LEARN

- Compound AI: models + retrieval + tools + checks.
- Generate-and-filter beats single-shot answers.
- Optimise the whole system for quality and cost.

### Compound system parts

**Models**

**Retrievers**

**Tools**

**Verifiers**

## KEY TERMS

**Sampling** Many candidates

**Verifier** Checks candidates

**Trade-off** Quality vs cost

# CONCEPT EXPLAINED: Design Smart Workflows

### In this chapter

State-of-the-art results increasingly come not from a single model call but from compound AI systems: pipelines and loops that combine models with retrieval, tools, verifiers and human checkpoints. Designing these workflows is now a core skill. The challenge is to improve quality and reliability while controlling cost and latency, and to evaluate the whole system rather than any single component.

## 1. Compound AI systems

A compound system breaks a task into components: a retriever finds relevant documents, a generator drafts an answer, a tool executes code or queries a database, and a verifier checks the result. Components can be swapped and improved independently, and each can be tested on its own metrics.

Compound designs often beat larger single models because they inject fresh knowledge, enforce structure and catch errors. But they also add failure points and latency. Good designs make interfaces explicit, log every intermediate output and fail gracefully, for example by returning 'insufficient evidence' rather than a fabricated answer.

**WORKED EXAMPLE**

Research assistant: query rewrite → retrieve 20 passages → rerank to 5 → draft answer with citations → verifier checks every claim has a supporting passage → unsupported claims removed or flagged.

## 2. Generate-and-filter

When verification is easier than generation, sampling many candidates and filtering with a verifier is powerful. DeepMind's AlphaCode (2022) generated very large numbers of candidate programs per problem, filtered them using example tests and clustered the survivors before submitting a handful, achieving roughly median performance among human competitors in simulated contests.

The arithmetic is simple: if each attempt independently succeeds with probability p, the chance that at least one of n attempts succeeds is 1 - (1 - p)^n. With p = 0.2 and n = 10, that is about 0.89. Real attempts are correlated, so gains are smaller, and a verifier with false positives can let wrong answers through.

🤔 **PAUSE & THINK**

In your domain, is checking an answer much easier than producing one? Give one task where generate-and-filter would help and one where it would not.

## 🧭 3. Routing, orchestration and agents

Routers send easy requests to small, cheap models and hard ones to larger models or human experts. Orchestrators manage multi-step plans, retries and parallel branches. Agentic loops let a model decide its own next step using tools, which adds flexibility but also unpredictability.

A useful rule is to use the simplest structure that works: a fixed pipeline before an agent, a single model before an ensemble. Add autonomy only where the task genuinely needs it, and cap steps, budget and tool permissions. Every additional stage should earn its place in measured quality or cost.

✏️ **WORKED EXAMPLE**

Routing maths: 70% of queries to a small model at ₹0.05, 30% to a large model at ₹0.60 gives an average of 0.7 × 0.05 + 0.3 × 0.60 = ₹0.215 per query, versus ₹0.60 for always-large.

## 📊 4. Evaluating and optimising the system

Evaluate end to end on realistic tasks, then attribute errors to components: was the right document retrieved, did the generator use it, did the verifier catch the mistake? Component metrics such as recall at k or verifier precision guide where to invest effort.

Optimisation is multi-objective. Plot quality against cost and latency for each configuration and choose a point on the Pareto frontier that suits the use case. Keep a regression suite so improvements to one component do not silently break another, and rerun it whenever a model version changes.

🖐️ **TRY IT**

Sketch a compound workflow for literature review in your discipline. Name each component, its metric, its cost, and the end-to-end evaluation you would run.

## ✅ Key takeaways

- Compound systems combine models, retrieval, tools, verifiers and humans.
- Generate-and-filter works when verifying is easier than generating.
- P(at least one success) = 1 - (1 - p)^n, reduced by correlation.
- Prefer simple pipelines; add routing and agents only when needed.
- Evaluate end to end, attribute errors to components and pick Pareto-optimal configs.

## ❓ Quick check: answer in one or two lines.

**1.** Why can a compound system outperform a larger single model?

**2.** Compute the chance of at least one success with p = 0.3 and n = 5.

**3.** How would you find which component causes most errors?

# 🔬 CONCEPT DEEP DIVE: Design Smart Workflows

🔷 Compound AI systems combine models, retrievers, tools, verifiers and control logic; quality comes from the system, not one model.

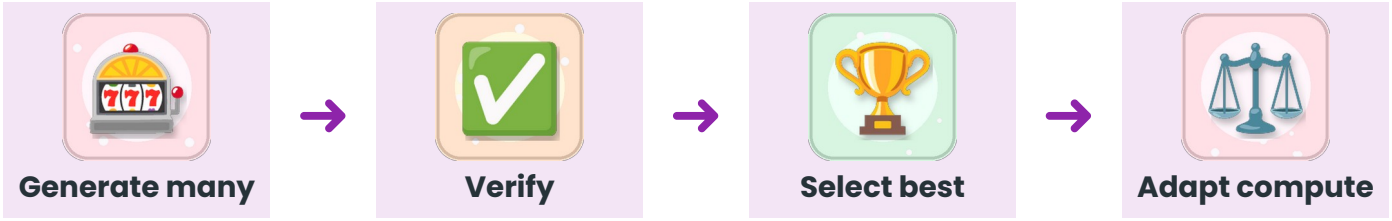
🔷 Inference-time scaling: sampling multiple candidates, self-consistency voting and verifier-guided selection can boost accuracy.

🔷 Verifiers range from unit tests and solvers (strong) to learned reward models or LLM judges (weaker, need calibration).

🔷 Adaptive computation spends more samples or larger models only on hard inputs, balancing quality, latency and cost.

🔷 System-level evaluation and optimisation: tune prompts, retrieval and routing jointly against end-to-end metrics.

## 🔄 How it works



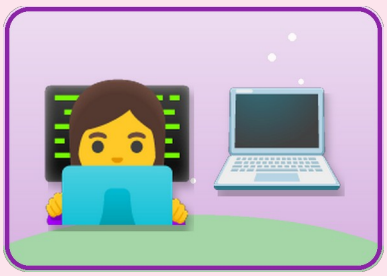

## 💻 Majority vote over samples

```
from collections import Counter
answers = ['42', '41', '42', '42', '40']
best, n = Counter(answers).most_common(1)[0]
print(best, n / len(answers))   # 42 0.6
```

## ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: Ten Tries and a Judge

**Characters:** *Sana (MSc CS), her lab's coding assistant project, Prof. Rao.*



SCENE 1

Sana's lab built an assistant to solve programming exercises for an online course. A single model call solved about 30% of problems correctly on the first try.

**Pause & predict:** Why might one attempt underperform?

SCENE 2

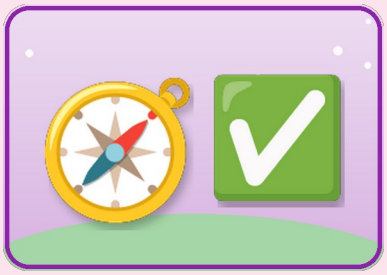
Inspired by AlphaCode, Sana generated ten candidate programs per problem and ran each against the example tests. The best passing candidate was submitted. Success rose to 71%.

**Pause & predict:** Why didn't success reach 1 - 0.7^10 ≈ 97%?

SCENE 3

Prof. Rao pointed out that costs had risen tenfold and responses took 40 seconds. Some 'passing' programs also failed hidden tests, because example tests were weak verifiers.

**Pause & predict:** What trade-offs did generate-and-filter introduce?



SCENE 4

Sana added a router: easy problems got one attempt, hard ones got ten. She strengthened tests with edge cases. Quality held at 69% while cost fell by 60%.

⏸ **Pause & predict:** How does routing improve the quality–cost balance?

## 🌟 The big idea

*Compound workflows boost quality, but every stage must earn its cost and needs strong verification.*

## 💬 Discuss with your classmates

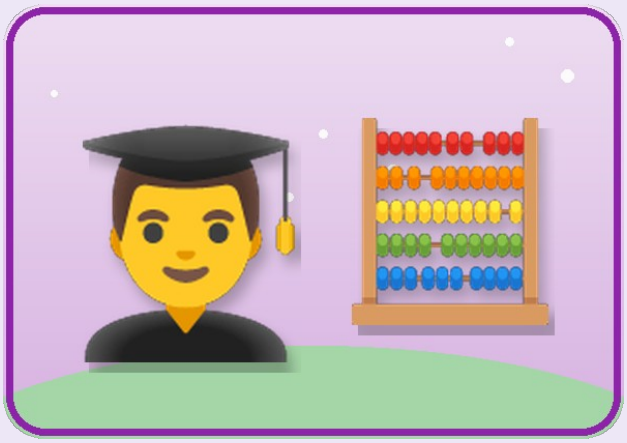
**1.** Where is checking easier than generating in your work?

**2.** How would you decide how many samples to draw?

## SCENARIO: Design Smart Workflows

Read the scenario. Think about the choices made at each step.

1



**Rohan's math tutor bot gets 60% right in one try.**

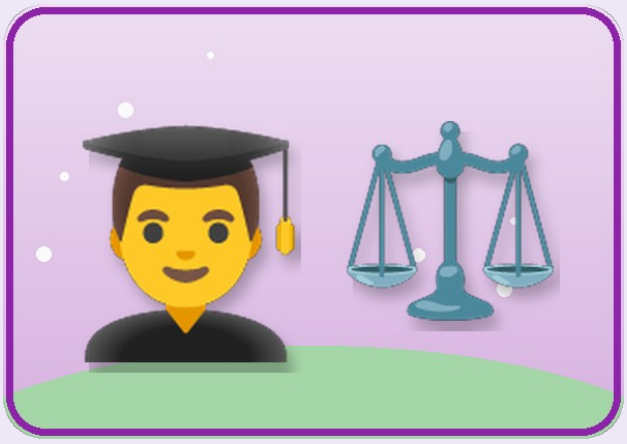
2

**He samples 8 answers and checks with code.**

3

**Accuracy hits 85%, but cost rises 8×.**

4



**He samples more only for hard questions.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

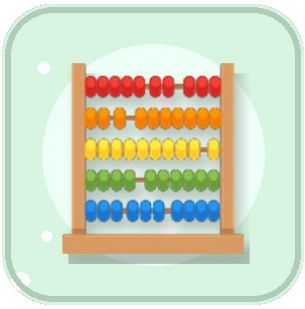
## LOOK AROUND YOU: Compound systems

**Code generation**

**Literature agents**

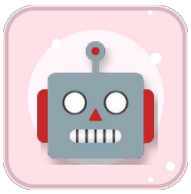
**Clinical summaries**

**Report writing**



**Math solvers**

**Grading aids**



**BOLT SAYS:**

One answer can be wrong. Ten answers plus a good checker? Much better!

**FUN FACT!**

DeepMind's AlphaCode (2022) generated many programs, filtered them and ranked near median coders.

**Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: AlphaCode's Generate-and-Filter

**CASE FILE**

In 2022, DeepMind described AlphaCode, which tackled competitive programming problems by generating very large numbers of candidate programs, filtering them using the problem's example tests, clustering similar programs and submitting a small number. In simulated Codeforces contests it ranked roughly in the top 54% of participants, around the median, showing how large-scale sampling plus filtering can solve complex tasks.

## 🔍 Analyse the case

**1. Why does generating many candidates help?**

**2. What served as AlphaCode's verifier?**

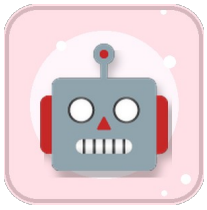
**3. What are the cost trade-offs of this approach?**

**⚖️ My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Generate & Filter

## You need:

5 puzzles



Chatbot

Pairs

## Steps:

1.
   Get 5 AI answers per puzzle.
2.
   Write a quick checker for each.
3.
   Keep only answers that pass.
4.
   Compare with first-answer accuracy.

**THINK & TALK**

When did filtering help most?

**Record your observations & results:**

# 💻 CODE LAB: Generate & Filter Odds

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  p = 0.05
2  for n in (1, 10, 100):
3      print(n, round(1 - (1 - p) ** ____, 2))
```

**COMPLETE** **T1. Fill the blank.**

**PREDICT** **T2. Values for n = 1, 10, 100?**

**EXPLAIN** **T3. Why do verifiers matter here?**

**EXTEND** **T4. Find n for 90% chance.**

# WORKSHEET: Design Smart Workflows (Part A)

Name: _____ Date: _____

## 1. Match the component to its role.

Generator ●

● Checks correctness

Retriever ●

● Picks the path

Verifier ●

● Proposes answers

Router ●

● Fetches evidence

## 2. Chance at least one of n is right (p = 0.05). Use the word box.

**WORD BOX:** 0.05 ✿ 0.40 ✿ 0.99

n = 1 → _____

n = 10 → ≈ _____

n = 100 → ≈ _____

# WORKSHEET: Design Smart Workflows (Part B)

Name: _____ Date: _____

## 1. Majority vote. Use the word box.

**WORD BOX: 42 ✿ 3 ✿ 60**

Answers 42, 41, 42, 42, 40 → winner ____

Votes for winner = ____

Agreement = ____%

## 2. Strong or weak verifier? Tick ✓.

| | Strong | Weak |
|---|---|---|
| Unit tests for code | | |
| 'Looks right' LLM judgement | | |
| Symbolic math check | | |
| Random spot-check of 1 in 100 | | |

# PUZZLE & DEFINE: Design Smart Workflows

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**Sampling**

Meaning: _____

Example: _____

**Verifier**

Meaning: _____

Example: _____

**Trade-off**

Meaning: _____

Example: _____

# ? QUIZ TIME: Design Smart Workflows

**Circle the right answer!**

**Q1. Systems of models + tools + checks:**

a) Compound AI

b) Art AI

c) Boxed AI

**Q2. A verifier...**

a) Draws

b) Checks answers

c) Advertises

**Q3. More samples generally raise...**

a) Sleep

b) Noise

c) Cost

**Q4. AlphaCode generated many...**

a) Programs

b) Songs

c) Images

**Q5. Optimise compound systems for quality and...**

a) Colour

b) Cost

c) Fun

**MY SCORE: ____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: Design Smart Workflows

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | Sampling many answers can raise accuracy. | | |
| | Extra samples are free. | | |
| | Unit tests can act as verifiers. | | |
| | Adaptive compute saves cost on easy inputs. | | |
| | LLM judges never need calibration. | | |
| | AlphaCode filtered candidates with example tests. | | |

**HOME CONNECT**

Ask a chatbot the same tricky question 5 times. Did the majority answer match the correct one?

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Design Smart Workflows

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why do strong verifiers matter?**

**COMPARE** **Q2. Self-consistency voting vs verifier selection.**

**APPLY** **Q3. p = 0.05 per sample; P(≥1 right) with 10 samples?**

**EVALUATE** **Q4. Always sample 100 candidates?**

**CREATE** **Q5. Design a compound system for exam-question generation.**

# 📝 EXAM PREP: Design Smart Workflows

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is a compound AI system?

**2.** Explain self-consistency.

**3.** What is adaptive computation?

**4.** Why calibrate LLM judges?

## PART B • Long answer (5 marks)

**5.** Design a compound AI system for a reasoning task and analyse its quality–cost trade-offs.

# UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

**Circle the right answer!**

**Q1. Most of an ML system is...**

a) Infra & data

b) Model code

c) UI

**Q2. Versioning datasets uses tools like...**

a) MP3

b) DVC

c) JPEG

**Q3. AlphaCode generated, then...**

a) Kept all

b) Posted

c) Filtered

**Q4. Compound AI adds tools and...**

a) Verifiers

b) Games

c) Stickers

**Q5. Shortcuts that cost later:**

a) Profit

b) Tech debt

c) Prizes

**Q6. System design balances quality and...**

a) Colour

b) Volume

c) Cost

# WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

DVC

TRACKING

TECH DEBT

SAMPLING

VERIFIER

TRADE-OFF

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| D | T | D | O | C | K | E | R | D | A |
| F | R | H | D | V | C | D | G | A | S |
| S | A | T | E | C | H | D | E | B | T |
| A | C | P | L | A | T | E | N | C | Y |
| M | K | P | K | T | N | G | G | O | O |
| P | I | P | I | P | E | L | I | N | E |
| L | N | A | G | E | N | T | S | A | U |
| I | G | T | R | A | D | E | O | F | F |
| N | V | E | R | I | F | I | E | R | U |
| G | B | R | E | T | R | I | E | V | E |

**I finished AI CONNECT! Teacher's sign: _____**

MONTH 3

# AI SOLVE

Problems • Solutions • Impact

## Area: AI Applications

### In this unit, you will explore:

Topic 5

**AI for Advanced Research**

Topic 6

**AI for Professional Innovation**

TOPIC 5 • MONTH 3 • AI APPLICATIONS

# AI for Advanced Research

## LET'S LEARN

- Good research: question, baseline, method, evidence.
- Ablations, seeds and statistics separate signal from luck.
- Leakage and benchmark overfitting fake progress.

### Research rigour checklist

**Baselines**

**Ablations**

**Multiple seeds**

**Leakage checks**

## KEY TERMS

**Ablation** Remove a part, test

**Seed** Randomness start

**Leakage** Test info in training

# 📖 CONCEPT EXPLAINED: AI for Advanced Research

### 💡 In this chapter

Postgraduate study is where you shift from learning results to producing them. In AI and in AI-enabled research across disciplines, the difference between a real finding and an artefact often comes down to method: a fair baseline, careful splits, enough runs, honest statistics and full reporting. This chapter covers the research practices that make AI results credible.

## ❓ 1. From question to claim

Good research starts with a precise, falsifiable question and a clear claim you hope to support, for example 'pre-training on regional news improves Tamil sentiment classification by at least three F1 points over multilingual baselines'. The claim determines datasets, baselines, metrics and the evidence needed.

Baselines must be strong and fairly tuned. Comparing a carefully tuned new method with an untuned old one exaggerates gains. Include simple baselines such as logistic regression or majority class, and the best published method you can reproduce. State compute budgets, because a method that wins only with ten times the compute is a different contribution.

**✏️ WORKED EXAMPLE**

Claim hygiene: replace 'our model understands sarcasm' with 'our model improves F1 on the X sarcasm benchmark from 0.71 to 0.76 (mean of 5 seeds, ±0.01)'.

## 🔬 2. Ablations, seeds and statistics

Ablation studies remove or replace one component at a time to show which parts drive performance. Without them, readers cannot tell whether the novelty or some incidental change, such as a better optimiser, produced the gain.

Neural network results vary with random seeds, data order and hardware. Report mean and spread across several seeds, use confidence intervals or appropriate significance tests, and avoid picking the best run. When many comparisons are made, correct for multiple testing or treat findings as exploratory rather than confirmed.

**🤔 PAUSE & THINK**

Your method beats the baseline by 0.4 points on one seed. Across five seeds the standard deviation is 0.6. What can you honestly conclude, and what would you do next?

## 💧 3. Leakage and benchmark overfitting

Data leakage occurs when information from the test set influences training: duplicates across splits, preprocessing fitted on all data, features that encode the label, or temporal splits violated. Kapoor and Narayanan's review found leakage-related errors affecting 294 papers across 17 fields using ML, often producing wildly over-optimistic results.

Benchmark overfitting happens when a community repeatedly tunes on the same test set until gains reflect quirks of the benchmark rather than real progress. For large language models, test items may also appear in web training data. Use held-out or newly collected test sets, check for contamination, and validate on external data where possible.

✏️ **WORKED EXAMPLE**

Illustrative leakage check: suppose a model predicting hospital readmission scores AUC 0.95 until you notice a feature recording discharge to hospice, decided after the outcome was known. Removing it drops AUC to 0.72.

## 📝 4. Reporting and reproducibility

Credible papers report data sources and licences, splits, preprocessing, hyperparameter search ranges, compute, seeds, evaluation code and limitations. Model cards and datasheets standardise this documentation. Releasing code and, where ethical, data allows others to verify and build on your work.

Negative and null results deserve reporting too; they save others time and correct publication bias. When using LLMs in research, for data annotation, analysis or writing, disclose how they were used, which versions, and how outputs were checked.

🖐️ **TRY IT**

Take a recent paper in your area. Score it against a checklist: baseline strength, ablations, seeds and spread, leakage risks, code availability and limitations. Write a one-paragraph reviewer comment.

## ✅ Key takeaways

- Start from a precise, falsifiable claim; it defines baselines and evidence.
- Use strong, fairly tuned baselines and report compute budgets.
- Ablations isolate contributions; multiple seeds and statistics separate signal from noise.
- Leakage and benchmark overfitting create false progress; use fresh held-out data.
- Report fully, release code, disclose LLM use and publish null results.

## ❓ Quick check: answer in one or two lines.

**1.** What is an ablation and why is it needed?

**2.** Give three common sources of data leakage.

**3.** Why report mean and spread across seeds?

# 🔬 CONCEPT DEEP DIVE: AI for Advanced Research

🔷 Formulate clear research questions and hypotheses; pick strong, fairly tuned baselines.

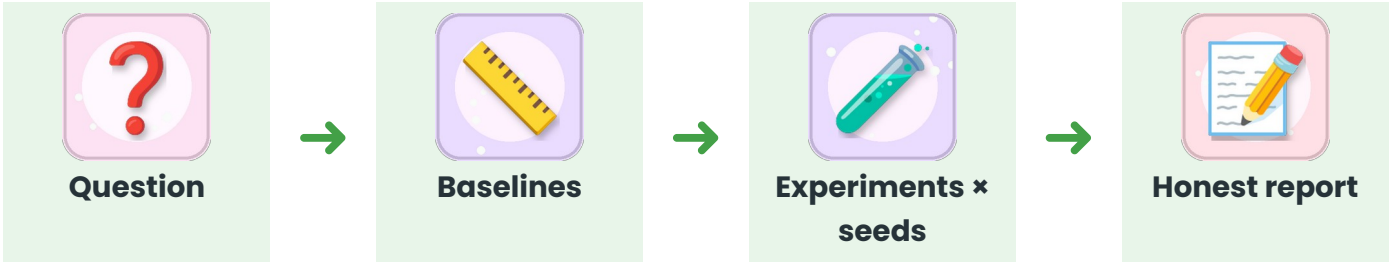
🔷 Use proper splits (train/validation/test), avoid tuning on test data, and check for leakage and duplicates.

🔷 Report variance across random seeds, confidence intervals or significance tests, not single best runs.

🔷 Ablation studies isolate which components cause gains; error analysis reveals failure modes.

🔷 Benchmarks saturate and can be contaminated; complement with new data, real-world tests and human evaluation.

## 🔄 How it works




## 💻 Mean ± std over seeds

```
import numpy as np
acc = np.array([0.81, 0.83, 0.82, 0.80, 0.84])
print(acc.mean().round(3), acc.std().round(3))
# 0.82 ± 0.014 → is a 0.3% gain real?
```

## ✍️ In your own words: explain this topic to a Class 10 student.

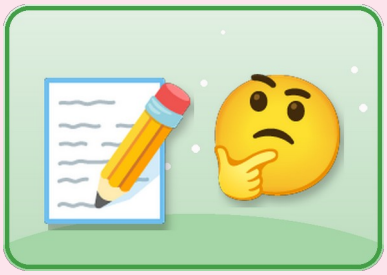
# AI STORY: The Result That Was Too Good

**Characters:** *Ishaan (MPhil Economics), co-author Leela, journal reviewer 2.*

SCENE 1

Ishaan's model predicted which villages would default on microfinance loans with 96% accuracy. Leela was thrilled; it beat every published baseline by twenty points. They submitted to a journal.

**Pause & predict:** What should a 20-point jump make a researcher suspect?



SCENE 2

Reviewer 2 asked: were features computed before or after the loan period? Were villages split, or just rows? Was the baseline tuned as carefully? Ishaan had no clear answers.

**Pause & predict:** Which leakage risks is the reviewer probing?

SCENE 3

Checking, Ishaan found a feature, 'number of recovery visits', recorded after defaults began. Rows from the same village also appeared in both train and test. Fixing both dropped accuracy to 74%.

**Pause & predict:** Why do these errors inflate results?

SCENE 4

They reran with village-level splits, five seeds and ablations. Their method still beat a tuned baseline by 3 points, reported with confidence intervals. The revised paper was accepted.

⏸ **Pause & predict:** Why is the smaller, honest gain more valuable?

### 🌟 The big idea

*Extraordinary results demand extraordinary checks: leakage, baselines, seeds and ablations.*

### 💬 Discuss with your classmates

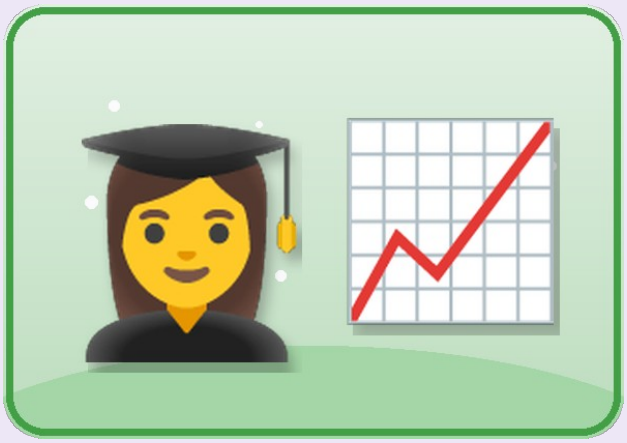
**1.** What would make you distrust your own best result?

**2.** How would you design splits for grouped data?

## SCENARIO: AI for Advanced Research

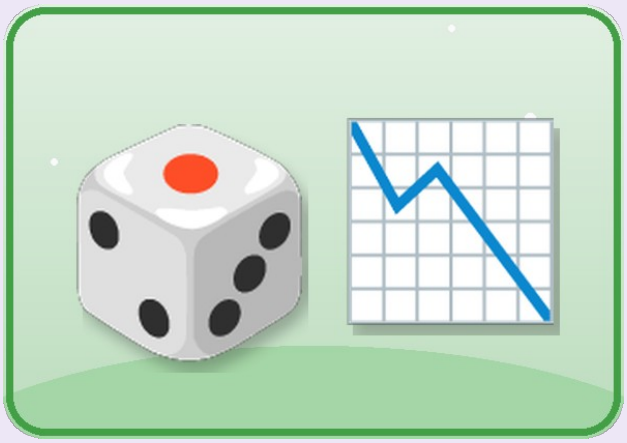
Read the scenario. Think about the choices made at each step.

**1**



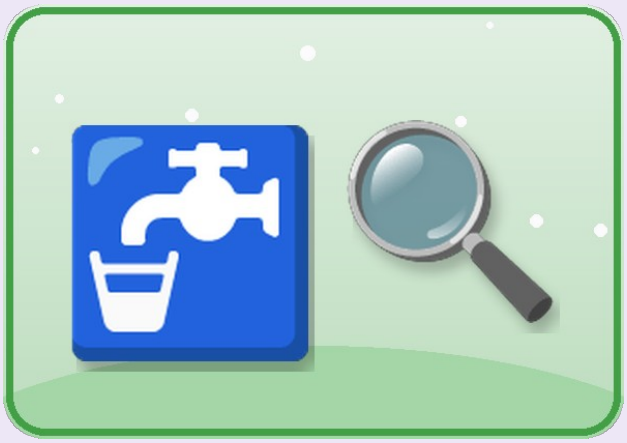
**Ananya's model beats SOTA by 2% on one run.**

**2**



**Over 5 seeds, the gain shrinks to 0.3% ± 0.8%.**

**3**



**She also finds near-duplicates across splits.**

**4**

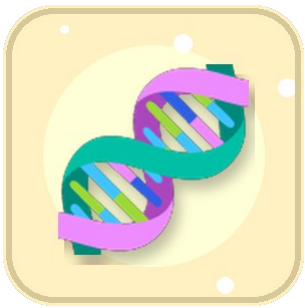
**Her honest paper is accepted with ablations.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

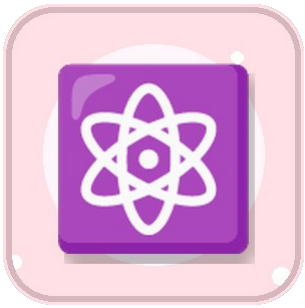
## LOOK AROUND YOU: AI in research



Genomics

Astronomy

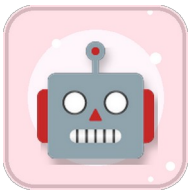
Climate



Materials

Neuroscience

Digital humanities



**BOLT SAYS:**

A 0.3% gain on one seed isn't a discovery. Show me baselines and error bars!

**FUN FACT!**

A 2023 review found data leakage affected 294 ML-based papers across 17 fields.

**Reflect: where have YOU seen this in real life? Note it down.**

# CASE STUDY: The Leakage and Reproducibility Crisis

**CASE FILE**

In 2023, researchers Sayash Kapoor and Arvind Narayanan at Princeton reviewed the use of machine learning across scientific fields and found that data leakage had affected at least 294 papers in 17 fields, sometimes producing wildly over-optimistic results. Examples included test data influencing training and features that would not be available at prediction time. They proposed 'model info sheets' to document how leakage was ruled out.

## Analyse the case

**1. Name two common forms of data leakage.**

**2. Why does leakage inflate results?**

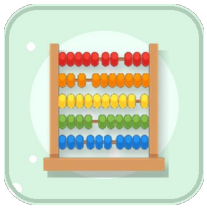
**3. What documentation would help reviewers?**

**My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Ablation Detectives

## You need:

Results table



Calculator

Teams of 3

## Steps:

1.
   Read a model's results with 4 add-ons.
2.
   Remove one add-on at a time.
3.
   Find which add-on mattered most.
4.
   Check gains against seed variation.

**THINK & TALK**

Which 'gain' was just noise?

**Record your observations & results:**

# 💻 CODE LAB: Seed Statistics

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import numpy as np
2  a = np.array([0.81, 0.83, 0.82, 0.80, 0.84])   # model A
3  b = np.array([0.82, 0.82, 0.83, 0.81, 0.83])   # model B
4  print(a.mean().round(3), a.____().round(3), b.mean().round(3))
```

**COMPLETE** **T1. Fill the blank for standard deviation.**

**PREDICT** **T2. What is printed?**

**EVALUATE** **T3. Is B better than A?**

**EXTEND** **T4. Run a paired t-test with scipy.**

# WORKSHEET: AI for Advanced Research (Part A)

Name: _____ Date: _____

## 1. Order a rigorous study. Write 1–4.

Report with error bars ☐

Define question & baseline ☐

Run ablations ☐

Run main experiment ☐

## 2. Circle signs of rigorous research.

Results over 5 seeds

Only best run reported

Strong baselines

Test data used to tune

Ablation table

No error bars

# ✏️ WORKSHEET: AI for Advanced Research (Part B)

Name: _____ Date: _____

## 1. Seed variance. Use the word box.

**WORD BOX:** **0.82** ✿ **0.04** ✿ **0.014**

Mean of 0.81, 0.83, 0.82, 0.80, 0.84 = ____

Range (max - min) = ____

Std (population) ≈ ____

## 2. Match the leakage type to an example.

**Duplicates** ●

**Future info** ●

**Preprocessing** ●

**Test tuning** ●

● **Scaler fitted on all data**

● **Same image in train & test**

● **Choosing model by test score**

● **Discharge date predicting admission**

# PUZZLE & DEFINE: AI for Advanced Research

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**Ablation**

Meaning: _____

Example: _____

**Seed**

Meaning: _____

Example: _____

**Leakage**

Meaning: _____

Example: _____

# ? QUIZ TIME: AI for Advanced Research

**Circle the right answer!**

**Q1. Removing a part to measure its effect:**

a) Ablation | b) Decoration | c) Packing

**Q2. Test information in training is...**

a) A bonus | b) Leakage | c) Validation

**Q3. Running several seeds shows...**

a) Colour | b) Cost | c) Variance

**Q4. Always compare against a...**

a) Baseline | b) Song | c) Rumour

**Q5. Leakage review counted papers in ... fields.**

a) 2 | b) 17 | c) 170

**MY SCORE: ____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: AI for Advanced Research

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | 👍 TRUE | 👎 FALSE |
|---|---|---|
| Results should be reported across seeds. | | |
| Tuning on test data is fine. | | |
| Ablations show which parts matter. | | |
| Leakage can inflate results. | | |
| Benchmarks never saturate. | | |
| The 2023 review found leakage in 294 papers. | | |

**HOME CONNECT**

Pick a published ML result. Check: baselines, seeds, ablations, leakage. Write a 5-line critique.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: AI for Advanced Research

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why report multiple seeds?**

**COMPARE** **Q2. Ablation vs error analysis.**

**APPLY** **Q3. Accuracies 0.81–0.84 over 5 seeds; mean and std?**

**EVALUATE** **Q4. Is beating SOTA by 0.2% on one run publishable?**

**CREATE** **Q5. Write a 5-point leakage checklist.**

# 📝 EXAM PREP: AI for Advanced Research

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is an ablation study?

**2.** Name three leakage sources.

**3.** Why report seeds and error bars?

**4.** What is benchmark contamination?

## PART B • Long answer (5 marks)

**5.** Plan a rigorous ML experiment and explain how you would ensure its findings are valid and reproducible.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

# AI for Professional Innovation

## LET'S LEARN

- AI reshapes law, medicine, finance and engineering.
- The 'jagged frontier': AI helps some tasks, harms others.
- Innovate with domain experts, ethics and regulation.

### Professional AI innovation

Law

Medicine

Finance

Engineering

## KEY TERMS

**Frontier** Where AI is reliable

**Compliance** Following rules

**Licensure** Right to practise

# 📖 CONCEPT EXPLAINED: AI for Professional Innovation

### 💡 In this chapter

AI is reshaping professional work in law, medicine, finance, engineering, consulting and education. Postgraduates entering these fields, or building AI for them, must understand where AI genuinely helps, where it quietly harms, and how professional duties of care, confidentiality and accountability apply. Innovation that ignores regulation and domain expertise rarely survives contact with practice.

## 🧭 1. The jagged frontier at work

A 2023 field experiment by Harvard researchers with Boston Consulting Group involved 758 consultants. For tasks inside the AI's capability frontier, consultants using GPT-4 completed more tasks, faster and at higher rated quality. For a task designed to lie outside the frontier, those using AI were less likely to reach the correct answer than those working without it.

The study also described two working styles. 'Centaurs' divided labour, giving clear sub-tasks to AI and keeping others; 'cyborgs' interleaved with AI continuously. Both can work, but professionals must learn where the frontier lies for their own tasks and keep the skills to check AI output critically.

### ✏️ WORKED EXAMPLE

Lesson for practice: before adopting AI for a professional workflow, run a small pilot on representative cases, including tricky ones, and compare quality with and without AI.

## 🏛️ 2. Domain applications

In law, AI assists with research, contract review and drafting, but there have been widely reported cases of lawyers submitting AI-generated citations to non-existent cases, leading to sanctions. In medicine, AI supports imaging, documentation and triage, but clinical tools may require regulatory approval and evidence from trials.

In finance, models support fraud detection, credit and compliance, under strict rules on fairness, explanation and model risk management. In engineering, AI speeds design exploration and inspection, yet safety-critical decisions must be verified against standards. Across domains the pattern repeats: AI drafts, experts verify, institutions remain accountable.

🤔 **PAUSE & THINK**

Pick a profession you might enter. Which tasks would you delegate to AI, which would you verify, and which would you never delegate? Justify each choice.

## 📜 3. Regulation, licensure and liability

Professions are regulated because errors harm the public. Licensed professionals remain responsible for advice given with AI assistance. Regulators increasingly classify AI by risk: the EU AI Act treats many uses in employment, credit, education and medical devices as high-risk with obligations for data quality, documentation, human oversight and accuracy.

In India, data handling falls under the Digital Personal Data Protection Act, 2023, and sector regulators issue their own guidance. Confidentiality is critical: entering client or patient data into public AI tools may breach professional duties. Innovators should involve compliance teams early and document design decisions.

✏️ **WORKED EXAMPLE**

Compliance checklist: data permissions and minimisation, approved tools only, human review before client delivery, audit logs, incident reporting and periodic review of model performance.

## 🚀 4. Innovating responsibly

Successful professional AI innovation pairs technologists with domain experts from the start. Experts define what 'good' means, provide realistic cases, identify edge cases and judge outputs. Technologists translate needs into systems, measurements and safeguards.

Value should be measured in professional outcomes: fewer errors, faster turnaround, better access, lower cost, improved satisfaction. Pilots should be designed to detect harm as well as benefit, and scaled only when evidence supports it. Transparency with clients about AI use builds trust and is increasingly expected by regulators.

🖐️ **TRY IT**

Propose one AI innovation for a profession in India. Name the expert partners, the outcome metric, the risk classification, the compliance steps and the pilot design.

## ✅ Key takeaways

- AI helps within its frontier and can harm just outside it.
- Centaur and cyborg styles both need critical checking skills.
- Pattern across professions: AI drafts, experts verify, institutions stay accountable.
- Regulation classifies risky uses; confidentiality and licensure still apply.
- Innovate with domain experts, measure professional outcomes and pilot carefully.

## ❓ Quick check: answer in one or two lines.

**1.** Summarise the main finding of the jagged frontier study.

**2.** Why are AI hiring or credit tools often classified as high-risk?

**3.** Name three items on a professional AI compliance checklist.

# 🔬 CONCEPT DEEP DIVE: AI for Professional Innovation

🔷 Professions combine codified knowledge, judgement, ethics and accountability; AI changes task mixes within them.

🔷 The jagged frontier: AI boosts quality and speed on some tasks yet degrades performance on tasks just beyond its capability.

🔷 Innovation requires domain experts, workflow redesign, validation studies and change management.

🔷 Regulation and professional duty: licensure, confidentiality, liability, medical-device rules and financial compliance.

🔷 Measure outcomes: quality, time, errors, client impact and equity across groups.

## 🔄 How it works

**Map tasks** →

**Test frontier** →

**Redesign workflow** →

**Validate & comply**

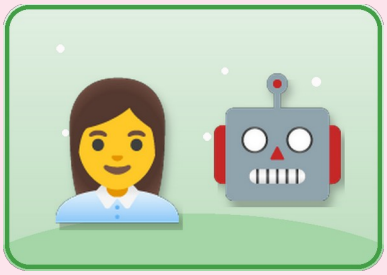
## 💻 Frontier map from pilot results

```
tasks = {'draft': (0.70, 0.88), 'review': (0.75, 0.86), 'novel case':
(0.80, 0.62)}
for t, (alone, with_ai) in tasks.items():
    print(t, 'inside' if with_ai > alone else 'OUTSIDE',
round(with_ai - alone, 2))
# novel case: AI hurts
```

## ✍️ In your own words: explain this topic to a Class 10 student.

# 📚 AI STORY: The Consultant's Shortcut

🎭 **Characters:** *Nisha (MBA, analytics), her manager Mr. Varghese, a client from a retail chain.*



SCENE 1

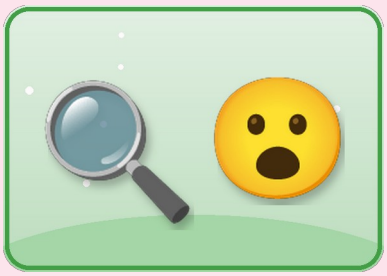
Nisha used an AI assistant to draft market reports. Her drafts became faster and more polished; Mr. Varghese praised her productivity in the quarterly review.

⏸ **Pause & predict:** Which tasks fall inside the AI's frontier?

SCENE 2

For a new client, she asked the AI to analyse store data and recommend which branches to close. It produced a confident answer, but missed that festive-season sales were stored in a separate sheet.

⏸ **Pause & predict:** Why is this task closer to the frontier's edge?



SCENE 3

Mr. Varghese checked the numbers manually and found two 'weak' stores were actually top performers in October. The recommendation would have closed profitable branches.

⏸ **Pause & predict:** What professional duty applies here?

SCENE 4

The firm adopted a rule: AI drafts, analysts verify data sources and numbers, and partners sign off. Nisha started a log of tasks where AI helped and where it misled her.

⏸ **Pause & predict:** How does the log help her learn the frontier?

## 🌟 The big idea

*Use AI inside its frontier; outside it, professional judgement and verification are essential.*

## 💬 Discuss with your classmates

**1.** Which of your tasks might sit just outside AI's frontier?

**2.** Who should sign off on AI-assisted advice in your field?

## SCENARIO: AI for Professional Innovation

Read the scenario. Think about the choices made at each step.

**1**

**Rohan, an LL.M student, tests AI on legal tasks.**

**2**

**AI speeds drafting and clause review.**

**3**

**On a novel case question, it confidently errs.**

**4**

**He maps tasks inside and outside AI's frontier.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Professional AI tools

**Contract drafting**

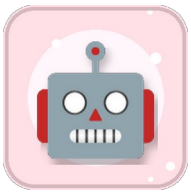
**Radiology triage**

**Risk models**

**Design copilots**

**Pharmacovigilance**

**Audit analytics**



**BOLT SAYS:**

Know my frontier. Inside it, I speed you up. Outside it, I can mislead you!

**FUN FACT!**

A 2023 study: GPT-4 made consultants faster and better, but worse on tasks beyond its reach.

**Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: The Jagged Frontier Study

**CASE FILE**

In a 2023 field experiment with Boston Consulting Group, researchers from Harvard and other universities gave 758 consultants realistic tasks with or without GPT-4. For tasks inside the AI's capability, consultants using AI completed about 12% more tasks, worked about 25% faster and produced results rated over 40% higher in quality. For a task designed to be outside the frontier, AI users were about 19 percentage points less likely to reach the correct answer.

## 🔍 Analyse the case

**1. What is the 'jagged frontier'?**

**2. Why did AI users do worse on one task?**

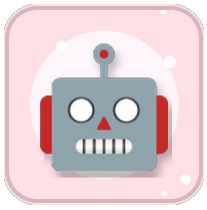
**3. How should firms train professionals to use AI?**

**⚖️ My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Map the Frontier

## You need:

**Task list**



**Chatbot**

**Teams of 3**

## Steps:

**1**

List 8 tasks in one profession.

**2**

Try each with AI; check results.

**3**

Mark inside or outside AI's frontier.

**4**

Note rules that apply to each task.

**THINK & TALK**

Which task surprised you?

**Record your observations & results:**

# 💻 CODE LAB: Frontier Mapper

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  tasks = {'draft': (0.70, 0.88), 'review': (0.75, 0.86), 'novel': (0.80, 0.62)}
 2  for t, (alone, ai) in tasks.items():
 3      print(t, round(ai - ____, 2))
```

**COMPLETE** **T1. Fill the blank for the gain.**

**PREDICT** **T2. Gains per task?**

**INTERPRET** **T3. Which task is outside the frontier?**

**EXTEND** **T4. Plot gains as a bar chart.**

# WORKSHEET: AI for Professional Innovation (Part A)

Name: _____ Date: _____

## 1. Match the profession to an AI innovation.

Law ●

Medicine ●

Finance ●

Engineering ●

● Generative structural design

● Contract clause review

● Fraud & risk models

● Scan triage

## 2. Productivity study arithmetic. Use the word box.

**WORD BOX: 25 ✿ 12 ✿ 19**

Task time 100 → 75 min = ____% faster

Tasks 50 → 56 = ____% more

Correct 84% → 65% = ____ points lower

# WORKSHEET: AI for Professional Innovation (Part B)

Name: _____ Date: _____

## 1. Inside or outside AI's frontier (typical today)? Tick ✓.

| | Inside | Outside |
|---|---|---|
| Drafting a standard NDA | | |
| Novel constitutional argument | | |
| Summarising a long report | | |
| Rare-disease diagnosis alone | | |

## 2. Order a professional AI pilot. Write 1–4.

**Measure vs control group**

☐

**Map tasks & risks**

☐

**Check rules & approvals**

☐

**Redesign workflow & train**

☐

# PUZZLE & DEFINE: AI for Professional Innovation

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**Frontier**

Meaning: _____

Example: _____

**Compliance**

Meaning: _____

Example: _____

**Licensure**

Meaning: _____

Example: _____

# ? QUIZ TIME: AI for Professional Innovation

**Circle the right answer!**

**Q1. Where AI is reliable is its...**

a) Frontier

b) Home

c) Palette

**Q2. Outside its frontier AI can...**

a) Always win

b) Mislead

c) Sleep

**Q3. Following professional rules is...**

a) Branding

b) Rhythm

c) Compliance

**Q4. The 2023 study involved...**

a) Consultants

b) Footballers

c) Singers

**Q5. Professional AI should be built with...**

a) Guesswork

b) Domain experts

c) Influencers

**MY SCORE: _____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: AI for Professional Innovation

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | AI capability is uneven across tasks. | | |
| | AI always improves professional accuracy. | | |
| | Professions have compliance duties. | | |
| | Domain experts are essential for AI innovation. | | |
| | Vendor demos replace local validation. | | |
| | The 2023 study involved 758 consultants. | | |

**HOME CONNECT**

Interview a professional. Which of their tasks might be inside AI's frontier, and which outside?

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: AI for Professional Innovation

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why can AI reduce accuracy on some tasks?**

**COMPARE** **Q2. Centaur vs cyborg styles of AI use.**

**APPLY** **Q3. Correct 84% → 65% with AI. Drop?**

**EVALUATE** **Q4. Should a hospital adopt an AI tool from a vendor demo?**

**CREATE** **Q5. Design a pilot for AI in audit firms.**

# 📝 EXAM PREP: AI for Professional Innovation

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is the jagged frontier?

**2.** Name two professional duties relevant to AI.

**3.** Why validate AI locally?

**4.** Give one outcome metric for professional AI.

## PART B • Long answer (5 marks)

**5.** Evaluate how AI can drive innovation in one profession, including risks, regulation and evaluation design.

# UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

**Circle the right answer!**

**Q1. Removing parts to test value:**

a) Ablation | b) Promotion | c) Guessing

**Q2. Test info sneaking into training:**

a) Privacy | b) Leakage | c) Balance

**Q3. Run many seeds to check...**

a) Style | b) Speed | c) Luck vs signal

**Q4. Jagged frontier study had ~...**

a) 758 staff | b) 12 staff | c) 1 million

**Q5. Every result needs a fair...**

a) Party | b) Baseline | c) Headline

**Q6. Medicine and law need AI with...**

a) No checks | b) Hype | c) Compliance

# WORD SEARCH: AI SOLVE

## Find these words. They go across → or down ↓.

ABLATION

SEED

LEAKAGE

FRONTIER

COMPLIANCE

LICENSURE

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| M | P | W | T | N | B | W | B | E | A |
| O | B | M | R | L | O | O | A | T | B |
| S | E | E | D | P | A | H | S | H | L |
| E | V | I | D | E | N | C | E | I | A |
| G | M | E | T | H | O | D | L | C | T |
| D | O | M | A | I | N | T | I | S | I |
| L | E | A | K | A | G | E | N | I | O |
| U | F | R | O | N | T | I | E | R | N |
| C | O | M | P | L | I | A | N | C | E |
| C | L | I | C | E | N | S | U | R | E |

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

# AI RISE

Skills • Careers • Future

**Area: AI & Career**

## In this unit, you will explore:

Topic 7

### AI Specialist Roadmap

Topic 8

### Research to Profession

TOPIC 7 • MONTH 4 • AI & CAREER

# AI Specialist Roadmap

## LET'S LEARN

- Go T-shaped: broad AI base, one deep specialism.
- Tracks: NLP, vision, RL, MLOps, safety, data.
- Prove depth with projects, papers, open source.

### Popular specialisations

NLP

Vision

RL

MLOps

## KEY TERMS

**T-shaped** Broad + deep skills

**Specialism** Focused expertise

**Open source** Public shared code

# 📖 CONCEPT EXPLAINED: AI Specialist Roadmap

### 💡 In this chapter

AI is now too broad for anyone to master entirely. Employers and research groups look for people with a solid general foundation plus genuine depth in one area. Your first postgraduate year is the right time to choose a direction deliberately, based on your interests, strengths and the problems you care about, and to build visible evidence of expertise.

## 🎓 1. The T-shaped specialist

A T-shaped profile combines breadth across the AI stack, the horizontal bar, with depth in one specialism, the vertical bar. Breadth means comfort with linear algebra, probability, optimisation, Python, data handling, core ML models, deep learning basics, evaluation and ethics. Depth means knowing the literature, tools, datasets and open problems of one area well enough to contribute something new.

Breadth lets you collaborate and switch when the field shifts; depth makes you valuable and lets you lead. Over a career many people become 'comb-shaped', adding a second or third depth. Start with one, chosen for a mix of passion, aptitude and opportunity rather than hype alone.

**✏️ WORKED EXAMPLE**

Self-audit: rate yourself 1–5 on maths, programming, classical ML, deep learning, data engineering, evaluation and responsible AI. Scores below 3 in breadth areas become first-semester goals.

## 🛤️ 2. Specialist tracks

Common tracks include natural language processing, computer vision, speech, reinforcement learning and robotics, recommender systems, ML engineering and MLOps, data engineering, AI safety and alignment, interpretability, AI policy and governance, and domain-applied AI in health, climate, finance or agriculture.

Each track has its own foundational papers, benchmarks, conferences and communities. NLP researchers follow ACL and EMNLP; vision researchers CVPR and ICCV; ML generalists NeurIPS, ICML and ICLR; applied fields have domain venues. Reading recent proceedings shows which problems are active and which skills are in demand.

🤔 **PAUSE & THINK**

Choose two tracks that interest you. For each, list two foundational papers, one benchmark, one conference and one open problem. Which list was easier to complete, and why?

## 📈 3. Learning plans and milestones

A roadmap converts aspiration into milestones. A typical first-year plan might include: complete core coursework; reproduce two recent papers in your track; contribute to an open-source library; write a short survey or blog series; complete a substantial project with a real dataset; and present at a seminar or workshop.

Reproducing papers is especially valuable. It forces you to understand details the paper omits, builds engineering skill and often reveals research ideas. Track your time and progress monthly; adjust the plan when you learn something about your interests, but avoid switching tracks every few weeks.

✏️ **WORKED EXAMPLE**

Skill gap code: need = set of required skills; have = set of current skills; gap = need - have. For need = 4 skills and overlap = 2, coverage is 50%; target 100% by semester end.

## 🌐 4. Proving depth publicly

Credentials matter less than visible evidence. A well-documented GitHub repository, a reproduced paper with honest findings, merged pull requests to a respected library, a Kaggle or shared-task result, a technical blog and a workshop paper all demonstrate competence to employers and admissions committees.

The history of deep learning shows the value of persistence. Hinton, Bengio and LeCun worked on neural networks for decades, often against prevailing opinion, before the 2018 Turing Award recognised their contributions; Hinton later shared the 2024 Nobel Prize in Physics with John Hopfield. Depth compounds over time when work is shared openly.

🖐️ **TRY IT**

Draft a 12-month roadmap with your chosen specialism, three breadth gaps, two papers to reproduce, one open-source target and one public output per quarter.

### ✅ Key takeaways

- T-shaped: broad AI foundation plus depth in one specialism.
- Tracks include NLP, vision, RL, MLOps, safety, policy and applied domains.
- Each track has its own papers, benchmarks and conferences.
- Reproducing papers builds depth and reveals research ideas.
- Prove expertise publicly through code, papers, open source and talks.

### ❓ Quick check: answer in one or two lines.

**1.** What does a T-shaped profile mean?

**2.** Why is reproducing a paper valuable for a new specialist?

**3.** List three forms of public evidence of depth.

## 🔬 CONCEPT DEEP DIVE: AI Specialist Roadmap

🔷 Specialisations include NLP, speech, computer vision, reinforcement learning, recommender systems, MLOps/ML platforms, AI safety and alignment, and AI policy.

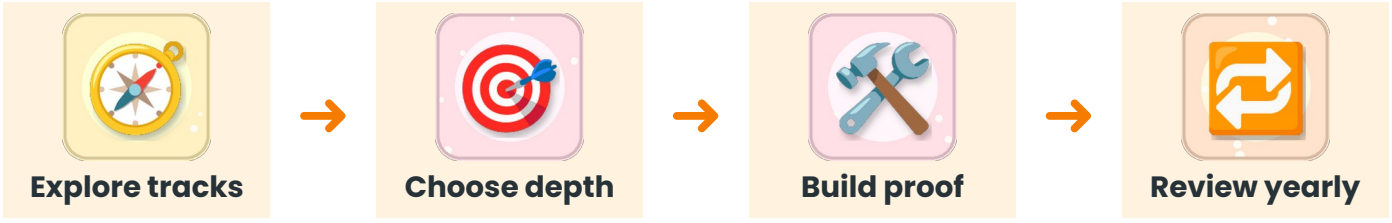
🔷 A T-shaped profile combines breadth (maths, ML fundamentals, software engineering, communication) with depth in one area.

🔷 Depth is demonstrated through substantial projects, publications, open-source contributions, competitions and internships.

🔷 Learning resources: graduate courses, research papers, reading groups, open courseware and mentorship.

🔷 Revisit your roadmap each semester as fields evolve; adjacent skills keep options open.

### 🔄 How it works



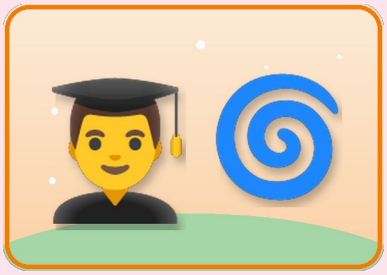

### 💻 Skill-gap analysis

```
need = {'pytorch', 'cv', 'statistics', 'docker'}
have = {'python', 'statistics', 'pytorch'}
gap = sorted(need - have)
print(gap, f'{len(need & have) / len(need):.0%}')   # ['cv',
'docker'] 50%
```

### ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: The Student Who Chose a Lane

**Characters:** *Rohan (MTech CSE), his advisor Dr. Fernandes, senior PhD student Ananya.*



SCENE 1

Rohan took every AI elective: vision, NLP, RL, robotics. By the end of the first semester he knew a little of everything but could not answer deep questions in any interview.

**Pause & predict:** What is missing from Rohan's profile?

SCENE 2

Dr. Fernandes asked him which problems excited him most. Rohan realised he loved speech, especially for Indian languages. Ananya suggested he become T-shaped: keep his breadth, but go deep in speech.

**Pause & predict:** How should he choose a specialism?

SCENE 3

Rohan reproduced two recent speech papers, found a bug in one open-source toolkit and submitted a fix that was merged. He started a blog explaining his experiments in simple language.

**Pause & predict:** Why do reproductions and open-source work signal depth?

SCENE 4

At an industry interview, Rohan was asked about low-resource speech recognition. He talked for twenty minutes, with examples from his own experiments. He received an internship offer the next week.

⏸ **Pause & predict:** How did public evidence help him?

## 🌟 The big idea

*Breadth helps you connect; depth, shown publicly, makes you valuable.*

## 💬 Discuss with your classmates

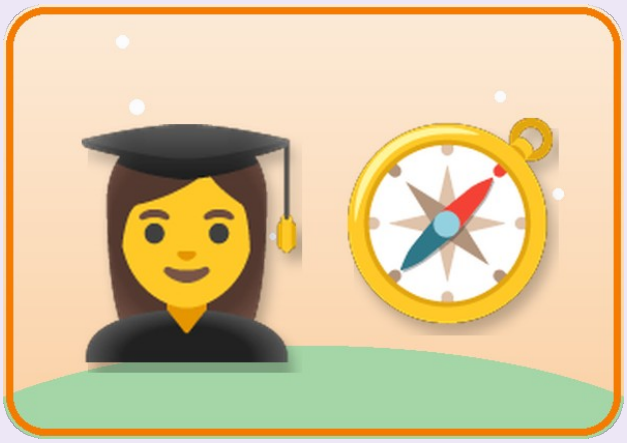
**1.** Which specialism would you choose today, and why?

**2.** What public evidence of depth could you build this year?

## SCENARIO: AI Specialist Roadmap

Read the scenario. Think about the choices made at each step.

**1**

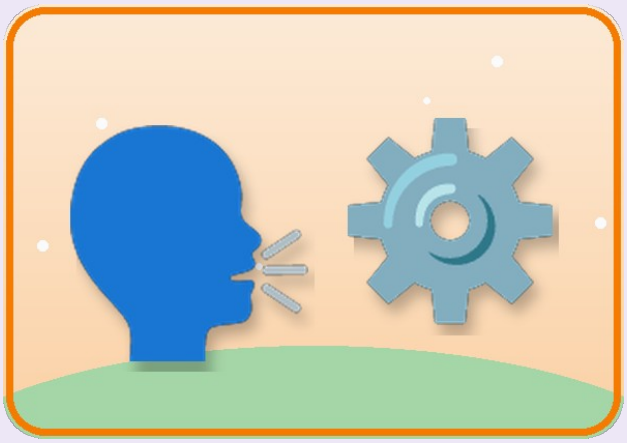


**Meera likes both NLP and MLOps.**

**2**

**Job-post analysis shows demand for both.**

**3**



**She picks NLP depth with MLOps breadth.**

**4**

**Her Tamil ASR project lands a lab role.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Specialist roles

Speech scientist

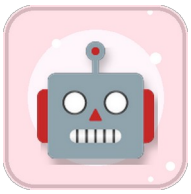
Vision engineer

AI safety researcher

ML platform engineer

Applied scientist

AI policy analyst



**BOLT SAYS:**

Breadth gets you in the room. Depth makes you the person they call!

**FUN FACT!**

Hinton, Bengio and LeCun shared the 2018 Turing Award for deep learning breakthroughs.

**Reflect: where have YOU seen this in real life? Note it down.**

# CASE STUDY: Deep Learning's Long Road

**CASE FILE**

Geoffrey Hinton, Yoshua Bengio and Yann LeCun pursued neural-network research for decades, including periods when the approach was unfashionable. Their work on backpropagation, convolutional networks and representation learning underpinned the deep-learning revolution, and they shared the 2018 ACM Turing Award. Hinton was also a co-recipient of the 2024 Nobel Prize in Physics with John Hopfield for foundational work on neural networks.

## Analyse the case

**1. What does their story teach about depth and persistence?**

**2. Why might an unfashionable specialism become important?**

**3. How do you balance depth with adaptability?**

**My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Specialism Speed-Dating

## You need:

**Track cards**

**Timer**

**Whole class**

## Steps:

| | | |
|---|---|---|
| 1 | | Six stations: one per specialism. |
| 2 | | Spend 4 minutes at each; read role tasks. |
| 3 | | Rate fit 1–5 for interest and skill. |
| 4 | | Pick top 2; list next 3 steps each. |

**THINK & TALK**

Which track surprised you most?

**Record your observations & results:**

# 💻 CODE LAB: Skill Gap

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
need = {'pytorch', 'cv', 'statistics', 'docker'}
have = {'python', 'statistics', 'pytorch'}
gap = sorted(need ____ have)
print(gap, f'{len(need & have) / len(need):.0%}')
```

**COMPLETE** **T1. Fill the blank for set difference.**

**PREDICT** **T2. What is printed?**

**MODIFY** **T3. Add 'git' to need.**

**EXTEND** **T4. Rank gaps by job-post frequency.**

# WORKSHEET: AI Specialist Roadmap (Part A)

Name: _____ Date: _____

## 1. Match the specialism to a core skill.

NLP ●

● Reward design & simulation

Vision ●

● Tokenisers & transformers

RL ●

● CI/CD & monitoring

MLOps ●

● CNNs & image data

## 2. Skill-gap check. Use the word box.

**WORD BOX: 4 ✿ 2 ✿ 50**

Role needs PyTorch, CV, stats, Docker → ____ skills

You have PyTorch and stats → have ____

Coverage = ____%

# WORKSHEET: AI Specialist Roadmap (Part B)

Name: _____ Date: _____

**1. Breadth or depth skill (for an NLP specialist)? Tick ✓.**

| | Breadth | Depth |
|---|---|---|
| Linear algebra | | |
| Tokeniser design | | |
| Software testing | | |
| Transformer fine-tuning | | |

**2. Job-post demand (50 posts). Use the word box.**

**WORD BOX: 60 ✿ 30 ✿ 10**

NLP in 30 posts → ____%

Vision in 15 posts → ____%

RL in 5 posts → ____%

# PUZZLE & DEFINE: AI Specialist Roadmap

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**T-shaped**

Meaning: _____

Example: _____

**Specialism**

Meaning: _____

Example: _____

**Open source**

Meaning: _____

Example: _____

# ? QUIZ TIME: AI Specialist Roadmap

**Circle the right answer!**

**Q1. Broad base plus one deep skill:**

a) T-shaped

b) O-shaped

c) X-shaped

**Q2. Reward design is core to...**

a) NLP

b) RL

c) Design

**Q3. CI/CD and monitoring belong to...**

a) Vision

b) Statistics

c) MLOps

**Q4. Publicly shared code is...**

a) Open source

b) Closed

c) Paid only

**Q5. The 2018 Turing Award honoured...**

a) Music AI

b) Deep learning

c) Cars

**MY SCORE: _____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: AI Specialist Roadmap

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | TRUE | FALSE |
|---|---|---|
| T-shaped profiles combine breadth and depth. | | |
| Specialisms should be chosen only by hype. | | |
| Open-source work can prove depth. | | |
| Roadmaps should be reviewed regularly. | | |
| Deep learning was always popular. | | |
| Hinton, Bengio and LeCun shared the 2018 Turing Award. | | |

**HOME CONNECT**

Read 3 job posts for your dream role. List the skills and mark which you have.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: AI Specialist Roadmap

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why be T-shaped rather than only broad?**

**COMPARE** **Q2. Applied scientist vs ML platform engineer.**

**APPLY** **Q3. Need 4 skills, have 2. Coverage?**

**EVALUATE** **Q4. Choose a specialism only by hype?**

**CREATE** **Q5. Write a 12-month roadmap for one specialism.**

# 📝 EXAM PREP: AI Specialist Roadmap

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is a T-shaped profile?

**2.** Name four AI specialisations.

**3.** How is depth demonstrated?

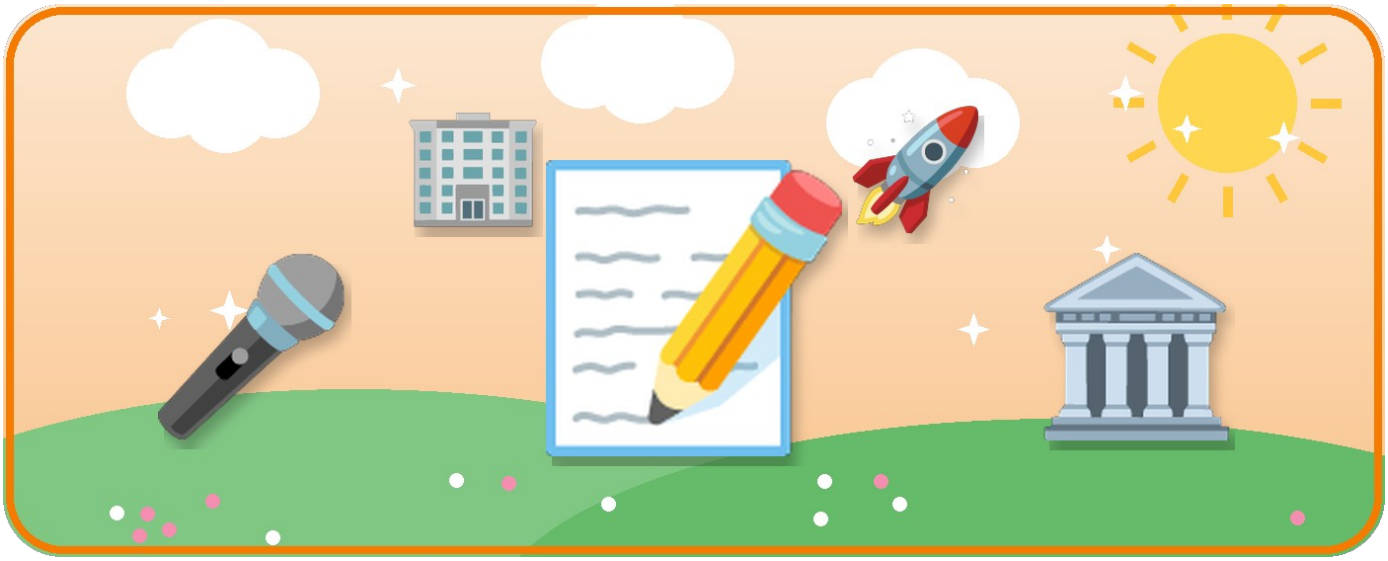
**4.** Why revisit roadmaps?

## PART B • Long answer (5 marks)

**5.** Design a two-year AI specialist roadmap and justify the specialism, skills and evidence you will build.

TOPIC 8 • MONTH 4 • AI & CAREER

# Research to Profession



## LET'S LEARN

- Research careers: academia, industry labs, start-ups.
- Publish, review, present and collaborate.
- Ideas become products, patents and policy.

### Research career paths

**Academia**

**Industry lab**

**Research start-up**

**Policy & public labs**

## KEY TERMS

**Peer review**
Expert checking

**Poster** Visual summary

**Tech transfer**
Lab to market

# 📖 CONCEPT EXPLAINED: Research to Profession

### 💡 In this chapter

Research skills open many professional paths: academic careers, industry research labs, applied science roles, start-ups, policy and consulting. Moving from research to profession involves learning how research is published, reviewed and communicated, and how ideas become products, patents, open-source tools or policy. This chapter maps that journey.

## 🏛️ 1. Career paths for researchers

Academia offers intellectual freedom and teaching, usually via PhD, postdoctoral and faculty positions, with competition for grants and posts. Industry research labs offer compute, data and engineering support, with research agendas aligned to company goals. Applied scientist and ML engineer roles turn research into products.

Start-ups let researchers commercialise ideas quickly but carry financial risk. Policy, think-tank and standards roles need people who understand technology deeply and can communicate with non-experts. Many careers move between these sectors; skills in writing, collaboration and rigorous evaluation transfer across all of them.

### ✏️ WORKED EXAMPLE

Indian pathways include research positions at IITs, IISc and IIITs, fellowships such as the Prime Minister's Research Fellowship, industry labs in Bengaluru and Hyderabad, and start-up incubators at universities.

## 📝 2. Publishing and peer review

In AI, conferences are the main venues. Authors submit papers, typically eight or nine pages plus appendices; anonymous reviewers score them on novelty, soundness, clarity and significance; an area chair recommends acceptance. Rebuttal periods let authors answer reviewer concerns. Preprints on arXiv spread work quickly but are not peer reviewed.

Good reviewing is a professional skill: summarise the paper fairly, identify strengths, raise specific concerns with evidence, and suggest improvements. Many students start by serving as sub-reviewers or reviewing for workshops. Learning to review sharpens your own writing, because you see what makes claims convincing.

🤔 **PAUSE & THINK**

A reviewer writes only 'not novel enough; reject'. What would make this review more useful to the authors and the area chair?

## 🎤 3. Communicating research

Researchers communicate through papers, posters, talks, blog posts, code and demos. Posters should state the question and main result in seconds, use figures over text and invite conversation. Talks should tell a story: the problem, why it matters, the key idea, evidence and limitations.

Communicating to non-specialists matters for funding, policy and impact. Explain without jargon, use concrete examples, and be honest about uncertainty. Collaboration, through co-authorship, open datasets and shared tasks, multiplies impact; the 2017 transformer paper had eight authors who went on to shape the field in many organisations.

✏️ **WORKED EXAMPLE**

Poster test: a passer-by should grasp your question, method and main number from two metres away within ten seconds.

## 🔄 4. From idea to impact

Technology transfer moves research into use through licensing, patents, spin-offs, open-source releases or collaboration with industry and government. Universities have technology transfer offices to help with intellectual property and contracts. Choosing between patenting and open release depends on goals: adoption, revenue, or public good.

Moving from prototype to product requires much more: reliability, scalability, user research, regulatory compliance, support and business models. Researchers who succeed in this transition partner with engineers, designers and business people, and accept that the product may look very different from the paper.

🖐️ **TRY IT**

Pick a research idea you have. Write a one-page impact plan: possible paper venue, open-source release, potential users, transfer route and the first collaborator you would approach.

### ✅ Key takeaways

- Research careers span academia, industry labs, applied roles, start-ups and policy.
- AI conferences use peer review with rebuttals; arXiv preprints are not reviewed.
- Reviewing well is a professional skill that improves your own writing.
- Communicate through papers, posters, talks, code and plain-language summaries.
- Tech transfer: patents, licensing, spin-offs and open source.

### ❓ Quick check: answer in one or two lines.

**1.** Contrast academic and industry research careers.

**2.** What makes a peer review useful?

**3.** Name three routes for transferring research to impact.

## 🔬 CONCEPT DEEP DIVE: Research to Profession

🔷 Research careers span academia (teaching + research), industry labs (applied and fundamental), start-ups and public research bodies.

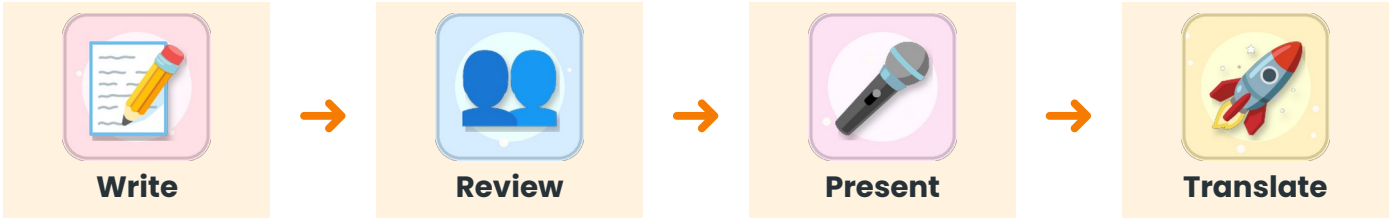
🔷 Publishing: write clearly, position against related work, submit to venues, respond to peer review, present posters and talks.

🔷 Open science: share code, data (where permitted) and models; cite properly; respect authorship and ethics.

🔷 Translation: patents, licensing, start-ups, open-source adoption and policy briefs move research into impact.

🔷 PhD vs industry: PhDs build independent research skills; industry offers scale and product impact; many move between them.

### 🔄 How it works



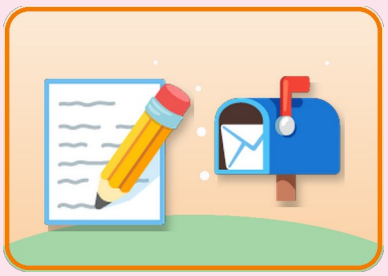

### 💻 h-index from citations

```
cites = [10, 8, 5, 4, 3]
c = sorted(cites, reverse=True)
h = sum(1 for i, x in enumerate(c, 1) if x >= i)
print(h)   # 4
```

### ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: Reviewer Number Two

**Characters:** *Priya (MSc AI), her co-author Kabir, Prof. Das (supervisor).*



SCENE 1

Priya's first paper was rejected. One reviewer wrote only 'not novel'. Another gave three pages of detailed concerns about baselines and missing ablations. Priya felt crushed.

**Pause & predict:** Which review is more useful, and why?

SCENE 2

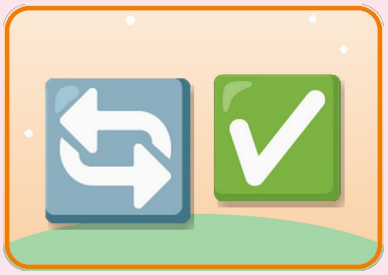
Prof. Das asked her to treat the detailed review as free consulting. They added a stronger baseline, two ablations and seed statistics. The gain shrank but became convincing.

**Pause & predict:** How can rejection improve research?

SCENE 3

The revised paper was accepted as a workshop poster. Kabir designed it so the main result was readable from two metres. A start-up founder stopped, asked questions and invited them for a chat.

**Pause & predict:** Why do posters create opportunities?



SCENE 4

The founder licensed their method through the university's technology transfer office. Priya also began reviewing for a workshop, writing the kind of careful reviews that once helped her.

⏸ **Pause & predict:** What routes from research to impact appear here?

## 🌟 The big idea

*Peer review, clear communication and tech transfer turn research into a profession.*

## 💬 Discuss with your classmates

**1.** How would you respond to a harsh review?

**2.** What would your research poster's one-line message be?

## SCENARIO: Research to Profession

Read the scenario. Think about the choices made at each step.

**1**

**Rohan's thesis improves Hindi OCR.**

**2**

**Reviewers ask for stronger baselines; he revises.**

**3**

**At a conference, an industry lab notices his work.**

**4**

**He joins the lab; his code powers a product.**

**Talk about it:**

What decision was made? What would YOU have done differently?

**What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Research outputs

Papers

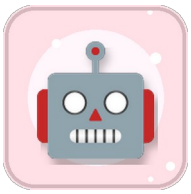
Talks & posters

Open code

Patents

Datasets

Policy briefs



**BOLT SAYS:**

A great idea unpublished, unshared and untested helps no one!

**FUN FACT!**

All eight authors of the 2017 Transformer paper later left Google, many to found start-ups.

**Reflect: where have YOU seen this in real life? Note it down.**

# CASE STUDY: The Transformer Authors

**CASE FILE**

The 2017 paper 'Attention Is All You Need', written by eight researchers at Google, introduced the Transformer architecture that underlies most modern language models. In the years that followed, all eight authors left Google. Several founded or co-founded AI start-ups, including companies working on language models and enterprise AI, illustrating how influential research can lead to diverse professional paths.

## Analyse the case

**1. Why did one paper have such large impact?**

**2. What paths did the authors take?**

**3. How can you position your research for impact?**

**My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Mini Conference

## You need:

1-page abstracts

Timer

Whole class

## Steps:

| | | |
|---|---|---|
| 1 | | Each student writes a 150-word abstract. |
| 2 | | Peers review 2 abstracts with a rubric. |
| 3 | | Top 5 give 3-minute talks. |
| 4 | | Q&A: one tough question each. |

**THINK & TALK**

What made an abstract convincing?

**Record your observations & results:**

# 💻 CODE LAB: h-index

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
def h_index(cites):
    c = sorted(cites, reverse=____)
    return sum(1 for i, x in enumerate(c, 1) if x >= i)
print(h_index([10, 8, 5, 4, 3]), h_index([6, 5, 3, 1]))
```

**COMPLETE** **T1. Sort descending.**

**PREDICT** **T2. Both h-indices?**

**EXPLAIN** **T3. Why is h-index an imperfect measure?**

**EXTEND** **T4. Compute i10-index.**

# WORKSHEET: Research to Profession (Part A)

Name: _____ Date: _____

## 1. Order the publication process. Write 1–4.

Present at conference ☐

Write & submit paper ☐

Revise after reviews ☐

Peer review ☐

## 2. Match the research output to its audience.

Paper ● ● Developers who build on it

Patent ● ● Researchers & reviewers

Policy brief ● ● Government decision-makers

Open code ● ● Companies & legal protection

# WORKSHEET: Research to Profession (Part B)

Name: _____ Date: _____

## 1. h-index practice. Use the word box.

**WORD BOX:** 4 ✿ 3 ✿ 2

Citations 10, 8, 5, 4, 3 → h = ____

Citations 6, 5, 3, 1 → h = ____

Citations 9, 2, 1 → h = ____

## 2. Ethical or unethical research practice? Tick ✓.

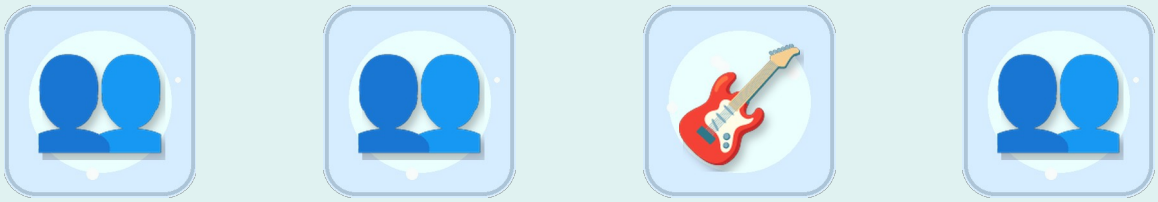
| | Ethical | Unethical |
|---|---|---|
| Citing all sources used | | |
| Gift authorship to a senior | | |
| Sharing code with paper | | |
| Dropping inconvenient results | | |

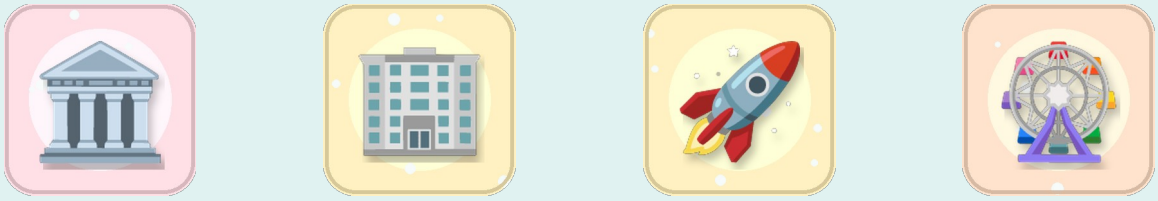
# PUZZLE & DEFINE: Research to Profession

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

A

B 

C 

## 2. Define it! Write each term in your own words with an example.

**Peer review**

Meaning: _____

Example: _____

**Poster**

Meaning: _____

Example: _____

**Tech transfer**

Meaning: _____

Example: _____

# ? QUIZ TIME: Research to Profession

**Circle the right answer!**

**Q1. Experts checking a paper:**

a) Peer review

b) Advert

c) Lottery

**Q2. Moving research into products is...**

a) Remixing

b) Tech transfer

c) Cleaning

**Q3. A visual research summary at events:**

a) Email

b) Game

c) Poster

**Q4. The Transformer paper appeared in...**

a) 2017

b) 2007

c) 2023

**Q5. Policy briefs mainly target...**

a) Gamers

b) Decision-makers

c) Children

**MY SCORE: ____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: Research to Profession

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| 👥 | Peer review checks research quality. | | |
| 👥 | Gift authorship is acceptable. | | |
| 💻 | Sharing code aids reproducibility. | | |
| 🔁 | Research can reach impact via patents and start-ups. | | |
| 🏛️ | Researchers can never move between academia and industry. | | |
| 📜 | All eight Transformer authors later left Google. | | |

**HOME CONNECT**

Find one paper in your area. Write its problem, method and result in 3 plain sentences.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Research to Profession

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why does peer review matter?**

**COMPARE** **Q2. Academic vs industry research careers.**

**APPLY** **Q3. Citations 10, 8, 5, 4, 3. h-index?**

**EVALUATE** **Q4. Should you add a friend as co-author for no work?**

**CREATE** **Q5. Write a 3-sentence lay summary of your research.**

# 📝 EXAM PREP: Research to Profession

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** Outline the publication process.

**2.** What is technology transfer?

**3.** Name two research-ethics principles.

**4.** Compare PhD and industry research.

## PART B • Long answer (5 marks)

**5.** Explain the pathways from research to professional impact, with examples and ethical considerations.

# UNIT REVIEW: AI RISE

Name: _____ Date: _____

**Circle the right answer!**

**Q1.** Broad base + one deep skill:

a) T-shaped

b) O-shaped

c) Flat

**Q2.** Papers are checked by...

a) Adverts

b) Peer review

c) Lottery

**Q3.** 'Attention Is All You Need' gave us...

a) CNNs

b) Trees

c) Transformers

**Q4.** Research into products is...

a) Tech transfer

b) Cleanup

c) Shutdown

**Q5.** Strong proof of depth:

a) Claims

b) Open source

c) Logos

**Q6.** 2018 Turing Award went to...

a) Singers

b) Players

c) DL pioneers

# WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

T-SHAPED

SPECIALISM

OPEN SOURCE

PEER REVIEW

POSTER

MENTOR

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| T | G | T | S | H | A | P | E | D | L |
| S | P | E | C | I | A | L | I | S | M |
| D | S | T | A | R | T | U | P | B | H |
| P | A | T | E | N | T | I | U | W | H |
| T | H | E | S | I | S | D | L | F | K |
| P | E | E | R | R | E | V | I | E | W |
| P | A | B | M | P | O | S | T | E | R |
| C | M | E | N | T | O | R | K | L | A |
| O | P | E | N | S | O | U | R | C | E |
| L | K | P | A | P | E | R | L | E | O |

**I finished AI RISE! Teacher's sign: _____**

MONTH 5

# AI CREATE

Try • Build • Innovate

**Area: AI Tools**

## In this unit, you will explore:

Topic 9

### AI Research Workbench

Topic 10

### Intelligent Prototype Lab

TOPIC 9 • MONTH 5 • AI TOOLS

# AI Research Workbench

## LET'S LEARN

- Workbench: notebooks, Git, tracking, compute, data.
- Reproducible by design: seeds, configs, envs.
- Organise experiments like a lab notebook.

### Research workbench kit

**Notebooks**

**Git + DVC**

**Tracking**

**GPU compute**

## KEY TERMS

**Config** Settings file

**Seed** Fix randomness

**Ref manager** Citation organiser

# 📖 CONCEPT EXPLAINED: AI Research Workbench

### 💡 In this chapter

A research workbench is the set of tools, habits and structures that let you run experiments efficiently and reproducibly. Postgraduate research involves hundreds of runs, many datasets and months of iteration. Without an organised workbench, results get lost, comparisons become unfair and papers cannot be reproduced. With one, you move faster and with confidence.

## 🧰 1. Core tools

A typical workbench includes Python with scientific libraries, Jupyter notebooks for exploration, an IDE for production code, Git for version control, an experiment tracker such as MLflow or Weights & Biases, a dataset versioning tool, and compute through university clusters, cloud credits or local GPUs.

Notebooks are excellent for exploration but risky for final results because of hidden state and out-of-order cells. A common practice is to explore in notebooks, then move stable code into modules and scripts run from configuration files. Keep notebooks short, restart and rerun them fully before trusting results.

**✏️ WORKED EXAMPLE**

Project layout: data/ (raw, processed), src/ (modules), configs/ (YAML per experiment), notebooks/ (exploration), runs/ (logs, tracked), paper/ (figures, draft), README with setup steps.

## 🔁 2. Reproducible by design

Reproducibility should be built in, not added at the end. Fix random seeds for Python, NumPy and your deep learning framework; record library versions in lock files or containers; store every experiment's configuration; and log the Git commit hash with each run.

Even with seeds, GPU non-determinism can cause small differences, so report spread across several seeds rather than a single run. The NeurIPS reproducibility checklist, introduced in 2019, asks authors about data, code, hyperparameters, compute and statistical reporting; using it as a personal checklist from day one makes final submission much easier.

🤔 **PAUSE & THINK**

Your run on a laptop and a cluster give results differing by 0.3%. Is this a reproducibility failure? How would you decide?

## 📓 3. The digital lab notebook

Experimental scientists keep lab notebooks; AI researchers should too. For each experiment, record the date, hypothesis, configuration, run identifiers, results and interpretation, including failures. This prevents repeating dead ends and helps write the methods section later.

Reference managers such as Zotero help you store papers, notes and citations. Tag papers by topic and keep short summaries: problem, method, result, limitation, relevance to your work. Over a year this becomes a personal literature map that speeds up related-work sections and idea generation.

✏️ **WORKED EXAMPLE**

Notebook entry: '12 Mar. H: data augmentation reduces overfitting on small set. Config aug_v2.yaml, runs 141–145, seeds 0–4. Val F1 0.78 ± 0.01 vs 0.74 ± 0.02. Supports H. Next: ablate each augmentation.'

## 🖥️ 4. Compute and collaboration

Plan compute like a budget. Estimate the GPU hours each experiment needs, run small pilot experiments first, and use job schedulers on clusters. Hyperparameter searches should use sensible ranges and early stopping to save resources and energy.

Collaboration needs shared conventions: branch naming, code review, shared trackers and data access rules. Sensitive data requires secure storage and ethics approval. A short onboarding document lets new lab members reproduce your main result within a day, which is a strong test of your workbench.

🖐️ **TRY IT**

Set up a research project template with folders, a config file, seed-fixing code, experiment tracking and a README. Ask a classmate to reproduce one run using only the README.

## ✅ Key takeaways

- Workbench: Python, notebooks, IDE, Git, tracker, data versioning, compute.
- Explore in notebooks; move final code to scripts with configs.
- Fix seeds, lock versions, log commits; report spread across seeds.
- Keep a digital lab notebook and a reference manager.
- Budget compute and share conventions for collaboration.

## ❓ Quick check: answer in one or two lines.

**1.** Why are notebooks risky for final results?

**2.** List four steps that make an experiment reproducible.

**3.** What should a lab notebook entry contain?

# 🔬 CONCEPT DEEP DIVE: AI Research Workbench

🔷 Workbench layers: data storage and versioning, code in Git, environments in containers, compute (GPUs/cloud), and experiment tracking.

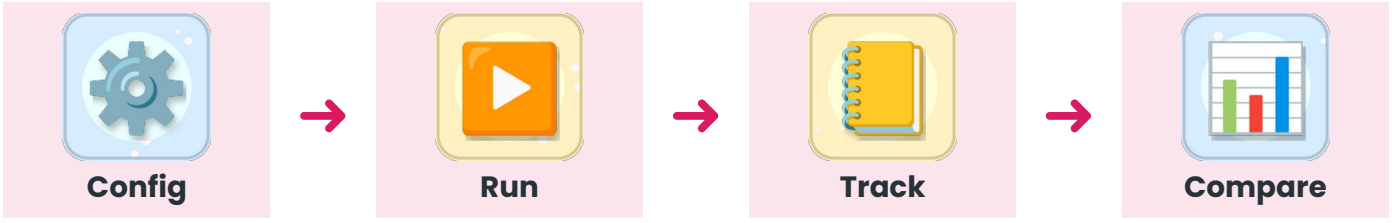
🔷 Configuration files separate settings from code; every run records config, seed, code commit and data version.

🔷 Notebooks are good for exploration; move stable code into modules and scripts with tests.

🔷 Compute planning: estimate GPU-hours, use smaller pilots, checkpoint long runs and monitor costs.

🔷 Research hygiene: lab notebook, reference manager, naming conventions and regular backups.

## 🔄 How it works




## 💻 Reproducible random runs

```
import random
random.seed(42); a = [random.random() for _ in range(3)]
random.seed(42); b = [random.random() for _ in range(3)]
print(a == b)   # True
```

## ✍️ In your own words: explain this topic to a Class 10 student.

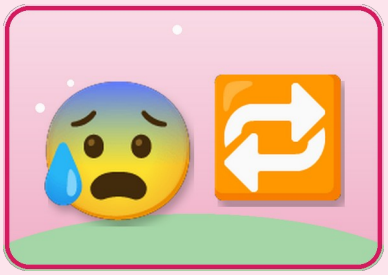
# AI STORY: The Lost Best Run

**Characters:** *Tanvi (MS by Research), her labmate Vivek, Dr. Pillai (supervisor).*

SCENE 1

Two weeks before a deadline, Tanvi remembered a run from months ago that had scored the best F1 of her project. She could not find which settings produced it.

**Pause & predict:** What workbench habit was missing?



SCENE 2

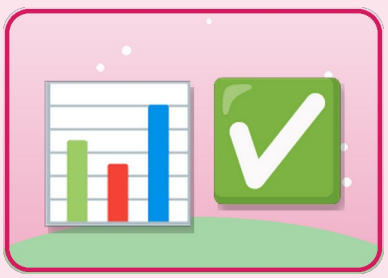
Her notebook had been edited many times. The seed was never set, and she had upgraded a library. Rerunning gave lower scores every time. Vivek suggested she stop chasing the ghost run.

**Pause & predict:** Why can't she reproduce it?

SCENE 3

Dr. Pillai helped her set up configs, seeds, Git commits per run, an experiment tracker and a digital lab notebook. They reran the key experiments across five seeds.

**Pause & predict:** Which tools solve which problems?



SCENE 4

Her final result was slightly lower than the lost run but reported with mean and spread. Using the reproducibility checklist, the paper went in on time. Her template is now used by the whole lab.

⏸ **Pause & predict:** Why is reporting spread more honest?

## 🌟 The big idea

*A good workbench makes every result traceable; luck is not a method.*

## 💬 Discuss with your classmates

**1.** How do you currently record your experiments?

**2.** What would a new labmate need to reproduce your work?

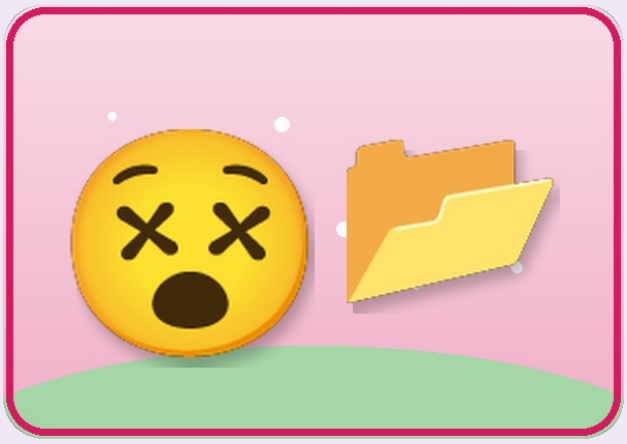
## SCENARIO: AI Research Workbench

Read the scenario. Think about the choices made at each step.

**1**

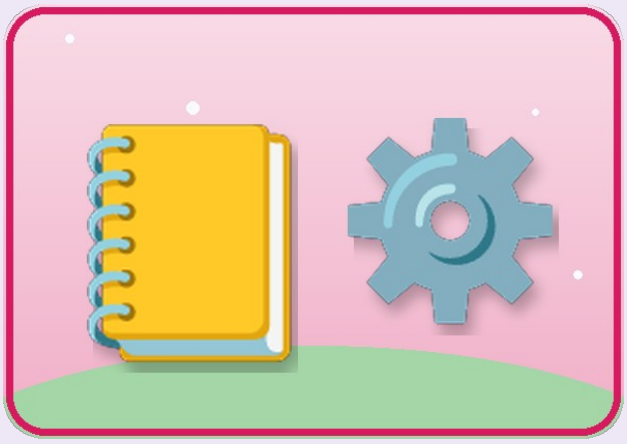
**Ananya runs 60 experiments in two weeks.**

**2**



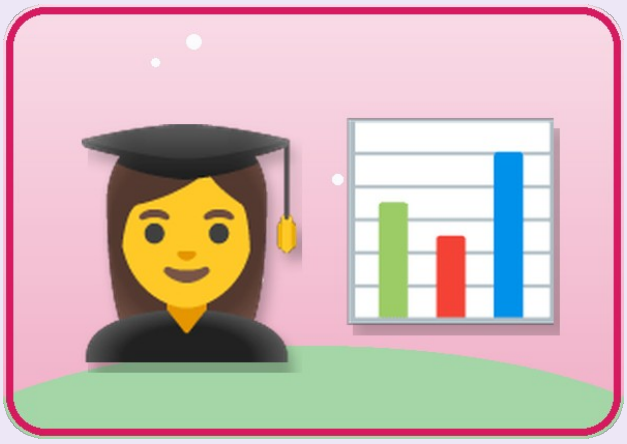
**She can't recall which settings gave the best result.**

**3**



**She adopts configs, tracking and seeds.**

**4**



**Her advisor reviews all runs in a dashboard.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Workbench tools

Jupyter / Colab

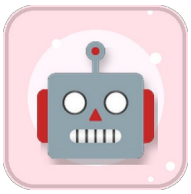
GitHub

W&B / MLflow

Containers

Cloud GPUs

Zotero



**BOLT SAYS:**

If you can't rerun it next month, it isn't a result yet!

**FUN FACT!**

NeurIPS introduced a reproducibility checklist for paper submissions in 2019.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: Checklists for Reproducible Research

**CASE FILE**

In 2019 the NeurIPS conference introduced a reproducibility programme, including a checklist that authors completed at submission covering items such as data, code availability, hyperparameters, compute and statistical reporting. Other venues adopted similar checklists. Organisers reported that more submissions began to include code, reflecting growing community attention to reproducibility.

## 🔍 Analyse the case

**1. Which checklist items would you find hardest?**

**2. How does sharing code help reviewers?**

**3. What belongs in your own lab's checklist?**

### ⚖️ My verdict: Was AI used well here? Why?

# LAB ACTIVITY: Reproduce-a-Friend

## You need:

**Laptops**

**Partner's repo**

**Pairs**

## Steps:

1. Swap repos with a partner.
2. Try to rerun their main result.
3. Log every missing step.
4. Fix your own repo from their notes.

**THINK & TALK**

What was the most common gap?

**Record your observations & results:**

# 💻 CODE LAB: Seeds & Reproducibility

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  import random
 2  random.seed(____); a = [round(random.random(), 3) for _ in
range(3)]
 3  random.seed(42); b = [round(random.random(), 3) for _ in
range(3)]
 4  print(a == b)
```

**COMPLETE** **T1. Make a equal b.**

**PREDICT** **T2. What is printed?**

**MODIFY** **T3. Use seed 7 for a. Result?**

**EXTEND** **T4. Also seed numpy.**

# WORKSHEET: AI Research Workbench (Part A)

Name: _____ Date: _____

## 1. Reproducible or not? Tick ✓.

| | Yes | No |
|---|---|---|
| Seed fixed and logged | | |
| 'Tuned by hand, not recorded' | | |
| Dockerfile in repo | | |
| Data only on a lost laptop | | |

## 2. GPU budget. Use the word box.

**WORD BOX: 40 ✿ 8 ✿ 320**

20 runs × 2 GPU-hours = ____ GPU-h

Price per GPU-h = ₹____ (given ₹320 ÷ 40)

Total cost at ₹8/h = ₹____

# WORKSHEET: AI Research Workbench (Part B)

Name: _____ Date: _____

## 1. Order a tracked experiment. Write 1–4.

Compare in dashboard ☐

Write config & seed ☐

Log params & metrics ☐

Launch run ☐

## 2. Circle good workbench habits.

Config files per run

Untitled notebooks only

Regular backups

No seeds

Reference manager

Delete failed runs' logs

# PUZZLE & DEFINE: AI Research Workbench

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

## 2. Define it! Write each term in your own words with an example.

**Config**

Meaning: _____

Example: _____

**Seed**

Meaning: _____

Example: _____

**Ref manager**

Meaning: _____

Example: _____

# ? QUIZ TIME: AI Research Workbench

**Circle the right answer!**

**Q1. A settings file for runs:**

a) Config

b) Theme

c) Playlist

**Q2. Fixing randomness uses a...**

a) Plant

b) Seed

c) Key

**Q3. Organises citations:**

a) Docker

b) Git

c) Ref manager

**Q4. NeurIPS added a reproducibility checklist in...**

a) 2019

b) 2009

c) 2025

**Q5. A result you can't rerun is...**

a) Final

b) Not yet reliable

c) Better

**MY SCORE: ____ / 5. Shade one star per correct answer.**

# ✅ TRUE OR FALSE: AI Research Workbench

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | Fixing seeds aids reproducibility. | | |
| | Notebooks alone are ideal for final pipelines. | | |
| | Configs record run settings. | | |
| | Research data should be backed up. | | |
| | Failed runs should be deleted without notes. | | |
| | NeurIPS added a reproducibility checklist in 2019. | | |

**HOME CONNECT**

Set up a project folder with data/, src/, configs/, results/ and a README. Commit it to Git.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: AI Research Workbench

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why separate config from code?**

COMPARE **Q2. Notebook vs script for experiments.**

APPLY **Q3. 20 runs × 2 GPU-h at ₹8/h. Cost?**

EVALUATE **Q4. Store data only on your laptop?**

CREATE **Q5. Design a run-naming convention.**

# EXAM PREP: AI Research Workbench

Name: _____ Date: _____

**Time: 30 min Max marks: 13 Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** List workbench components.

**2.** Why use config files?

**3.** How do seeds aid reproducibility?

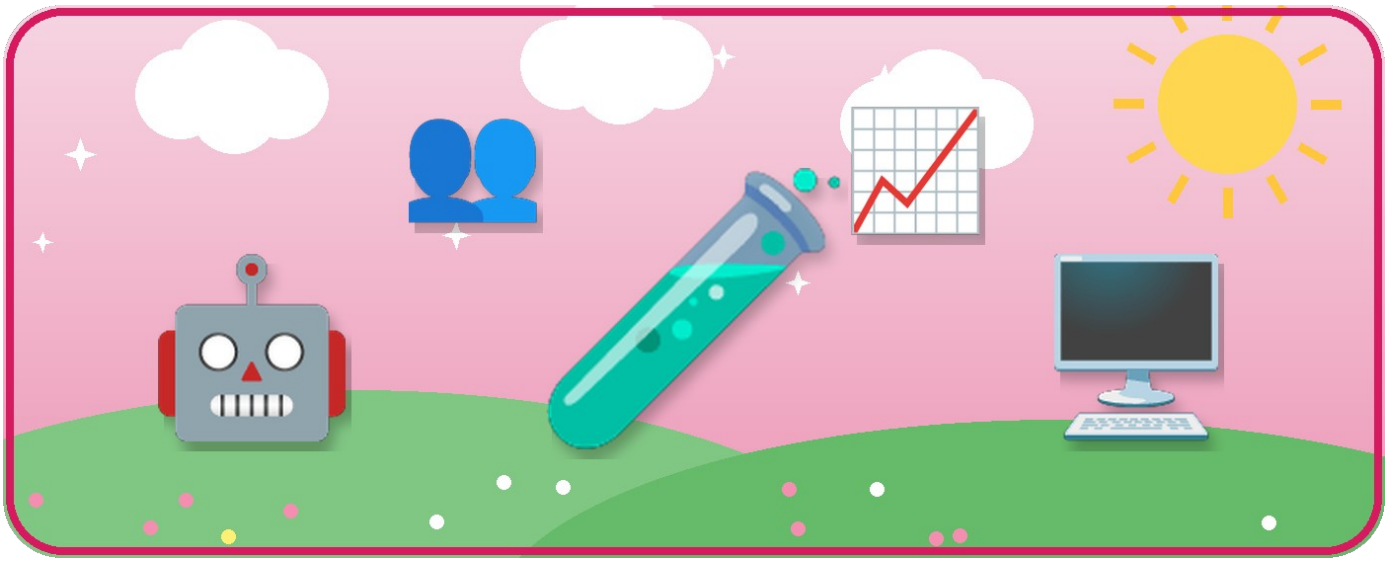
**4.** Notebook vs script?

## PART B • Long answer (5 marks)

**5.** Design a reproducible research workbench for a PG AI lab.

TOPIC 10 • MONTH 5 • AI TOOLS

# Intelligent Prototype Lab



## LET'S LEARN

- Research prototypes test ideas on real data and users.
- Build an evaluation harness before polishing.
- Validate on realistic cases, not synthetic demos.

### Research prototype loop

**Hypothesis** **Prototype** **Eval harness** **User study**

## KEY TERMS

**Harness**
Automated eval suite

**Real cases**
Actual-world data

**User study** Test with people

# 📖 CONCEPT EXPLAINED: Intelligent Prototype Lab

### 💡 In this chapter

A research prototype tests whether an idea works for real users on real data. It is not a product, but it must be honest: built on realistic inputs, evaluated rigorously and designed to reveal failure. Postgraduates often build prototypes for theses, start-up ideas or industry projects; doing it well saves months and avoids misleading conclusions.

## 🎯 1. Define the hypothesis

Every prototype should test a clear hypothesis: 'an AI summariser will cut case-note writing time by 30% without reducing accuracy', for example. The hypothesis defines users, tasks, metrics and the minimum evidence needed to proceed. Without one, prototypes drift into polished demos that prove little.

Keep scope narrow. One user group, one workflow, one key metric. Decide in advance what result would cause you to stop, pivot or continue. This discipline protects you from sunk-cost bias after weeks of building.

**✏️ WORKED EXAMPLE**

Pre-registered criteria: continue if time saved ≥ 25% and error rate ≤ baseline; pivot if time saved < 10%; stop if any serious safety error occurs.

## 🧪 2. Build the evaluation harness first

An evaluation harness is code that runs the system on a fixed set of test cases and computes metrics automatically. Building it before polishing the prototype lets you measure every change and prevents optimising for demos. Include typical, edge and adversarial cases, with expected outputs or grading rubrics agreed with domain experts.

For generative systems, combine automatic metrics, model-based grading checked against human judgements, and periodic expert review. Track results over versions in a table; regressions should block release just as failing tests do in software engineering.

**🤔 PAUSE & THINK**

Your harness has 50 cases. A change improves 30, worsens 15 and leaves 5 unchanged. Is it an improvement? What else would you need to know?

## 🩺 3. Real cases, not synthetic demos

Prototypes often look excellent on synthetic or hand-picked examples and disappoint on real data. IBM's Watson for Oncology illustrates the risk: internal documents reported by STAT News in 2018 indicated that it had been trained partly on synthetic cases and gave some unsafe or incorrect treatment recommendations, which undermined trust among clinicians.

Use real, representative data, appropriately consented and protected. Include messy inputs: typos, mixed languages, missing fields, unusual formats. Involve real users early, even with rough interfaces, because their behaviour reveals failure modes no developer imagines.

✏️ **WORKED EXAMPLE**

Data realism check: compare the distribution of input length, language mix and missing fields in your test set with a sample of genuine user inputs.

## 👥 4. User studies and iteration

User studies measure whether the prototype helps people. Choose appropriate designs: think-aloud sessions for usability, within-subject comparisons for speed, randomised trials for outcomes. Obtain ethics approval and informed consent where required, and protect participant data.

Iterate in short cycles: build, measure, learn. Record every change and its effect in the harness and user metrics. When evidence supports the hypothesis, document limitations clearly before scaling; when it does not, a well-documented negative result is still a valuable research outcome.

🖐️ **TRY IT**

Plan a two-week prototype sprint: hypothesis, 30 real test cases, harness metrics, five users, a think-aloud protocol and stop or continue criteria.

### ✅ Key takeaways

- Start with a clear hypothesis and pre-agreed decision criteria.
- Build an evaluation harness before polishing the prototype.
- Use real, representative, messy data; synthetic demos mislead.
- Run ethical user studies with appropriate designs.
- Iterate in short cycles; document negative results honestly.

### ❓ Quick check: answer in one or two lines.

**1.** What is an evaluation harness?

**2.** What lesson does Watson for Oncology offer prototype builders?

**3.** Name two user study designs and what each measures.

# 🔬 CONCEPT DEEP DIVE: Intelligent Prototype Lab

🔷 Research prototypes test whether an idea works under realistic conditions before investing in products.

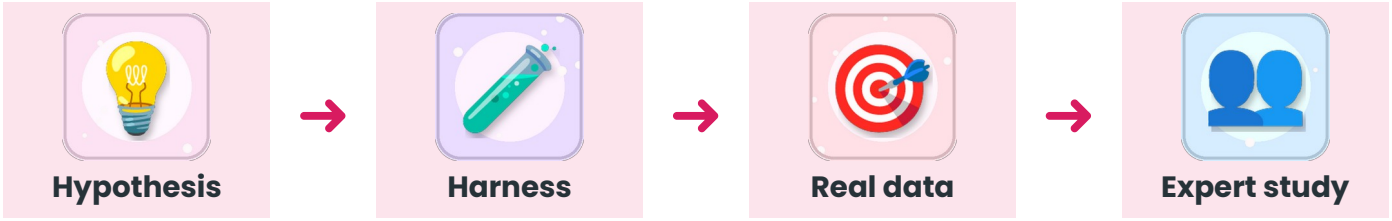
🔷 Build an evaluation harness first: curated real cases, expected outcomes, metrics and slices for rare or hard cases.

🔷 Synthetic or hand-picked data inflates results; validate on representative, consented, real-world data.

🔷 User studies with domain experts reveal workflow fit, trust and failure modes that metrics miss.

🔷 Document limitations and intended use; iterate with experts before claiming readiness.

## 🔄 How it works




## 💻 A tiny evaluation harness

```
cases = [('fever cough', 'flu'), ('chest pain', 'urgent'), ('rash',
'derm')]
results = [(want, model(text)) for text, want in cases]
acc = sum(w == g for w, g in results) / len(results)
fails = [r for r in results if r[0] != r[1]]
```

## ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: The Demo That Charmed Everyone

**Characters:** *Karthik (MTech, health informatics), Dr. Lakshmi (physician), hospital IT team.*

SCENE 1

Karthik built a prototype that suggested diagnoses from patient notes. On ten examples he had written himself, it was impressively accurate. The hospital board loved the demo.

**Pause & predict:** Why are self-written examples weak evidence?

SCENE 2

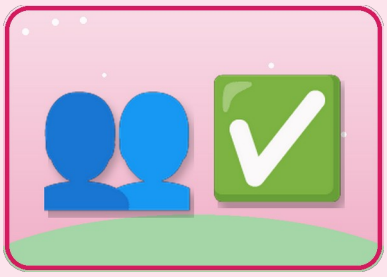
Dr. Lakshmi brought 50 real, anonymised notes with abbreviations, Tamil words and missing vitals. Accuracy fell to 52%, and one suggestion missed a serious condition.

**Pause & predict:** What does real data reveal that demos hide?

SCENE 3

Karthik paused polishing and built an evaluation harness with 200 real cases graded by two doctors. He pre-registered success criteria and added an 'uncertain, refer to doctor' output.

**Pause & predict:** Why build the harness before polishing?



SCENE 4

A small user study showed the tool helped junior doctors draft notes faster but was not reliable for diagnosis. The team re-scoped it as a documentation aid. Dr. Lakshmi called it honest science.

⏸ **Pause & predict:** Why is re-scoping a success, not a failure?

## 🌟 The big idea

*Prototypes must face real cases and real users; honest evaluation beats charming demos.*

## 💬 Discuss with your classmates

**1.** How would you collect realistic test cases ethically?

**2.** What would make you stop a prototype project?

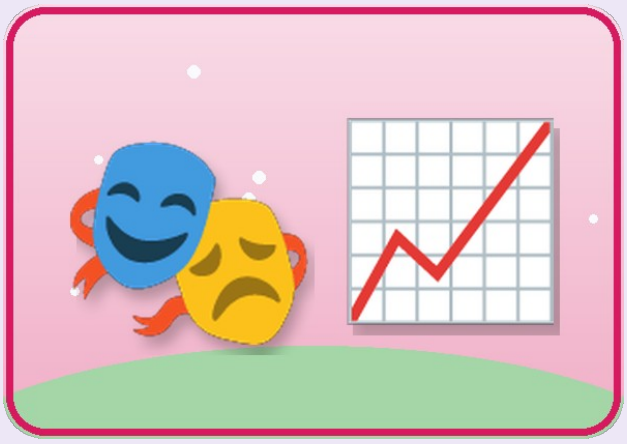
## SCENARIO: Intelligent Prototype Lab

Read the scenario. Think about the choices made at each step.

**1**

**Rohan prototypes a symptom-triage aid.**

**2**

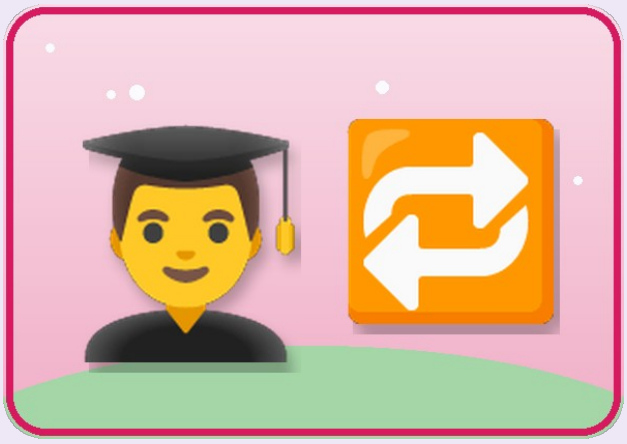


**On synthetic cases it scores 95%.**

**3**

**On 200 real, anonymised cases: 78%.**

**4**



**Doctors help fix gaps; it reaches 88%.**

**Talk about it:**

What decision was made? What would YOU have done differently?

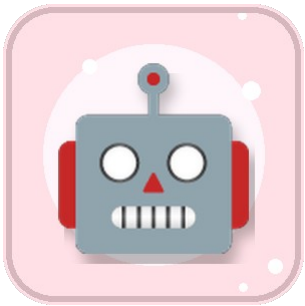
**What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Prototype types

Speech demo

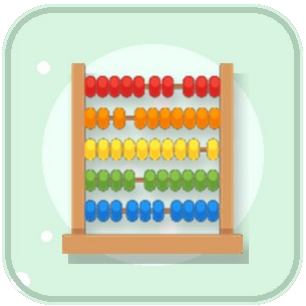
Clinical decision aid

Search ranker

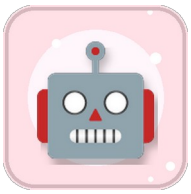


Robot controller

Summariser



Tutor agent



**BOLT SAYS:**

A flashy demo proves little. A harness of real test cases proves a lot!

**FUN FACT!**

Reports in 2018 said IBM Watson for Oncology gave unsafe advice in some internal tests.

**Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: Watson for Oncology

**CASE FILE**

IBM's Watson for Oncology aimed to recommend cancer treatments. In 2018, STAT News reported internal IBM documents indicating that the system had given unsafe and incorrect treatment recommendations in some test cases, and that it had been trained partly on a small number of hypothetical (synthetic) cases rather than large sets of real patient data. Several hospital partnerships were scaled back.

## 🔍 Analyse the case

**1. Why can synthetic training cases mislead?**

**2. What evaluation should precede clinical use?**

**3. What role should oncologists play in development?**

### ⚖️ My verdict: Was AI used well here? Why?

# LAB ACTIVITY: Build an Eval Harness

## You need:

**20 test cases**

**Notebook**

**Teams of 3**

## Steps:

| | | |
|---|---|---|
| 1 | | Write 20 real test cases with answers. |
| 2 | | Include 5 hard or rare cases. |
| 3 | | Run the prototype on all cases. |
| 4 | | Report accuracy overall and on hard cases. |

**THINK & TALK**

Did hard cases change your view?

## Record your observations & results:

# 💻 CODE LAB: Evaluation Harness

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  cases = [('fever', 'flu'), ('chest pain', 'urgent'), ('rash',
'derm'), ('cough', 'flu')]
 2  pred = {'fever': 'flu', 'chest pain': 'urgent', 'rash': 'flu',
'cough': 'flu'}
 3  ok = [want == pred[x] for x, want in cases]
 4  print(sum(ok) / len(____))
```

**COMPLETE** **T1. Fill the blank.**

**PREDICT** **T2. Accuracy?**

**MODIFY** **T3. List failed cases.**

**EXTEND** **T4. Report accuracy by category.**

# ✏️ WORKSHEET: Intelligent Prototype Lab (Part A)

Name: _____ Date: _____

## 1. Harness results. Use the word box.

**WORD BOX:** 85 ✿ 40 ✿ 20

17 of 20 correct → ____%

Hard cases 2 of 5 → ____%

Total cases = ____

## 2. Circle good prototype-validation practices.

Test on real cases

Only synthetic demos

Include rare cases

Hide failures

Expert review

Ship after one demo

# WORKSHEET: Intelligent Prototype Lab (Part B)

Name: _____ Date: _____

## 1. Real vs synthetic. Use the word box.

**WORD BOX: 95 ✿ 78 ✿ 17**

Synthetic score = _____%

Real-case score = _____%

Gap = _____ points

## 2. Order prototype validation. Write 1–4.

**Expert user study** ☐

**Define hypothesis & metrics** ☐

**Run on real cases** ☐

**Build eval harness** ☐

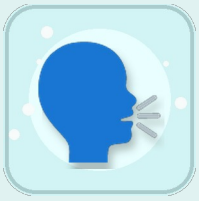
## 3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

# PUZZLE & DEFINE: Intelligent Prototype Lab

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | |  | |

## 2. Define it! Write each term in your own words with an example.

**Harness**

Meaning: _____

Example: _____

**Real cases**

Meaning: _____

Example: _____

**User study**

Meaning: _____

Example: _____

# ? QUIZ TIME: Intelligent Prototype Lab

**Circle the right answer!**

**Q1. An automated evaluation suite:**

a) Harness

b) Horse

c) Canvas

**Q2. Prototypes should be validated on...**

a) Synthetic only

b) Real cases

c) Random words

**Q3. 17 of 20 correct =**

a) 17%

b) 20%

c) 85%

**Q4. Testing with people is a...**

a) User study

b) Package

c) Song

**Q5. Watson for Oncology reports raised...**

a) Awards

b) Safety concerns

c) Prices

**MY SCORE: _____ / 5. Shade one star per correct answer.**

# ✅ TRUE OR FALSE: Intelligent Prototype Lab

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | An eval harness should come early. | | |
| | Synthetic demos prove real-world readiness. | | |
| | Rare cases belong in test sets. | | |
| | Expert user studies reveal workflow issues. | | |
| | Limitations should be hidden from users. | | |
| | Watson for Oncology reports raised safety concerns. | | |

**HOME CONNECT**

Write 10 realistic test cases for an AI tool you use, including 3 hard ones. Try them and record results.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Intelligent Prototype Lab

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why build the eval harness first?**

**COMPARE** **Q2. Synthetic vs real-world validation.**

**APPLY** **Q3. Hard cases 2 of 5 correct. Accuracy?**

**EVALUATE** **Q4. Is a 95% synthetic score evidence of readiness?**

**CREATE** **Q5. List 5 slices for a triage harness.**

# EXAM PREP: Intelligent Prototype Lab

Name: _____ Date: _____

**Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is an evaluation harness?

**2.** Why validate on real data?

**3.** What do expert user studies reveal?

**4.** Why test rare cases?

## PART B • Long answer (5 marks)

**5.** Plan the validation of an AI research prototype for a high-stakes domain.

# UNIT REVIEW: AI CREATE

Name: _____ Date: _____

**Circle the right answer!**

**Q1. Settings saved in a file:**

a) Config | b) Guess | c) Photo

**Q2. Build this before polishing:**

a) Logo | b) Eval harness | c) Advert

**Q3. NeurIPS 2019 introduced a...**

a) Prize | b) Game | c) Checklist

**Q4. Watson for Oncology lacked...**

a) Real cases | b) Lights | c) Colour

**Q5. Tests with real users:**

a) Launch | b) User study | c) Cleanup

**Q6. Tool to organise citations:**

a) Playlist | b) Camera | c) Ref manager

# WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

CONFIG

SEED

REF MANAGER

HARNESS

REAL CASES

USER STUDY

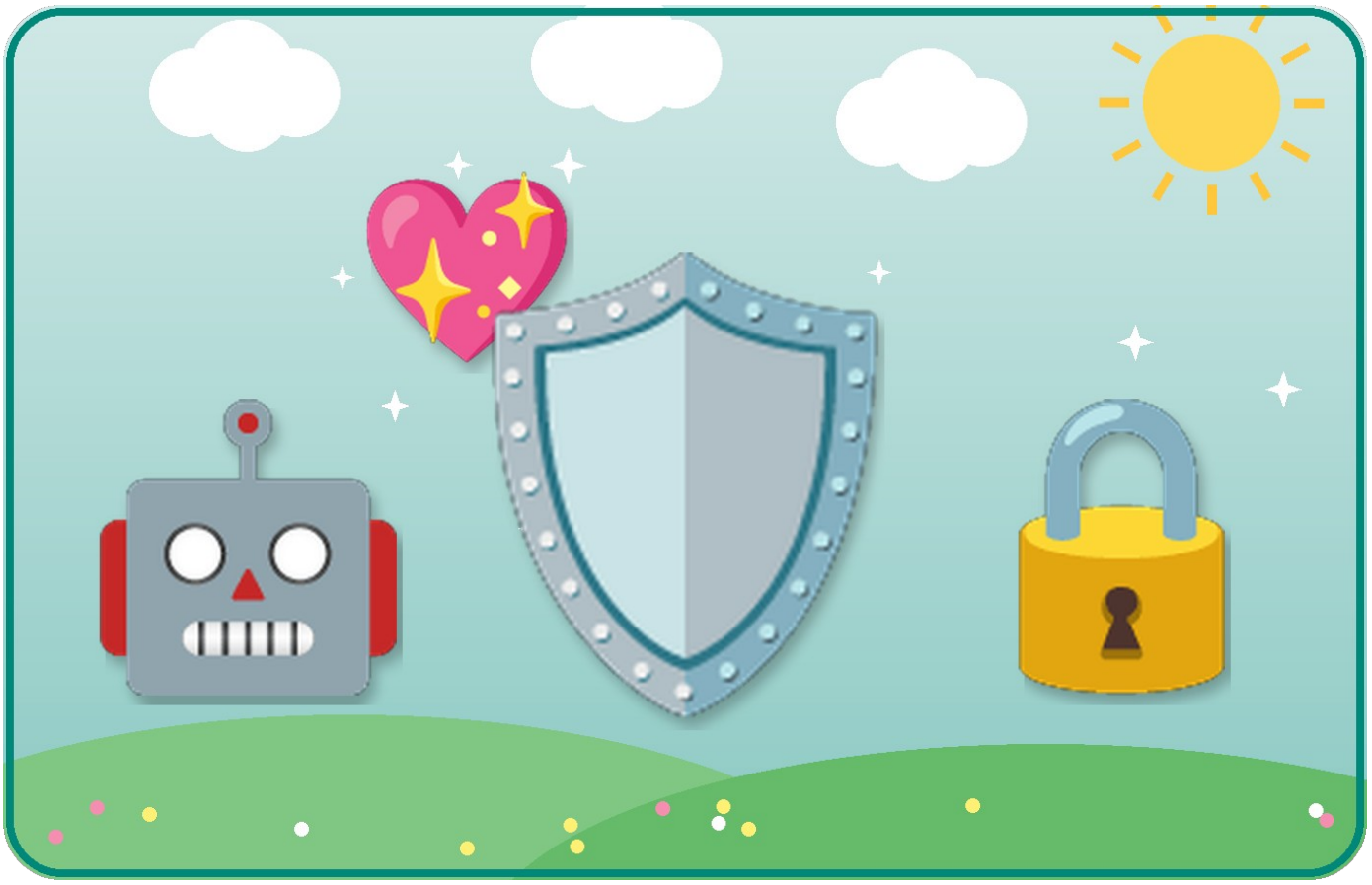
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| T | R | U | C | F | R | S | P | R | H |
| D | E | S | O | L | N | E | R | E | A |
| A | A | E | N | C | O | E | O | F | R |
| T | L | R | F | O | T | D | T | M | N |
| A | C | S | I | M | E | F | O | A | E |
| S | A | T | G | P | B | C | T | N | S |
| E | S | U | R | U | O | I | Y | A | S |
| T | E | D | S | T | O | T | P | G | S |
| M | S | Y | C | E | K | M | E | E | U |
| B | G | I | T | H | U | B | K | R | M |

**I finished AI CREATE! Teacher's sign: _____**

MONTH 6

# AI CARE

Verify • Protect • Respect



**Area: Responsible AI**

## In this unit, you will explore:

Topic 11

### Transparent AI

Topic 12

### Trustworthy AI Systems

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

# Transparent AI

## LET'S LEARN

- Transparency: explain, document and disclose AI.
- Methods: attribution, probing, counterfactuals.
- Explanations must be faithful, not just plausible.

### Transparency toolkit

**Attribution**

**Probing**

**Counterfactual**

**Documentation**

## KEY TERMS

**SHAP** Feature attribution

**Faithful** True to the model

**Disclosure** Telling users

# 📖 CONCEPT EXPLAINED: Transparent AI

**💡 In this chapter**

People affected by AI decisions increasingly expect, and are often legally entitled, to know when AI is used and why it reached a decision. Transparency covers explanation methods, documentation and disclosure. For postgraduates, the challenge is to provide transparency that is faithful to how the model actually works, useful to its audience and honest about its limits.

## 🔍 1. Levels of transparency

Transparency operates at several levels. System transparency discloses that AI is used and for what purpose. Process transparency documents data, training, evaluation and governance. Decision transparency explains individual outputs to the people affected. Each serves different audiences: users, regulators, auditors, developers.

Documentation tools include model cards, which report intended use, performance across groups and limitations, and datasheets for datasets, which describe how data was collected, labelled and should be used. Publishing these builds accountability and helps others decide whether a model fits their context.

**✏️ WORKED EXAMPLE**

Model card fields: model details, intended use and out-of-scope uses, training data, evaluation data, metrics by subgroup, ethical considerations, caveats and recommendations.

## 📊 2. Explanation methods

Feature attribution methods estimate how much each input contributed to an output. SHAP uses ideas from cooperative game theory; LIME fits a simple local model around one prediction; integrated gradients accumulate gradients along a path from a baseline input. Counterfactual explanations state what minimal change would alter the outcome: 'approval if income were ₹5,000 higher'.

For deep networks, probing tests whether concepts are linearly decodable from internal representations, and mechanistic interpretability tries to identify circuits that implement specific behaviours. Inherently interpretable models, such as sparse linear models or small decision trees, may be preferable in high-stakes settings when they perform comparably.

🤔 **PAUSE & THINK**

A counterfactual explanation tells an applicant to 'reduce age by 10 years'. Why is this explanation unhelpful, and how should counterfactual methods be constrained?

## 🎯 3. Faithful versus plausible

An explanation can sound convincing yet not reflect the model's real reasoning. Saliency maps sometimes highlight similar regions even for randomised models, and language models' stated reasons can differ from the factors that actually drove their answers. Plausibility is about human acceptance; faithfulness is about truth to the model.

Test faithfulness: remove features ranked important and check whether predictions change more than when removing random features; compare explanations across models with known behaviour; and check stability under small input changes. Report explanation limitations alongside the explanations themselves.

✏️ **WORKED EXAMPLE**

Deletion test: removing the top 5 SHAP features drops accuracy from 0.86 to 0.61, while removing 5 random features drops it only to 0.84; this supports the attributions' faithfulness.

## 📢 4. Disclosure and accountability

Laws increasingly require transparency. The EU AI Act includes obligations to inform people when they interact with AI systems and to label certain synthetic content, plus documentation duties for high-risk systems. Credit and employment decisions often require reasons that applicants can understand and contest.

In 2019, public complaints alleged that the Apple Card offered lower credit limits to some women than to their husbands. The New York State Department of Financial Services investigated; its 2021 report found no unlawful discrimination but criticised poor transparency and customer service. The case shows that inability to explain decisions damages trust even without proven bias.

🖐️ **TRY IT**

Write a model card and a one-paragraph plain-language explanation for a model you built. Ask a non-specialist to read it and tell you what they understood.

### ✅ Key takeaways

- Transparency: system, process and decision levels for different audiences.
- Model cards and datasheets document models and datasets.
- Methods: SHAP, LIME, integrated gradients, counterfactuals, probing.
- Explanations must be faithful, not merely plausible; test them.
- Disclosure duties are growing; unexplained decisions erode trust.

### ❓ Quick check: answer in one or two lines.

**1.** What is the difference between faithful and plausible explanations?

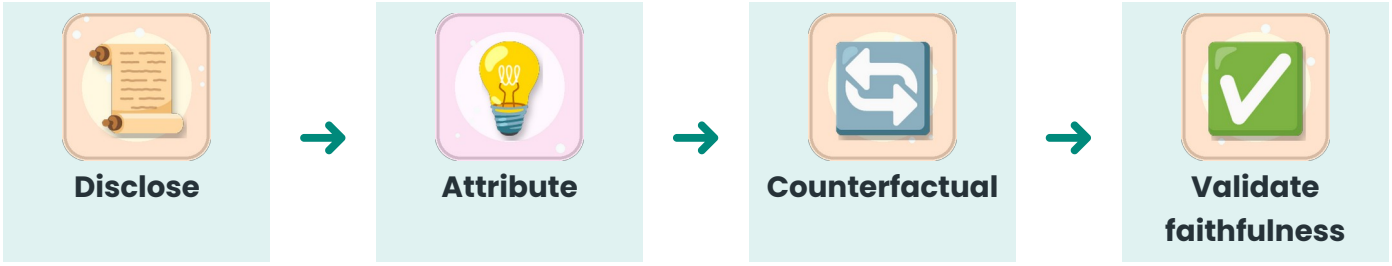
**2.** Name two feature attribution methods.

**3.** What lesson about transparency does the Apple Card case offer?

## CONCEPT DEEP DIVE: Transparent AI

- Transparency spans system-level disclosure (that AI is used, for what), documentation (model cards, datasheets) and decision-level explanations.
- Post-hoc attribution (SHAP, LIME, gradients, occlusion) estimates feature influence; it can be unstable or misleading.
- Counterfactual explanations state what minimal change would alter the outcome, useful for recourse.
- Mechanistic interpretability and probing study internal representations to understand how models compute.
- Faithfulness vs plausibility: explanations must reflect the model's real reasoning, not just sound convincing.

### How it works



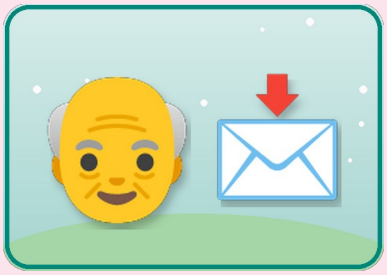

### Occlusion attribution

```
f = lambda x: 3 * x[0] + 1 * x[1] + 0 * x[2]
x = [2, 4, 5]; base = f(x)
drops = [base - f([0 if j == i else v for j, v in enumerate(x)]) for
i in range(3)]
print(base, drops)   # 10 [6, 4, 0]
```

### In your own words: explain this topic to a Class 10 student.

# AI STORY: Why Was I Rejected?

**Characters:** *Fatima (MBA, fintech), a loan applicant Mr. Gowda, her team's data scientist Neel.*



SCENE 1

Mr. Gowda, a small shopkeeper, was refused a loan by Fatima's fintech app. He called asking why. The support team could only say 'the system decided'.

**Pause & predict:** Why is this answer unacceptable?

SCENE 2

Neel ran SHAP on the decision. The top factor was 'irregular monthly income'. A counterfactual showed approval if he provided six months of UPI sales records.

**Pause & predict:** How do attributions and counterfactuals help?

SCENE 3

Fatima asked whether the explanation was faithful. Neel ran a deletion test; removing top SHAP features did change predictions sharply. He also wrote a model card listing known limitations.

**Pause & predict:** Why test faithfulness?

SCENE 4

The app now discloses AI use, gives plain reasons and offers an appeal. Mr. Gowda shared his UPI records and was approved. Complaints fell, and the regulator praised the transparency.

⏸ **Pause & predict:** How does transparency build trust?

## 🌟 The big idea

*Transparent AI gives faithful reasons, honest documentation and a path to appeal.*

## 💬 Discuss with your classmates

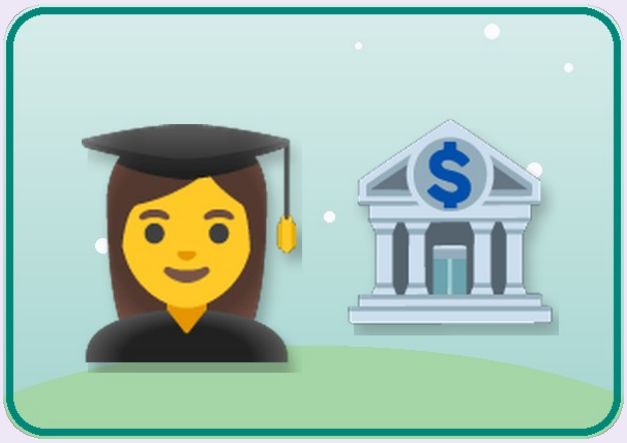
**1.** What explanation would you want from an AI decision about you?

**2.** When could an explanation mislead?

## SCENARIO: Transparent AI

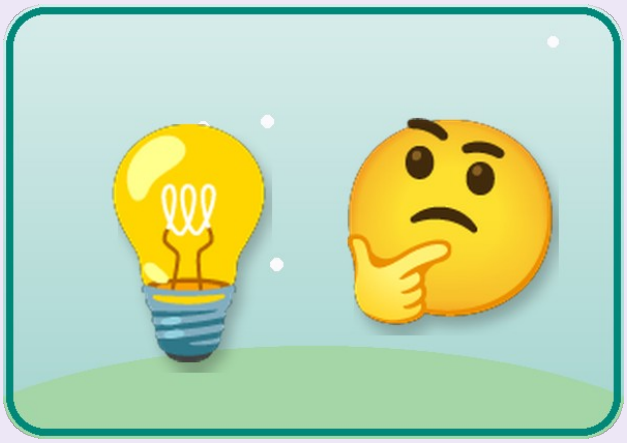
Read the scenario. Think about the choices made at each step.

1



**Ananya audits a loan model's explanations.**

2



**SHAP shows pin code drives many rejections.**

3

**Counterfactuals and a model card reveal a proxy.**

4

**The bank removes it and explains decisions clearly.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Where transparency matters

**Credit decisions**

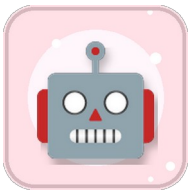
**Diagnosis aids**

**Hiring tools**

**Public services**

**News ranking**

**Content moderation**



**BOLT SAYS:**

A pretty explanation that isn't true is worse than none. Make it faithful!

**FUN FACT!**

In 2019 New York regulators examined Apple Card credit-limit gaps between spouses.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

# CASE STUDY: The Apple Card Inquiry

**CASE FILE**

In November 2019, a technology entrepreneur publicly complained that Apple Card, issued by Goldman Sachs, gave him a much higher credit limit than his wife despite shared finances, and others reported similar experiences. New York's Department of Financial Services investigated. In 2021 it found no unlawful discrimination, but noted problems with transparency and customer service, saying applicants deserved clearer explanations of credit decisions.

## Analyse the case

**1. Why did lack of explanation fuel public distrust?**

**2. What explanations should applicants receive?**

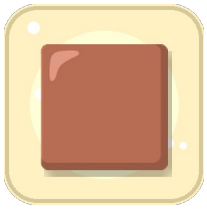
**3. How can transparency coexist with protecting proprietary models?**

**My verdict: Was AI used well here? Why?**

# LAB ACTIVITY: Occlusion Explainer

## You need:

Printed image



Paper masks

Pairs

## Steps:

1. Partner guesses the object in an image.
2. Cover one region at a time.
3. Note when the guess changes.
4. Draw a 'heat map' of key regions.

**THINK & TALK**

Which region mattered most?

**Record your observations & results:**

# 💻 CODE LAB: Occlusion Attribution

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  f = lambda x: 3 * x[0] + 1 * x[1] + 0 * x[2]
 2  x = [2, 4, 5]; base = f(x)
 3  drops = [base - f([0 if j == i else v for j, v in
enumerate(____)]) for i in range(3)]
 4  print(base, drops)
```

**COMPLETE** **T1. Fill the blank.**

**PREDICT** **T2. Base and drops?**

**EXPLAIN** **T3. Why is feature 3 zero?**

**EXTEND** **T4. Try x = [1, 9, 9].**

# WORKSHEET: Transparent AI (Part A)

Name: _____ Date: _____

## 1. Match the method to what it shows.

SHAP ●

● What change flips the decision

Counterfactual ●

● Feature contributions

Probing ●

● What a layer encodes

Occlusion ●

● Which region matters

## 2. Occlusion on f = 3a + 1b + 0c with a=2, b=4, c=5. Use the word box.

**WORD BOX:** 10 ✿ 6 ✿ 0

Prediction f = ____

Drop when a is removed = ____

Drop when c is removed = ____

# WORKSHEET: Transparent AI (Part B)

Name: _____ Date: _____

## 1. Faithful or merely plausible? Tick ✓.

| | Faithful | Plausible |
|---|---|---|
| Verified by removing the feature | | |
| Sounds right, never tested | | |
| Counterfactual actually flips output | | |
| Generic 'many factors' text | | |

## 2. Match the document to its content.

**Model card** ●

**Datasheet** ●

**System card** ●

**AI disclosure** ●

● **How data was collected**

● **Notice that AI is used**

● **Model use, metrics, limits**

● **Whole system risks & safeguards**

# PUZZLE & DEFINE: Transparent AI

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**SHAP**

Meaning: _____

Example: _____

**Faithful**

Meaning: _____

Example: _____

**Disclosure**

Meaning: _____

Example: _____

# ? QUIZ TIME: Transparent AI

**Circle the right answer!**

## Q1. Feature attribution method:

a) SHAP

b) CSS

c) SMTP

## Q2. True to how the model works:

a) Plausible

b) Faithful

c) Pretty

## Q3. Shows what change flips a decision:

a) Probe

b) Card

c) Counterfactual

## Q4. Telling users AI is involved:

a) Disclosure

b) Hiding

c) Guessing

## Q5. The Apple Card inquiry examined...

a) Music

b) Credit limits

c) Phone sales

**MY SCORE: _____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: Transparent AI

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | SHAP estimates feature contributions. | | |
| | Plausible explanations are always faithful. | | |
| | Counterfactuals suggest what would change a decision. | | |
| | Model cards document limits. | | |
| | Users need not know AI is used. | | |
| | New York regulators examined Apple Card. | | |

**HOME CONNECT**

Find an app decision that affected you. What explanation was given? Rewrite it to be clearer.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Transparent AI

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why can a plausible explanation be harmful?**

**COMPARE** **Q2. Attribution vs counterfactual explanations.**

**APPLY** **Q3. f = 3a + b, a = 2, b = 4. Contribution of a by occlusion?**

**EVALUATE** **Q4. Is publishing full model weights required for transparency?**

**CREATE** **Q5. Write a 2-line decision explanation for a loan applicant.**

# 📝 EXAM PREP: Transparent AI

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** What is a faithful explanation?

**2.** Compare SHAP and counterfactuals.

**3.** What is a model card?

**4.** Why disclose AI use?

## PART B • Long answer (5 marks)

**5.** Evaluate transparency methods for a credit-decision model, including their limitations.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

# Trustworthy AI Systems

## LET'S LEARN

- Trust needs robustness, calibration and privacy.
- Test shifts, adversarial inputs and rare cases.
- Assure with evidence, monitoring and governance.

### Pillars of trustworthy AI

**Robustness**

**Calibration**

**Privacy**

**Fairness**

## KEY TERMS

**Calibration**
Confidence = accuracy

**Drift** Data changes over time

**DP** Differential privacy

# CONCEPT EXPLAINED: Trustworthy AI Systems

### In this chapter

Trustworthy AI is AI that deserves trust, not merely AI that people happen to trust. It must be robust to changing and adversarial conditions, honest about its uncertainty, protective of privacy, fair, secure and governed by accountable processes. Postgraduates designing high-impact systems need both the technical methods and the assurance practices that turn claims of trustworthiness into evidence.

## 1. Robustness

Robustness means performance holds under realistic variation: different devices, lighting, accents, languages, populations and time periods. Distribution shift, where deployment data differs from training data, is a leading cause of failure. Stress tests on shifted datasets, subgroup analysis and monitoring for drift reveal fragility before it harms users.

Adversarial robustness concerns deliberate manipulation. Eykholt and colleagues showed in 2018 that small black-and-white stickers on a stop sign caused a road-sign classifier to misread it as a speed-limit sign in physical tests. Defences include adversarial training, input checks, ensembles and human fallback, but no method is complete.

**WORKED EXAMPLE**

Robustness report: accuracy on clean test data 0.94; on blurred images 0.88; on night images 0.79; on a new city's signs 0.83; under a known patch attack 0.41.

## 2. Calibration and uncertainty

A calibrated model's confidence matches its accuracy: of predictions made with 80% confidence, about 80% should be correct. Modern deep networks are often overconfident. Calibration can be measured with reliability diagrams and expected calibration error, and improved with temperature scaling or ensembles.

Uncertainty enables safe behaviour. Systems can abstain or defer to humans when uncertain, and conformal prediction offers prediction sets with statistical coverage guarantees under assumptions. Communicating uncertainty clearly to users, without false precision, is as important as computing it.

🤔 **PAUSE & THINK**

A model says it is 90% confident but is right only 65% of the time at that level. What harm could follow, and how would you fix it?

## 🔐 3. Privacy and security

Models can leak training data through memorisation, membership inference or model inversion attacks. Differential privacy adds carefully calibrated noise during training or analysis so that the output barely changes whether any one person's data is included, with a privacy budget, epsilon, quantifying protection. Federated learning keeps data on devices and shares only model updates.

Security threats include data poisoning, model theft, prompt injection for LLM applications and supply-chain risks in pre-trained models and packages. Threat modelling, access control, red-teaming and secure development practices belong in every AI project, alongside compliance with data protection law.

✏️ **WORKED EXAMPLE**

Privacy trade-off: training with differential privacy at a strict epsilon may reduce accuracy by a few points; teams must decide what level of privacy the use case requires and document the choice.

## 📋 4. Assurance and governance

Assurance turns trustworthiness into evidence. An assurance case states claims ('the system is safe for triage use'), arguments and supporting evidence such as test results, audits and monitoring data. Frameworks such as the NIST AI Risk Management Framework (Govern, Map, Measure, Manage) and ISO/IEC 42001 help organisations structure this work.

Governance assigns responsibility: who approves deployment, who monitors, who handles incidents, and who can shut the system down. Independent audits, incident reporting, and channels for affected people to appeal decisions complete the picture. Trust is maintained through continuous evidence, not a one-time certificate.

🖐 **TRY IT**

Write a one-page assurance case for an AI system you know: top claim, three sub-claims (robustness, calibration, privacy), the evidence for each and how it will be monitored.

## ✅ Key takeaways

- Robustness covers natural shift and adversarial manipulation.
- Calibrated confidence enables abstention and safe deferral.
- Differential privacy and federated learning protect personal data.
- Threat-model for poisoning, theft and prompt injection.
- Assurance cases and governance turn trust claims into evidence.

## ❓ Quick check: answer in one or two lines.

**1.** Define calibration and explain why it matters.

**2.** What does differential privacy guarantee, informally?

**3.** List the parts of an assurance case.

# 🔬 CONCEPT DEEP DIVE: Trustworthy AI Systems

🔷 Trustworthy AI combines validity, reliability, robustness, safety, security, privacy, fairness, transparency and accountability.

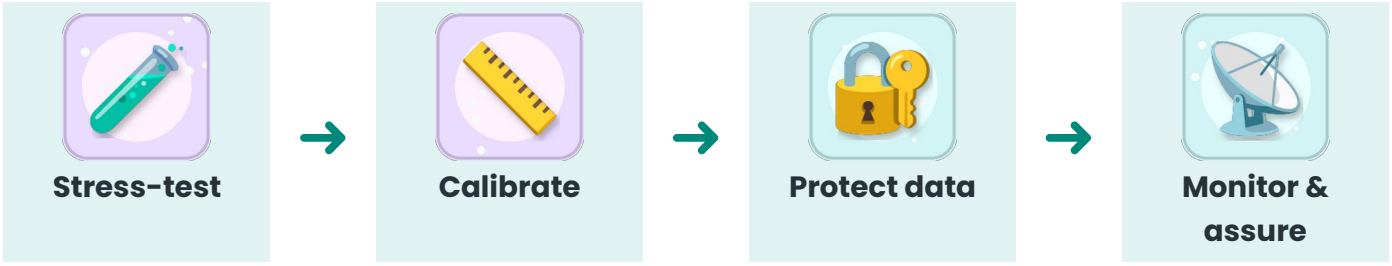
🔷 Robustness: test distribution shift, corruptions, rare cases and adversarial examples; use augmentation and adversarial training.

🔷 Calibration and uncertainty: confidence should match accuracy; models should abstain or defer when uncertain.

🔷 Privacy: data minimisation, differential privacy, federated learning and protection against membership inference.

🔷 Assurance: evidence-based safety cases, independent evaluation, monitoring for drift and incident response.

## 🔄 How it works



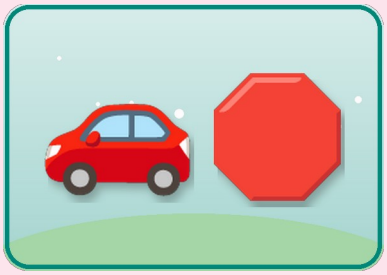

## 💻 A simple calibration check

```
conf = [0.9, 0.9, 0.9, 0.9]
correct = [1, 1, 1, 0]
acc = sum(correct) / len(correct)
print(acc, round(sum(conf) / len(conf) - acc, 2))   # 0.75 0.15 → overconfident
```

## ✍️ In your own words: explain this topic to a Class 10 student.

# AI STORY: The Sign That Lied

**Characters:** *Aditi (MTech, autonomous systems), Prof. Kulkarni, test driver Mr. Singh.*



SCENE 1

Aditi's lab tested a sign-reading system for driver assistance. It scored 97% on the test set. Prof. Kulkarni asked how it behaved at dusk, in rain and with dirty or defaced signs.

**Pause & predict:** Why is one accuracy number not enough?

SCENE 2

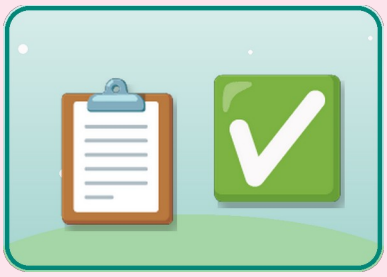
Field tests showed 81% at dusk and 74% in heavy rain. Worse, the model was 95% confident on many wrong readings. Aditi recalled the stop-sign sticker research.

**Pause & predict:** What problem does overconfidence create?

SCENE 3

She applied temperature scaling to calibrate confidence, added augmentation for weather and defaced signs, and made the system defer to the driver when uncertain. Mr. Singh tested it on new routes.

**Pause & predict:** How do calibration and deferral improve safety?



SCENE 4

Aditi wrote an assurance case with robustness tests, calibration plots and a monitoring plan. The review board approved a limited pilot with clear stop conditions.

⏸ **Pause & predict:** Why is an assurance case better than a claim of 'safe'?

## 🌟 The big idea

*Trustworthy AI is earned through robustness, calibration and evidence, not assumed from accuracy.*

## 💬 Discuss with your classmates

**1.** What conditions could break a model in your field?

**2.** Who should be able to stop an AI system?

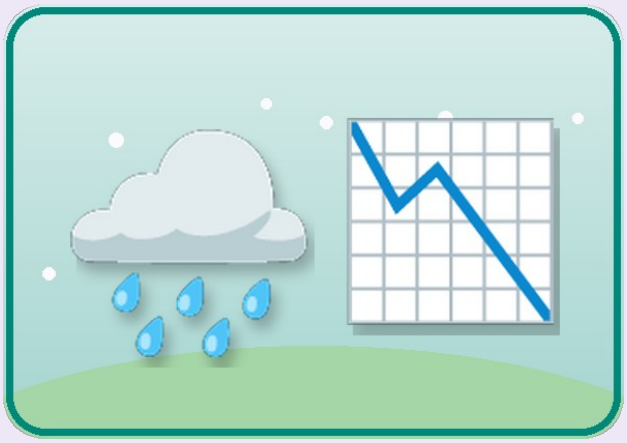
## SCENARIO: Trustworthy AI Systems

Read the scenario. Think about the choices made at each step.

**1**

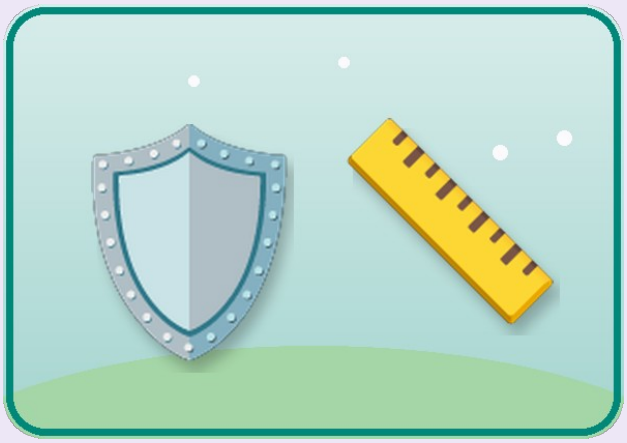
**Rohan tests a campus delivery robot's vision.**

**2**



**Accuracy drops from 96% to 71% in rain.**

**3**



**He adds rainy data, calibration and a safe stop.**

**4**

**Robot now pauses when unsure; no collisions.**

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

## LOOK AROUND YOU: Trust engineering

**Perception testing**

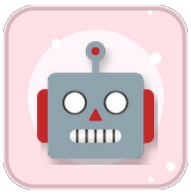
**Calibrated risk scores**

**Differential privacy**

**Fairness audits**

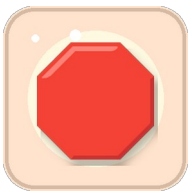
**Stress tests**

**Drift monitoring**



**BOLT SAYS:**

Trust isn't claimed, it's earned with tests, evidence and honesty!



**FUN FACT!**

In 2018 researchers fooled a stop-sign classifier with a few stickers on the sign.

**Reflect: where have YOU seen this in real life? Note it down.**

# 📁 CASE STUDY: Stickers That Fooled a Stop-Sign Classifier

**CASE FILE**

In 2018, researchers led by Kevin Eykholt showed that placing a few black and white stickers on a real stop sign could cause a deep-learning image classifier to misread it, for example as a speed-limit sign, across many viewing angles and distances in their tests. The work demonstrated that physical-world adversarial examples are possible and motivated research into robust perception for safety-critical systems.

## 🔍 Analyse the case

**1. Why are adversarial examples a safety concern?**

**2. Which defences could reduce the risk?**

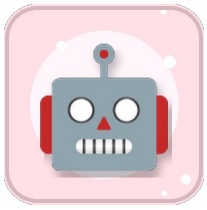
**3. How should autonomous systems behave when unsure?**

### ⚖️ My verdict: Was AI used well here? Why?

# LAB ACTIVITY: Stress-Test Lab

## You need:

Image set



Vision demo

Teams of 3

## Steps:

1. Test a vision demo on clean images.
2. Add blur, dark light and stickers.
3. Record accuracy for each condition.
4. Propose 2 robustness fixes.

**THINK & TALK**

Which condition hurt most?

**Record your observations & results:**

# 💻 CODE LAB: Calibration Check

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
conf = [0.9, 0.9, 0.9, 0.9, 0.6, 0.6]
correct = [1, 1, 1, 0, 1, 0]
for c in (0.9, 0.6):
    idx = [i for i, v in enumerate(conf) if v == c]
    acc = sum(correct[i] for i in idx) / len(____)
    print(c, acc)
```

**COMPLETE** **T1. Fill the blank.**

**PREDICT** **T2. Accuracy per confidence bin?**

**INTERPRET** **T3. Over- or under-confident?**

**EXTEND** **T4. Compute average gap (simple ECE).**

# WORKSHEET: Trustworthy AI Systems (Part A)

Name: _____ Date: _____

**1. Match the threat to a safeguard.**

Adversarial input ●

Data drift ●

Privacy leak ●

Overconfidence ●

● Calibration & abstention

● Adversarial training & tests

● Monitoring & retraining

● Differential privacy

**2. Calibration check: 4 predictions at 0.9 confidence, 3 correct. Use the word box.**

**WORD BOX: 0.75 ✿ 0.15 ✿ over**

Accuracy = ____

Gap (0.9 - accuracy) = ____

The model is ____-confident

# WORKSHEET: Trustworthy AI Systems (Part B)

Name: _____ Date: _____

## 1. Robustness results. Use the word box.

**WORD BOX: 96 ✿ 71 ✿ 25**

Clean accuracy = ____%

Rain accuracy = ____%

Drop = ____ points

## 2. Robustness or privacy safeguard? Tick ✓.

| | Robustness | Privacy |
|---|---|---|
| Train with corrupted images | | |
| Add DP noise to training | | |
| Adversarial training | | |
| Federated learning on devices | | |

# PUZZLE & DEFINE: Trustworthy AI Systems

Name: _____ Date: _____

## 1. Odd one out! Circle it and justify your choice.

| | | | | |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |

## 2. Define it! Write each term in your own words with an example.

**Calibration**

Meaning: _____

Example: _____

**Drift**

Meaning: _____

Example: _____

**DP**

Meaning: _____

Example: _____

# ? QUIZ TIME: Trustworthy AI Systems

**Circle the right answer!**

**Q1. Confidence matching accuracy is...**

a) Calibration

b) Colouring

c) Volume

**Q2. Data changing over time:**

a) Freeze

b) Drift

c) Float

**Q3. Adds noise to protect individuals:**

a) Music

b) Dice

c) Diff. privacy

**Q4. Stickers fooled a ... classifier.**

a) Stop-sign

b) Dog

c) Song

**Q5. Trust is best earned with...**

a) Claims

b) Evidence

c) Gifts

**MY SCORE: ____ / 5. Shade one star per correct answer.**

☆ ☆ ☆ ☆ ☆

# ✅ TRUE OR FALSE: Trustworthy AI Systems

**Look at the picture. Tick ✓ TRUE or FALSE.**

| | | 👍 TRUE | 👎 FALSE |
|---|---|---|---|
| | Calibrated models' confidence matches accuracy. | | |
| | Adversarial examples only exist in theory. | | |
| | Distribution shift can lower accuracy. | | |
| | Differential privacy protects individuals in data. | | |
| | High test accuracy guarantees safety. | | |
| | Stickers fooled a stop-sign classifier in 2018. | | |

**HOME CONNECT**

Test a phone's voice assistant in a noisy place and a quiet one. Note the difference in accuracy.

## 🔢 Self-assessment: tick one.

**I know it well!**

**I know some.**

**I need practice.**

# CHALLENGE PROBLEMS: Trustworthy AI Systems

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

**EXPLAIN** **Q1. Why must AI abstain sometimes?**

**COMPARE** **Q2. Distribution shift vs adversarial attack.**

**APPLY** **Q3. Confidence 0.9, accuracy 0.75. Gap?**

**EVALUATE** **Q4. Is high test accuracy enough for a medical device?**

**CREATE** **Q5. Write 5 stress tests for a speech assistant.**

# 📝 EXAM PREP: Trustworthy AI Systems

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

## PART A • Short answers (2 marks each)

**1.** Define calibration.

**2.** Distribution shift vs adversarial attack?

**3.** What is differential privacy?

**4.** What is an assurance case?

## PART B • Long answer (5 marks)

**5.** Describe how to engineer and demonstrate trustworthiness in a safety-critical AI system.

# UNIT REVIEW: AI CARE

Name: _____ Date: _____

**Circle the right answer!**

**Q1. SHAP gives feature...**

a) Attributions

b) Colours

c) Sounds

**Q2. Explanation true to the model:**

a) Plausible

b) Faithful

c) Pretty

**Q3. Confidence matching accuracy:**

a) Drift

b) Privacy

c) Calibration

**Q4. Stop-sign stickers were a...**

a) Attack

b) Party

c) Fix

**Q5. Adds noise to protect people:**

a) SHAP

b) DP

c) Seed

**Q6. Apple Card was probed by...**

a) Gamers

b) Film fans

c) NY DFS

# WORD SEARCH: AI CARE

🔍 **Find these words. They go across → or down ↓.**

SHAP

FAITHFUL

DISCLOSURE

DRIFT

DP

ROBUST

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| F | M | R | D | P | G | S | D | U | S |
| A | R | O | B | U | S | T | I | A | M |
| I | H | W | O | K | P | D | S | M | G |
| T | A | U | D | C | R | R | C | M | O |
| H | U | S | S | G | I | I | L | O | V |
| F | D | D | W | S | V | F | O | N | E |
| U | I | I | K | H | A | T | S | I | R |
| L | T | G | N | A | C | R | U | T | N |
| H | E | W | G | P | Y | S | R | O | A |
| G | F | A | L | T | W | W | E | R | K |

I finished AI CARE! Teacher's sign: _____

# AI GLOSSARY (1/2)

**2nd reader**
AI double-check

**Ablation**
Remove a part, test

**Augment**
Boost human work

**Calibration**
Confidence = accuracy

**Compliance**
Following rules

**Config**
Settings file

**Disclosure**
Telling users

**DP**
Differential privacy

**Drift**
Data changes over time

**DSS**
Decision support

**DVC**
Data version control

**Embedding**
Learned vector

**Faithful**
True to the model

**Frontier**
Where AI is reliable

**Harness**
Automated eval suite

**Leakage**
Test info in training

**Licensure**
Right to practise

**Multimodal**
Many data types

# AI GLOSSARY (2/2)

**Open source**
Public shared code

**Peer review**
Expert checking

**Poster**
Visual summary

**Real cases**
Actual-world data

**Ref manager**
Citation organiser

**Sampling**
Many candidates

**Seed**
Randomness start

**SHAP**
Feature attribution

**Specialism**
Focused expertise

**SSL**
Self-supervised learning

**T-shaped**
Broad + deep skills

**Tech debt**
Costly shortcuts

**Tech transfer**
Lab to market

**Tracking**
Log runs & results

**Trade-off**
Quality vs cost

**User study**
Test with people

**Verifier**
Checks candidates

# ? OLYMPIAD PRACTICE TEST • Part 1 of 4

**Read each question. Circle the right answer. 20 questions in all!**

## Q1. Embeddings turn items into...

a) Vectors

b) Pictures

c) Songs

## Q2. Cosine of identical vectors =

a) 0

b) 1

c) -1

## Q3. Move 37 was played by...

a) Lee Sedol

b) A fan

c) AlphaGo

## Q4. Decision support systems...

a) Aid decisions

b) Ban people

c) Play games

## Q5. Track experiments with tools like...

a) Spotify

b) MLflow

c) Camera

# ? OLYMPIAD PRACTICE TEST • Part 2 of 4

**Q6.** 10 samples, each 20% right → any right ≈

a) 20%

b) 50%

c) 89%

**Q7.** A verifier's job is to...

a) Check outputs

b) Draw

c) Advertise

**Q8.** Accuracy 0.80, 0.84, 0.82 → mean

a) 0.80

b) 0.82

c) 0.84

**Q9.** Leakage review covered ~...

a) 3 papers

b) 29 papers

c) 294 papers

**Q10.** Doctors and lawyers need...

a) Licensure

b) Gaming

c) Fame

# ? OLYMPIAD PRACTICE TEST • Part 3 of 4

**Q11.** Specialism example:

a) Cooking

b) NLP

c) Painting

**Q12.** Research shown on a board:

a) Email

b) Film

c) Poster

**Q13.** Same seed + config gives...

a) Same result

b) Chaos

c) Errors

**Q14.** Watson concerns were reported by...

a) Gamers

b) STAT News

c) Radio

**Q15.** Prototype success is judged by...

a) Looks

b) Hype

c) Evaluation

# ? OLYMPIAD PRACTICE TEST • Part 4 of 4

**Q16.** Telling users AI is used is...

a) Disclosure

b) Hiding

c) Guessing

**Q17.** Says 90% sure, right 60% → model is

a) Calibrated

b) Overconfident

c) Shy

**Q18.** Data slowly changing over time:

a) DP

b) SHAP

c) Drift

**Q19.** Explain 'what if input changed':

a) Counterfactual

b) Random

c) Batch

**Q20.** Structured safety evidence:

a) Advert

b) Assurance case

c) Game

**MY SCORE: I got ____ / 20. Colour a star for every 4 right!**

# CERTIFICATE

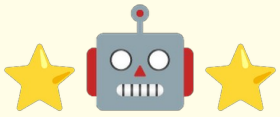
## of AI Explorer

This is to say that

_____

of PG 1st Year finished the 6-month AI journey:

**DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE**



Teacher: _____ Date: _____

# ? ANSWER KEY (for teachers & parents)

## 1. Modern Intelligence Explained

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** Supervised - Human labels; Self-supervised - Data-made labels; Reinforcement - Rewards; Unsupervised - Structure
**Part A 2:** Plays only Go: Narrow; Writes, codes and translates: General; Reads one scan type: Narrow; Captions, answers, draws: General
**Part B 1:** 15, 85, 40
**Part B 2:** Fine-tuning - Update weights; LoRA - Small matrices; Prompting - Instructions; RAG - Retrieve documents
**Challenge hints:** Q1: Labels come from data; learns general features first. • Q2: Fine-tune: updates weights, task-specific; prompt: no training. • Q3: $2 \div (1.41 \times 1.73) \approx 0.82$. • Q4: No: check shifts, robustness, real tasks. • Q5: Mask audio segments; predict them from context.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: norm; 0.82 and 0.0; vectors are orthogonal (unrelated); compute all pairs with itertools.combinations.
**Exam prep key points:** 1: Learning from labels derived from the data itself. • 2: Broadly pre-trained model adapted to many tasks. • 3: Strong on some tasks, brittle on others. • 4: Fine-tuning, LoRA, prompting, RAG. • 5: Representation learning, SSL, multimodal, RL, foundation models; jaggedness, shift, evaluation.

## 2. AI Beyond Automation

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** Auto-file invoices: Automate; Flag scans for a doctor: Augment; Auto-sort spam: Automate; Suggest cases to a lawyer: Augment
**Part A 2:** 20000, 11000, 45
**Part B 1:** Accepting every AI flag unchecked: Bias; Ignoring accurate AI advice: Aversion; Skipping review when AI agrees: Bias; Distrusting AI after one error: Aversion
**Part B 2:** 8, 4, 96
**Challenge hints:** Q1: Over-relying on automated advice, missing its errors. • Q2: Automation does the task; augmentation improves the human. • Q3: 8,800. • Q4: No: calibrated confidence supports trust. • Q5: AI finds cases + reasons; lawyer verifies, decides.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: 0.1; reads 11000, saving 0.45; at 20%: 12000, 0.40; (20000 - 11000) ÷ 60 = 150 hours.
**Exam prep key points:** 1: Doing the task vs improving the human. • 2: Over-reliance on automated advice. • 3: Calibrates human trust. • 4: Team outcomes: accuracy, time, safety. • 5: Roles, uncertainty display, trust calibration, RCT-style evaluation, accountability.

### 3. Develop with Intelligence

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** Git - Code; DVC - Data & models; MLflow - Experiments; Docker - Environment
**Part A 2:** Function returns right type: Code; No nulls in key columns: Data/model; Unit test for a parser: Code; Accuracy ≥ baseline on test set: Data/model
**Part B 1:** 1-Run unit & data tests, 2-Train model, 3-Check performance gates, 4-Deploy if gates pass
**Part B 2:** Circle: Versioned datasets, Container image, Logged experiments
**Challenge hints:** Q1: Same code + different data = different model. • Q2: Unit: code logic; gates: metrics on held-out data. • Q3: lr 0.01. • Q4: No: containerise and test in CI. • Q5: Code commit, data version, env, seed, config.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: 1; (0.01, 0.86); test must stay untouched for the final estimate; use csv.writer.
**Exam prep key points:** 1: Costly maintenance from data deps, glue code, config. • 2: Code commit, data version, environment, seed. • 3: Params, metrics, artefacts, code version. • 4: Differences between training and production data/processing. • 5: Versioning, tracking, testing, CI/CD, containers, monitoring.

### 4. Design Smart Workflows

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** Generator - Proposes; Retriever - Fetches evidence; Verifier - Checks; Router - Picks path
**Part A 2:** 0.05, 0.40, 0.99
**Part B 1:** 42, 3, 60
**Part B 2:** Unit tests for code: Strong; 'Looks right' LLM judgement: Weak; Symbolic math check: Strong; Random spot-check of 1 in 100: Weak
**Challenge hints:** Q1: Selection is only as good as the checker. • Q2: Voting: majority agreement; verifier: explicit checks. • Q3: $1 - 0.95^{10} \approx 0.40$. • Q4: No: cost/latency; adapt to difficulty. • Q5: Generate → check syllabus → solve → verify → human review.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: n; 0.05, 0.4, 0.99; you must identify the correct candidate; n ≈ 45 ($1 - 0.95^n \geq 0.9$).
**Exam prep key points:** 1: Models + retrieval + tools + verifiers working together. • 2: Majority vote over sampled answers. • 3: Spend more compute only on hard inputs. • 4: They may be biased or unreliable. • 5: Generation, verification, selection, routing, metrics, cost.

| | |
|---|---|
| **5. AI for Advanced Research** | **Quiz:** 1-a 2-b 3-c 4-a 5-b<br>**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T<br>**Odd one out:** A: picture 4; B: picture 3; C: picture 4<br>**Part A 1:** 1-Define question & baseline, 2-Run main experiment, 3-Run ablations, 4-Report with error bars<br>**Part A 2:** Circle: Results over 5 seeds, Strong baselines, Ablation table<br>**Part B 1:** 0.82, 0.04, 0.014<br>**Part B 2:** Duplicates - Same image; Future info - Discharge date; Preprocessing - Scaler on all data; Test tuning - Choosing by test<br>**Challenge hints:** Q1: Shows variance; separates real gains from luck. • Q2: Ablation: which parts help; error analysis: where it fails. • Q3: 0.82 ± 0.014. • Q4: Weak: needs seeds, significance, ablations. • Q5: Split before preprocessing, dedupe, time order, feature timing, test isolation.<br>**Case study:** Open answers. Reward clear reasoning and examples.<br>**Code lab:** Blank: std; 0.82 0.014 0.822; difference 0.002 is well within noise; scipy.stats.ttest_rel(a, b).<br>**Exam prep key points:** 1: Removing components to measure their contribution. • 2: Duplicates, future features, preprocessing on all data. • 3: Show variance and reliability. • 4: Test data appearing in training. • 5: Question, baselines, splits, seeds, statistics, ablations, leakage checks, reporting. |
| **6. AI for Professional Innovation** | **Quiz:** 1-a 2-b 3-c 4-a 5-b<br>**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T<br>**Odd one out:** A: picture 4; B: picture 3; C: picture 4<br>**Part A 1:** Law - Clause review; Medicine - Scan triage; Finance - Fraud/risk; Engineering - Generative design<br>**Part A 2:** 25, 12, 19<br>**Part B 1:** Drafting a standard NDA: Inside; Novel constitutional argument: Outside; Summarising a long report: Inside; Rare-disease diagnosis alone: Outside<br>**Part B 2:** 1-Map tasks & risks, 2-Check rules & approvals, 3-Redesign workflow & train, 4-Measure vs control group<br>**Challenge hints:** Q1: Plausible but wrong outputs; over-trust outside frontier. • Q2: Centaur: divide tasks; cyborg: tightly interleave. • Q3: 19 percentage points. • Q4: No: local validation, regulation, monitoring. • Q5: Tasks, control group, metrics, compliance, review.<br>**Case study:** Open answers. Reward clear reasoning and examples.<br>**Code lab:** Blank: alone; draft 0.18, review 0.11, novel -0.18; 'novel' is outside; matplotlib bar chart.<br>**Exam prep key points:** 1: Uneven boundary of tasks AI does well. • 2: Confidentiality, liability, licensure, compliance. • 3: Performance varies by setting and population. • 4: Quality, time, errors, client impact. • 5: Task mapping, frontier, workflow, compliance, controlled pilots, outcomes. |

**7. AI Specialist Roadmap**

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** NLP - Tokenisers/transformers; Vision - CNNs/images; RL - Rewards/simulation; MLOps - CI/CD/monitoring
**Part A 2:** 4, 2, 50
**Part B 1:** Linear algebra: Breadth; Tokeniser design: Depth; Software testing: Breadth; Transformer fine-tuning: Depth
**Part B 2:** 60, 30, 10
**Challenge hints:** Q1: Depth creates distinctive value; breadth enables collaboration. • Q2: Models & experiments vs infrastructure & reliability. • Q3: 50%. • Q4: No: interest, aptitude, demand, longevity. • Q5: Courses, project, paper/OSS, internship, review.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: -; ['cv', 'docker'] 50%; with git: ['cv', 'docker', 'git'] 40%; multiply by counts from posts.
**Exam prep key points:** 1: Broad base + deep specialism. • 2: NLP, vision, RL, MLOps, safety, policy. • 3: Projects, papers, open source, internships. • 4: Fields and goals change. • 5: Track choice, breadth/depth, resources, projects, milestones, review.

**8. Research to Profession**

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** 1-Write & submit paper, 2-Peer review, 3-Revise after reviews, 4-Present at conference
**Part A 2:** Paper - Researchers; Patent - Companies/legal; Policy brief - Government; Open code - Developers
**Part B 1:** 4, 3, 2
**Part B 2:** Citing all sources used: Ethical; Gift authorship to a senior: Unethical; Sharing code with paper: Ethical; Dropping inconvenient results: Unethical
**Challenge hints:** Q1: Checks validity, clarity and novelty before publication. • Q2: Academia: freedom, teaching; industry: scale, product impact. • Q3: 4. • Q4: No: authorship requires real contribution. • Q5: Problem, method, impact in plain words.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: True; 4 and 3; ignores field differences, author roles, quality; sum(x >= 10).
**Exam prep key points:** 1: Write, submit, review, revise, present. • 2: Moving research into products or practice. • 3: Honest reporting, proper authorship, citation. • 4: Independence vs scale/product impact. • 5: Papers, open source, patents, start-ups, policy; authorship, integrity.

**9. AI Research Workbench**

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** Seed fixed and logged: Yes; 'Tuned by hand, not recorded': No; Dockerfile in repo: Yes; Data only on a lost laptop: No
**Part A 2:** 40, 8, 320
**Part B 1:** 1-Write config & seed, 2-Launch run, 3-Log params & metrics, 4-Compare in dashboard
**Part B 2:** Circle: Config files per run, Regular backups, Reference manager
**Challenge hints:** Q1: Change settings without code edits; record exact runs. • Q2: Notebook: exploration; script: repeatable, testable. • Q3: ₹320. • Q4: No: back up, version, document. • Q5: date_model_dataset_seed_commit.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: 42; True; False; np.random.seed(42).
**Exam prep key points:** 1: Notebooks, Git/DVC, tracking, containers, compute. • 2: Separate settings; record runs. • 3: Same randomness → same results. • 4: Exploration vs repeatable pipelines. • 5: Data, code, environments, compute, tracking, hygiene, checklists.

**10. Intelligent Prototype Lab**

**Quiz:** 1-a 2-b 3-c 4-a 5-b
**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T
**Odd one out:** A: picture 4; B: picture 3; C: picture 4
**Part A 1:** 85, 40, 20
**Part A 2:** Circle: Test on real cases, Include rare cases, Expert review
**Part B 1:** 95, 78, 17
**Part B 2:** 1-Define hypothesis & metrics, 2-Build eval harness, 3-Run on real cases, 4-Expert user study
**Challenge hints:** Q1: Defines success; prevents demo-driven development. • Q2: Synthetic: easy, unrealistic; real: representative, harder. • Q3: 40%. • Q4: No: validate on real data and with experts. • Q5: Age, rare conditions, languages, severity, comorbidity.
**Case study:** Open answers. Reward clear reasoning and examples.
**Code lab:** Blank: ok (or cases); 0.75; [('rash', 'derm')]; group by want.
**Exam prep key points:** 1: Automated suite of real test cases and metrics. • 2: Synthetic data can inflate results. • 3: Workflow fit, trust, failure modes. • 4: Failures often hide there. • 5: Hypothesis, harness, slices, real data, experts, limits, ethics.

| | |
|---|---|
| **11. Transparent AI** | **Quiz:** 1-a 2-b 3-c 4-a 5-b<br>**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T<br>**Odd one out:** A: picture 4; B: picture 3; C: picture 4<br>**Part A 1:** SHAP - Contributions; Counterfactual - What flips it; Probing - Layer encoding; Occlusion - Key regions<br>**Part A 2:** 10, 6, 0<br>**Part B 1:** Verified by removing the feature: Faithful; Sounds right, never tested: Plausible; Counterfactual actually flips output: Faithful; Generic 'many factors' text: Plausible<br>**Part B 2:** Model card - Use/metrics/limits; Datasheet - Data collection; System card - System risks; Disclosure - AI notice<br>**Challenge hints:** Q1: Misleads users and auditors about real model behaviour. • Q2: Attribution: what mattered; counterfactual: what would change it. • Q3: 6. • Q4: Not always: documentation, explanations, audits can suffice. • Q5: Main reasons + what could change the decision.<br>**Case study:** Open answers. Reward clear reasoning and examples.<br>**Code lab:** Blank: x; 10 [6, 4, 0]; weight 0; base 12, drops [3, 9, 0].<br>**Exam prep key points:** 1: One that reflects the model's real reasoning. • 2: Contributions vs minimal changes for a different outcome. • 3: Document of use, metrics and limits. • 4: Informed users, trust, accountability. • 5: Disclosure, documentation, attribution, counterfactuals, faithfulness, recourse. |
| **12. Trustworthy AI Systems** | **Quiz:** 1-a 2-b 3-c 4-a 5-b<br>**True/False:** 1-T 2-F 3-T 4-T 5-F 6-T<br>**Odd one out:** A: picture 4; B: picture 3; C: picture 4<br>**Part A 1:** Adversarial - Adversarial training/tests; Drift - Monitoring/retraining; Privacy - DP; Overconfidence - Calibration<br>**Part A 2:** 0.75, 0.15, over<br>**Part B 1:** 96, 71, 25<br>**Part B 2:** Train with corrupted images: Robustness; Add DP noise to training: Privacy; Adversarial training: Robustness; Federated learning on devices: Privacy<br>**Challenge hints:** Q1: Prevents confident errors when uncertain. • Q2: Shift: natural change; attack: deliberate manipulation. • Q3: 0.15, overconfident. • Q4: No: robustness, calibration, safety case, monitoring. • Q5: Noise, accents, low volume, code-mixing, adversarial audio.<br>**Case study:** Open answers. Reward clear reasoning and examples.<br>**Code lab:** Blank: idx; 0.9 → 0.75, 0.6 → 0.5; overconfident in both bins; gaps 0.15 and 0.1 weighted (4/6, 2/6) ≈ 0.13.<br>**Exam prep key points:** 1: Confidence matches actual accuracy. • 2: Natural change vs deliberate manipulation. • 3: Noise that limits individual information leakage. • 4: Structured evidence that a system is safe. • 5: Robustness, calibration, privacy, fairness, assurance, monitoring. |

## Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

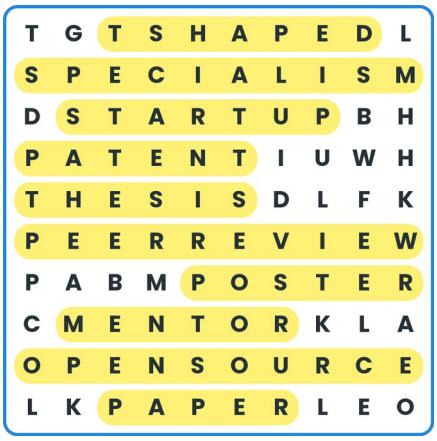
Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

# WORD SEARCH ANSWERS

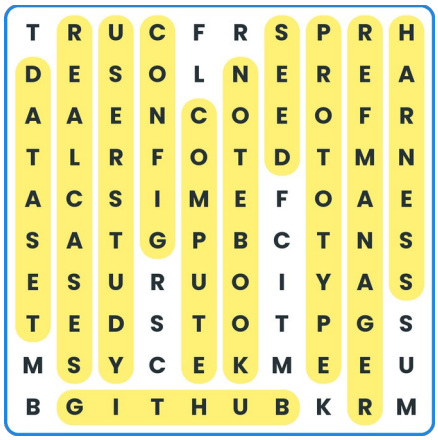
**AI DISCOVER**

**AI CONNECT**

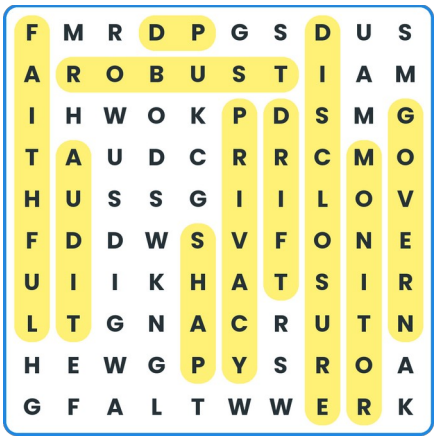
**AI SOLVE**




**AI RISE**




**AI CREATE**

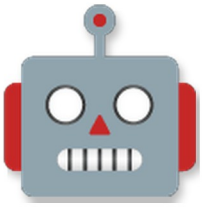



**AI CARE**

# MAMMA ADA CENTRE

## GAIO • Global AI Olympiad

From Intelligence to Impact Beyond Automation



## About this book

A postgraduate AI book for PG 1st Year students: concepts, research labs, stories & exam prep.

- 📖 Concept Explained: in-depth chapters
- 🔬 Research methods & workbench labs
- 📚 AI Stories & real-world case studies
- 🧭 Specialist roadmaps & research careers
- 🛡️ Transparent & trustworthy AI

**AI Basics** | **Communicate with AI** | **AI Applications**

**AI & Career** | **AI Tools** | **Responsible AI**

**Think | Explore | Create | A Better Tomorrow**

Mamma Ada Centre • In collaboration with FACM, Italy

Space for barcode / price / ISBN